Complex manifolds and Arithmetic varieties

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1 Some recollections on manifolds

We recall that a *nice* topological space that is locally (not necessarily globally) homeomorphic to n-dimensional Euclidean space is called a *topological* manifold. To be useful for geometrical and analytic purposes we impose some extra conditions to it i.e. we equip structures on the topological manifold. The most straight forward way to sew analytic structures with the manifold is to declare it's transition functions on local charts to be analytic functions. So for example, as we recall that, a C^∞-structure on a topological manifold is to declare it's transition functions to C^∞ functions on n-dimensional Euclidean spaces which are just automorphisms of the Euclidean space (right??). Therefore, the most educated guess we can make about the definition of a complex manifold is that it's a topological manifold that is locally homeomorphic to $\mathbb{C}^n$ and the transition functions are complex analytic or holomorphic in n complex variables. It's interesting that the very notion of holomorphicity of complex functions can be rewritten in the language of complex manifold theory. Namely, suppose we have a complex function $f : \mathbb{C}^m \to \mathbb{C}^n$. It's holomorphic as function it and only if the graph of this function that is the set

$$\Gamma(f) = \{(z, f(z)) : z \in \mathbb{C}^m\} \subset \mathbb{C}^{m+n}$$

is a *complex* submanifold of the Euclidean space $\mathbb{C}^{m+n}$. We have cheated in the sense that we have jumped from topological manifolds to the complex ones directly, of course there are topological manifolds that cannot be declared to be complex. But in this section, we want to discuss some philosophy of equipping structures of various types on topological manifolds that appeal us. Above, we have had a glimpse of how to sew analytic structures on manifolds. What about geometrical structures? As the reader knows, to do geometry on manifolds that first tool we need is a *metric* on (really the tangents spaces at points of) the manifold. The principle here (see [4]) is that:
A topological manifold M admits a geometrical structure if it admits a complete, locally

homogeneous metric
We recall that a metric is locally homogeneous if any two points x, y having neighborhoods $U, V \subset M$, there exists an isometry $h : U \to V$. If this holds globally on M, then we say that this metric is homogeneous. And being complete means complete as a metric space. Clearly the Riemannian metric does have these properties, for which we can use the Riemannian metric to do geometry on the manifold i.e. we can compute geometrical quantities like the curvature tensor using this metric. In fact, if we have a Riemannian manifold (M, g) with it's Riemannian metric $g_p : T_pM \otimes T_pM \to \mathbb{R}$, a smoothly varying (over points $p \in M$) $(0, 2)$ tensor field, X it's universal cover and Γ is the Deck transformation group of X (a subgroup of the group of isometries of X) something more elegant is true: **M is isometrically isomorphic to the quotient** X/Γ For its we refer to the exposition [4].

2 Complexification of vector spaces and complex manifolds

(In the first few sections, we will closely be following chapter 7 and 8 of [7] and also [10])
We will deliberately assume that the reader is familiar with smooth manifolds. Complex geometry is essentially doing everything you did with smooth manifolds via holomorphic functions instead of the smooth ones. And all the linear algebraic constructions for smooth manifolds, for example, tangent spaces are complexified (which we define below) in to complex vector spaces, and from these complexified (tangent) vector spaces we construct complex metric tensors, complexified covariant differentiations etc using which we can do geometry. There's a peculiar condition; not every even dimensional smooth manifold can be "made" into a complex manifold. The almost complex structure (which we will define) must satisfy the so called integrability condition. Without further ado, we start with the notion of complexification of real vector spaces.
Let V be a vector space over $\mathbb{R}$. We define the complexification of a real vector space as the space

$$V^{\mathbb{C}} := \{X + iY | X, Y \in V\}$$

which is clearly a vector space over $\mathbb{C}$. Interestingly, its complex dimension is same as the real dimension of V: $\forall X, Y \in V$, write them as

$$X = X^k e_k$$

$$Y = Y^k e_k$$

where e_k are the basis of V. So, in its complexified vector space lives the vector

$$X + iY = (X^k + iY^k)e_k$$

So, we can write $dim_{\mathbb{R}} V = dim_{\mathbb{C}} V^{\mathbb{C}}$. We often denote complexification as $V \otimes_{\mathbb{R}} \mathbb{C}$.

Complexification allows us to extend the linear operators A acting on V to act on $V^{\mathbb{C}}$ as

$$A(X+iY) = A(X) + iA(Y)$$

as well as to extend the linear functionals (belonging to the dual space V^*) $A : V \to \mathbb{R}$ to $A^{\mathbb{C}} : V^{\mathbb{C}} \to \mathbb{C}$ belonging to the complexification of the dual $(V^*)^{\mathbb{C}}$. This construction of complexifying real vector spaces to make complex vector spaces will be extremely important throughout out the lectures. You can verify whether complexification is functorial.

Now we turn our focus towards manifolds. All our manifolds will be even dimensional smooth manifolds in the first place, it will turn out that many of them are in fact higher dimensional generalizations of the complex plane and we can define holomorphic functions on these manifolds. At first we just naively make some replacements in the definition of a smooth manifold; namely we replace the word (and also the notion of course) of "smooth" with the word "holomorphic" to get the following:

With its underlying differentiable manifold M_0 of dimension $dim_{\mathbb{R}} M_0 = 2m$, we call a topological space M a complex manifold of complex dimension $dim_{\mathbb{C}} = m$ if it satisfies the following:

(i) If $\{U_i\}_{i\in I}$ is an open covering of M, there is a family of pairs $\{(U_i, \phi_i)\}_{i\in I}$ where ϕ_i is a homeomorphism from U_i to an open set $\phi_i(U_i) \subset \mathbb{C}^m$

(ii) Given U_i and U_j with $U_i \cap U_j \neq \emptyset$, the map $\psi_{ij} = \phi_j \circ \phi_i^{-1} : \phi_i(U_i \cap U_j) \to \phi_j(U_i \cap U_j)$ is holomorphic.

The most obvious example of a complex manifold is $\mathbb{C}^n$, which is often called the affine complex n-space, with a single chart consisting of the identity map $id : \mathbb{C}^n \to \mathbb{C}^n$. But for future needs, we now address an extremely important example of a complex manifold, called the complex projective space, denoted $\mathbb{CP}^n$; intuitively it's a space parametrizing (complex) one dimensional subspaces of $\mathbb{C}^{n+1}$. That is

$$\mathbb{CP}^n := \frac{\mathbb{C}^{n+1} - \{0\}}{\sim} \tag{1}$$

where the equivalence relation is defined as the following: two points lying on the same (complex) line through the origin of $\mathbb{C}^{n+1}$ are equivalent. So, if $z_1, z_2 \in \mathbb{C}^{n+1}$, then we will say $z_1 \sim z_2$ iff $z_1 = \lambda z_2$ for some $\lambda \in \mathbb{C}^*$.

Let us go some steps backwards; viewing the complex projective space $\mathbb{CP}^n$ as a topological space at first. It inherits the quotient topology via the quotient map

$$\pi : \mathbb{C}^{n+1} - \{0\} \to \mathbb{CP}^n \tag{2}$$

The points in topological space $\mathbb{CP}^n$ are equivalence classes $[z] \in \mathbb{CP}^n$ with $z = (z_0, z_1, ..., z_n) \in \mathbb{C}^{n+1}$ representing the class. The class $[z] = [z_0, z_1, ..., z_n]$ is actually all points lying on a particular line through the origin of $\mathbb{C}^{n+1}$ "compressed" into a single point. Here's another rule of the game: among the z_i's, at least one of them should be non zero. For example, if $z_0 \neq 0$, then

$$[z_0, z_1, ..., z_n] = [1, \frac{z_1}{z_0}, ..., \frac{z_n}{z_0}] = (\frac{z_1}{z_0}, ..., \frac{z_n}{z_0}) \in \mathbb{C}^n \tag{3}$$

So the point $[z] = [z_0, z_1, ..., z_n]$ then represents the line through the origin in $\mathbb{C}^{n+1}$ intersection the affine space $\mathbb{C}^n$ at the point $(\frac{z_1}{z_0}, ..., \frac{z_n}{z_0})$.
We can define open sets on it

$$U_i := \{[z] : z_i \neq 0\} \tag{4}$$

which is just the set of lines not contained or intersecting the complex hyperplane $z_i = 0$. Then the chart maps should be of the form

$$\phi_i : U_i \to \mathbb{C}^n$$

in which for any point $[z] = [z_0, ..., z_n] \in U_i$

$$\phi_i([z_0, ..., z_n]) := (\frac{z_0}{z_i}, ..., \frac{z_{i-1}}{z_i}, \frac{z_{i+1}}{z_i}, ..., \frac{z_n}{z_i}) \tag{5}$$

We can think of set of lines not intersecting more than just one complex hyperplane, let another such complex hyperplane be $z_j = 0$, so that the open set

$$U_j := \{[z] : z_j \neq 0\}$$

has non empty intersection with U_i. On the open set $U_i \cap U_j$, we can define the maps

$$\Phi_{ij} := \phi_j \circ \phi_i^{-1} : \phi_i(U_i \cap U_j) \to \phi_j(U_i \cap U_j) \tag{6}$$

which is essentially a holomorphic map $\mathbb{C}^n$ to $\mathbb{C}^n$. Pick a point from the image (under the chart map) of the intersection, $(z_1, ..., z_n) \in \phi(U_i \cap U_j) \subset \mathbb{C}^n$ where we are not caring about the index i or j for the moment,let the map act on it

$$\phi_j \circ \phi_i^{-1}(z_1, ..., z_n) = \phi_j([z_1, ..., z_i, 1, z_{i+1}, ..., z_n]) \tag{7}$$

$$= (\frac{z_1}{z_j}, .., \frac{z_i}{z_j}, \frac{1}{z_j}, ..., \frac{z_{j-1}}{z_j}, \frac{z_{j+1}}{z_j}, ..., \frac{z_n}{z_j})$$

So the map Φ_{ij} is just the transition map if we consider the space $\mathbb{CP}^n$ as a manifold. We adivce the reader to convince herself about the holomorphicity of the maps (4),(5).
So with these open sets U_i's, the complex projective space can be seen as an overlapping union

$$\mathbb{CP}^n = \bigcup_{i=0}^{n} U_i$$

Also, from (3), we can see that $\mathbb{CP}^n$ contains affine spaces of lower dimensions

$$\mathbb{CP}^n = \mathbb{C}^n \cup \mathbb{C}^{n-1} \cup ... \cup \mathbb{C}^0$$

Before actually stating (because we won't prove) the integrability condition, let us just stick to the "naive" version of the definition. With such a leap of faith, we will meet another important class of complex manifolds, known as smooth

projective algebraic varieties over the field of complex numbers. Algebraic varieties are (central objects in algebraic geometry of course, defined as) zero sets of irreducible polynomials over an algebraically closed field which in our exposition will be $\mathbb{C}$, on which complex rational functions are defined. Surely we won't dig deeper into the vast sea of algebraic geometry here, we will only introduce the concepts we will be in need for doing complex geometry, in cases where differential geometric methods might be useless (or too cumbersome to use). If these commutative algebraic stuffs frustrate the reader because of unfamiliarity, we recommend the reader to skip this part and make a safe jump to page:
The polynomial ring

$$\mathbb{C}[z_0, ..., z_n]$$

is the ring of polynomials in the $n+1$ variables $z_0, ..., z_n$ with coefficients in the field $\mathbb{C}$. Consider a finite subset of polynomials

$$S = \{f_1, ...f_s\} \subset \mathbb{C}[z_0, ..., z_n]$$

define the set

$$Z(S) := \{(z_0, ..., z_n) \in \mathbb{C}^{n+1} : f_i(z_0, ..., z_n) = 0\} \subset \mathbb{C}^{n+1} \tag{8}$$

with $i = 1, .., s$, as an algebraic set. If each of the polynomials are irreducible (that is, cannot be factorized), then this zero set is called an algebraic variety. It is obvious that the ideal generated by the elements of the set S would give us the same algebraic set i.e. $Z(\langle S \rangle) = Z(S)$.
The keen reader might have observed that the space $\mathbb{C}^{n+1}$ is itself an algebraic variety since its the zero set of the polynomial

$$f(z_0, ..., z_n) = 0$$

as it vanishes everywhere. The ring $\mathbb{C}[z_0, ..., z_n]$ contains all the polynomial functions that can be defined on the variety $\mathbb{C}^{n+1}$; technically its called the coordinate ring of $\mathbb{C}^{n+1}$. Similarly taking the quotient

$$\frac{\mathbb{C}[z_0, ..., z_n]}{\langle f_1, ..., f_s \rangle} \tag{9}$$

gives us a ring of polynomials that can be defined on the variety $Z(S)$, called the coordinate ring of it.
But what happens, when we consider zero sets of polynomials in the complex projective space $\mathbb{CP}^n$?
Pick a polynomial

$$f([z_0, ..., z_n]) = \sum_{i=0}^{n} a_i z_i^{n-i} = a_0 z_0^n + a_1 z_1^{n-1} + ... + a_n$$

where the coefficients $a_0, ..., a_n$ are complex numbers, evaluated at a point $[z_0, ..., z_n] \in \mathbb{CP}^n$. But the class $[z_0, ..., z_n]$ is not just a single point! This class contains other points of the form

$$(\lambda z_0, ..., \lambda z_n) \in \mathbb{C}^{n+1}$$

for some $\lambda \in \mathbb{C}^*$, for which the polynomial evaluates to a different complex value

$$f([z_0, ..., z_n]) = \sum_{i=0}^{n} a_i \lambda^{n-i} z_i^{n-i} = a_0 \lambda^n z_0^n + a_1 \lambda^{n-1} z_1^{n-1} + ... + a_n$$

giving us multivariate polynomials! Thus, studying zero sets of polynomials in $\mathbb{CP}^n$ with non-homogeneous polynomials is not sensible. But the good news is, homogeneous polynomials give us a remedy of this absurd situation. Homogeneous polynomials have the delicate feature that for equivalent points belonging to a particular class $[z_0, ..., z_n] \in \mathbb{CP}^n$ they give us "proportional" values.

Let us now perform some complexifications that we just defined. Recall that if $U, V \subseteq M_0$ are neighbourhoods of a point $p \in M_0$, with functions $f : U \to \mathbb{R}$ and $g : V \to \mathbb{R}$ we say that $f \sim_p g$ iif there exists $W \subseteq U \cap V$ containing p, such that $f|_W = g|_W$.
And the equivalence classes

$$[f]_p := \{g : g \sim_p f\}$$

are called the germs of functions at the point p. If we consider the pointwise addition of functions, germs can form a real vector space under the operations

$$[f_1]_p + [f_2]_p := [f_1 + f_2]_p$$

$$r.[f_1]_p := [r.f_1]_p$$

We will just write f instead of $[f]_p$ if the meaning is clear. We let $\mathcal{C}^\infty_p$ be the $\mathbb{R}$-vector space of germs of smooth (C^∞) real valued functions at p. We recall that the tanget space is then defined as the space

$$T_pM_0 := \{v : \mathcal{C}^\infty_p \to \mathbb{R}\}$$

where v satisfies the $\mathbb{R}$ linearity and Leibnitz product rule and is called a derivation. Its an elementary fact from smooth differential geometry that the tangent space is isomorphic to $\mathbb{R}^{2m}$ i.e. $T_pM_0 \simeq \mathbb{R}^{2m}$, which we won't prove here. For future usage, we mention the basis

$$\{\frac{\partial}{\partial x^1}|_p, ..., \frac{\partial}{\partial x^m}|_p, \frac{\partial}{\partial y^1}|_p, ..., \frac{\partial}{\partial y^m}|_p\} .....(*)$$

of the tangent space T_pM_0.
Let A^0_p be the $\mathbb{C}$-linear space of germs of smooth complex valued functions at p, constructed as

$$A^0_p = \mathcal{C}^\infty_p + i\mathcal{C}^\infty_p$$

whose elements look like $f = g + ih$, where $g, h \in \mathcal{C}^\infty_p, f \in A^0_p$
Then any $v \in T_pM_0$ naturally extends to a derivation on A^0_p which we write as $v^{\mathbb{C}} : A^0_p \to \mathbb{C}$. It operates as

$$f \mapsto v^{\mathbb{C}}(f) = v(g) + iv(h)$$

where $f = g + ih \in A^0_p$.
The construction above indicates to the complexification

$$T_pM_0 \otimes_{\mathbb{R}} \mathbb{C} = (T_pM_0)^{\mathbb{C}}$$

for which we have the direct sum decomposition

$$T_pM_0 \oplus iT_pM_0$$

We point out that for any $v^{\mathbb{C}} \in (T_pM_0)^{\mathbb{C}}$ we could write it as $v^{\mathbb{C}} = v_1 + iv_2$, where $v_1, v_2 \in T_pM_0$.

Clearly,

$$v^{\mathbf{C}}(f) = v_1(g) - v_2(h) + i[v_1(h) + v_2(g)]$$

as you can check.
As for the dimension, we have

$$dim_{\mathbf{C}}(T_pM_0)^{\mathbf{C}} = dim_{\mathbb{R}}(T_pM_0) = 2m$$

which is twice the dimension of the complex manifold which was assumed to be m. So we try to cut some part of the complexified tangent space in order to have something similar to our good old notion for tangent spaces for differentiable manifolds.
In order to do that, we define
Definition 1: *A complex structure on an* $\mathbb{R}$*-vector space* V *is an* $\mathbb{R}$ *linear endomorphism* $J : V \to V$ *such that* $J^2 = -Id_V$.
So using the basis $(*)$ of the tangent space $T_pM_0 \simeq \mathbb{R}^{2m}$, we immediately obtain a map $J_p : T_pM_0 \to T_pM_0$

$$\frac{\partial}{\partial x^j}|_p \mapsto J_p(\frac{\partial}{\partial x^j}|_p) = \frac{\partial}{\partial y^j}|_p \tag{10}$$

$$\frac{\partial}{\partial y^j}|_p \mapsto J_p(\frac{\partial}{\partial y^j}|_p) = -\frac{\partial}{\partial x^j}|_p \tag{11}$$

It is obvious that acting via J_p is essentially multiplying by i , $-i$, the imaginary unit. Implying that J_p has the following two eigenspaces

$$T_p^{1,0}(M) := \{v^{\mathbf{C}} \in (T_pM_0)^{\mathbf{C}} : J_p(v^{\mathbf{C}}) = iv^{\mathbf{C}}\} \tag{12}$$

$$T_p^{0,1}(M) := \{v^{\mathbf{C}} \in (T_pM_0)^{\mathbf{C}} : J_p(v^{\mathbf{C}}) = -iv^{\mathbf{C}}\} \tag{13}$$

Note that we have the direct sum decomposition

$$(T_pM_0)^{\mathbf{C}} = T_p^{1,0}(M) \oplus T_p^{0,1}(M) \tag{14}$$

and the dimensions are

$$dim_{\mathbf{C}}(T_p^{1,0}(M)) = dim_{\mathbf{C}}(T_p^{1,0}(M)) = m$$

To make things much more obvious, we define the bases vectors

$$\frac{\partial}{\partial z^j}|_p := \frac{1}{2}\{\frac{\partial}{\partial x^j}|_p - i\frac{\partial}{\partial y^j}|_p\} \tag{15}$$

$$\frac{\partial}{\partial \bar{z}^j}|_p := \frac{1}{2}\{\frac{\partial}{\partial x^j}|_p + i\frac{\partial}{\partial y^j}|_p\} \tag{16}$$

spanning the spaces $T_p^{1,0}(M)$ and $T_p^{0,1}(M)$ respectively.

Following equations (1), (2), we can easily see that

$$j_p(\frac{\partial}{\partial z^j}|_p) = i\frac{\partial}{\partial z^j}|_p \tag{17}$$

$$j_p(\frac{\partial}{\partial \bar{z}^j}|_p) = -i\frac{\partial}{\partial \bar{z}^j}|_p \tag{18}$$

The space $T_p^{1,0}(M)$ is called the holomorphic tangent space of the complex manifold M. This nomenclature is motivated by the following observation:
Take $f = g + ih \in A_p^0$, and apply (7) to get

$$2\frac{\partial}{\partial \bar{z}^j}|_p f = (\frac{\partial}{\partial x^j}|_p - i\frac{\partial}{\partial y^j}|_p)(g + ih) \tag{19}$$

$$= (\frac{\partial}{\partial x^j}g(p) - \frac{\partial}{\partial y^j}h(p)) + i(\frac{\partial}{\partial x^j}h(p) + \frac{\partial}{\partial y^j}g(p))$$

Now, the condition

$$\frac{\partial}{\partial \bar{z}^j}|_p f = 0 \tag{20}$$

yields

$$\frac{\partial g}{\partial x^j} = \frac{\partial h}{\partial y^j} \tag{21}$$

$$\frac{\partial h}{\partial x^j} = -\frac{\partial g}{\partial y^j} \tag{22}$$

which are the so called Cauchy Riemann equations.
Thus if f is the germ of a holomorphic function, $\frac{\partial}{\partial \bar{z}^j} = 0, \forall j$, so $\omega(f) = 0$ (which is the same as the condition (11)) $\forall \omega \in T_p^{0,1}M$
The space $T_p^{0,1}M$ is called the anti-holomorphic tangent space of the complex manifold M for similar reasons.

3 Hermitian manifold

We start this section with the following
Definition 2: *A hermitian form on a complex vector space V is defined to be a map $h : V \times V \to \mathbb{C}$ so that*
(i) $h(u, v)$ is $\mathbb{C}$ linear in u
(ii) $h(u, v)$ is $\mathbb{C}$ antilinear in v
(iii) $h(v, u) = \overline{h(u, v)}$
Note that (iii) implies that $h(v, v) \in \mathbb{R}$. The form h said to be *positive definite* if $h(v, v) > 0$ for all $v \neq 0$. A positive definite hermitian form on V is called a hermitian metric.

We now consider for the moment the underlying M_0 and define the Riemannian metric g on it as a type $(0, 2)$ tensor field on M_0 satisfying the following at each point $p \in M_0$:
(a) $g_p(u, v) = g_p(v, u)$
(b) $g_p(u, u) \geq 0$ where the equality holds iff $u = 0$
for each pair $u, v \in T_pM_0$. A Riemannian metric for each $p \in M_0$ is a smoothly varying map $g_p : T_pM_0 \otimes T_pM_0 \to \mathbb{R}$. If $p \in U \subseteq M_0$, then on the chart (U, ϕ) and the local cooridinates $\{x_\mu\}$ we can write

$$g_p = g_{\mu\nu}(p) dx^\mu \otimes dx^\nu \tag{23}$$

where

$$g_{\mu\nu}(p) = g_p(\frac{\partial}{\partial x^\mu}, \frac{\partial}{\partial x^\nu}) = g_{\mu\nu}(p)$$

We want to do something similar with our complex manifold M, *i.e.* defining a smoothly variyng Hermitian metric on each holomorphic tangent space $T_p^{1,0}M$. We will see that this is equivalent to defining a Riemannian metric on M_0 with the property that it preserves the complex structure at each point $p \in M_0$ i.e. the metric g_p is J_p invariant

$$g_p(J_pX, J_pY) = g_p(X, Y) \tag{24}$$

for any $X, Y \in T_pM_0$. The pair (M, g) where g satisfies (14) at each point $p \in M$ is called a Hermitian manifold. We can easily show that any complex manifold M is also a Hermitian manifold:
Let g be any Riemannian metric of M. Then we can define the new metric $\hat{g}$ by

$$\hat{g}_p(X, Y) \equiv \frac{1}{2}[g_p(X, Y) + g_p(J_pX, J_pY)]$$

Now

$$\hat{g}_p(J_pX, J_pY) = \frac{1}{2}[g_p(J_pX, J_pY) + g_p(J_p^2X, J_p^2Y)]$$

So $\hat{g}_p(J_pX, J_pY) = \hat{g}_p(X, Y)$, moreover, since g_p is positive definite so is $\hat{g}_p$.

We now show that only the mixed components of the Hermitian metric are non vanishing by the following observation

$$g_{\mu\nu} = g(\frac{\partial}{\partial z^\mu}, \frac{\partial}{\partial z^\nu})$$

By the J-invariance property of the Hermitian metric

$$g_{\mu\nu} = g(J\frac{\partial}{\partial z^\mu}, J\frac{\partial}{\partial z^\nu}) = -g(\frac{\partial}{\partial z^\mu}, \frac{\partial}{\partial z^\nu}) = -g_{\mu\nu} \tag{25}$$

implying that $g_{\mu\nu} = 0$. By similar arguments it can be shown that $g_{\bar{\mu}\bar{\nu}} = g(\frac{\partial}{\partial \bar{z}^\mu}, \frac{\partial}{\partial \bar{z}^\nu})$ also vanishes. So following (14) we write

$$\begin{aligned} g &= g_{\mu\nu} dz^\mu \otimes dz^\nu + g_{\bar{\mu}\nu} d\bar{z}^\mu \otimes dz^\nu + g_{\mu\bar{\nu}} dz^\mu \otimes d\bar{z}^\nu + g_{\bar{\mu}\bar{\nu}} d\bar{z}^\mu \otimes d\bar{z}^\nu \\ &= g_{\bar{\mu}\nu} d\bar{z}^\mu \otimes dz^\nu + g_{\mu\bar{\nu}} dz^\mu \otimes d\bar{z}^\nu \end{aligned} \tag{26}$$

(the p's in the subscripts are suppressed)
As an important remark, we want to show that this construction is indeed compatible with **Definition 2** of Hermitian metric. Recall that the holomorpic tangent space at the point p of the complex manifold M, $T_p^{1,0}M$ is spanned by the vectors $\{\frac{\partial}{\partial z^1}|_p, ..., \frac{\partial}{\partial z^m}|_p\}$. We define a Hermitian metric on this space

$$h_p : T_p^{1,0}M \times T_p^{1,0}M \to \mathbb{C}$$

as

$$h_p(X, Y) \equiv g_p(X, \overline{Y}) \tag{27}$$

for any two $X, Y \in T_p^{1,0}M$. Obviously

$$h_p(\alpha X, Y) = g_p(\alpha X, \overline{Y}) = \alpha g_p(X, \overline{Y})$$

$$h_p(X, \alpha Y) = g_p(X, \overline{\alpha Y}) = \overline{\alpha} g_p(X, \overline{Y})$$

$$\overline{h_p(X, Y)} = \overline{g_p(X, \overline{Y})} = g_p(\overline{X}, Y) = g_p(Y, \overline{X}) = h_p(Y, X)$$

verifying (i),(ii) and (iii) respectively of **Definition 2**. Positive definiteness of h_p follows from the positive definiteness of the metric g_p. We should also mention the J invariance property holds : writing

$$X = X^\mu \frac{\partial}{\partial z^\mu}$$

$$Y = Y^\nu \frac{\partial}{\partial z^\nu}$$

we see that

$$h_p(J_pX, J_pY) = g_p(X^\mu J_p(\frac{\partial}{\partial z^\mu}), Y^\nu J_p(\frac{\partial}{\partial \bar{z}^\nu}))$$

By equations (8),(9) this equals

$$g_p(iX^\mu \frac{\partial}{\partial z^\mu}, -iY^\nu \frac{\partial}{\partial \bar{z}^\nu})$$

$$= g_p(X^\mu \frac{\partial}{\partial z^\mu}, Y^\nu \frac{\partial}{\partial \bar{z}^\nu})$$

$$= g_p(X, \overline{Y})$$

$$= h_p(X, Y)$$

This justifies our claim that giving a Hermitian metric on each holomorphic tangent space $T_p^{1,0}M$ of the complex manifold M is equivalent to giving a J invariant Riemannian metric on the underlying differentiable manifold M_0.

4 Kähler Form and metric

We define a tensor field Ω on the Hermitian manifold (M, g) such that it acts on $X, Y \in T_pM_0$ as

$$\Omega_p(X, Y) = g_p(J_pX, Y) \tag{28}$$

Defining a it in this fashion makes it an anti symmetric tensor field in the following way:

$$\Omega_p(X, Y) = g_p(J_pX, Y)$$

as a Hermitian metric on M, g_p is J invariant so

$$g_p(J_pX, Y) = g_p(J_p^2X, J_pY)$$

$$= -g_p(X, J_pY)$$

$$= -g_p(J_pY, X)$$

$$= -\Omega_p(Y, X)$$

But this tensor itself is J invariant:

$$\Omega_p(J_pX, J_pY) = g_p(J_p^2X, J_pY)$$

$$= g_p(J_p^3X, J_p^2Y)$$

$$= g_p(J_pX, Y)$$

$$= \Omega_p(X, Y)$$

We now extend the domain to the complexified tangent space $(T_pM_0)^{\mathbb{C}}$ as follows

$$\Omega_{\mu\nu} = \Omega(\frac{\partial}{\partial z^\mu}, \frac{\partial}{\partial z^\nu}) = g(J\frac{\partial}{\partial z^\mu}, \frac{\partial}{\partial z^\nu}) = ig_{\mu\nu}$$

By (16), $\Omega_{\mu\nu} = 0$. Similarly, $\Omega_{\bar{\mu}\bar{\nu}} = 0$. So only the mixed index components survive which are

$$\begin{aligned}\Omega(\frac{\partial}{\partial \bar{z}^{\mu}}, \frac{\partial}{\partial z^{\nu}}) &= g(J\frac{\partial}{\partial \bar{z}^{\mu}}, \frac{\partial}{\partial z^{\nu}}) \\ &= g(-i\frac{\partial}{\partial \bar{z}^{\mu}}, \frac{\partial}{\partial z^{\nu}}) \\ &= -ig_{\bar{\mu}\nu}\end{aligned}$$

$$\begin{aligned}\Omega(\frac{\partial}{\partial z^\mu}, \frac{\partial}{\partial \bar{z}^\nu}) &= g(J\frac{\partial}{\partial z^\mu}, \frac{\partial}{\partial \bar{z}^\nu}) \\ &= g(i\frac{\partial}{\partial z^\mu}, \frac{\partial}{\partial \bar{z}^\nu}) \\ &= ig_{\mu\bar{\nu}}\end{aligned}$$

Since $g_{\bar{\mu}\nu} = g_{\nu\bar{\mu}}$ we see that

$$\Omega_{\mu\bar{\nu}} = -\Omega_{\bar{\nu}\mu} = ig_{\mu\bar{\nu}} \tag{29}$$

Writing the tensor in local coordinates

$$\begin{aligned}\Omega &= \Omega_{\bar{\mu}\nu} d\bar{z}^\mu \otimes dz^\nu + \Omega_{\mu\bar{\nu}} dz^\mu \otimes d\bar{z}^\nu \\ &= \Omega_{\bar{\mu}\nu} d\bar{z}^\mu \otimes dz^\nu - \Omega_{\bar{\nu}\mu} dz^\mu \otimes d\bar{z}^\nu\end{aligned}$$

Then by (20)

$$\begin{aligned}\Omega &= ig_{\mu\bar{\nu}}(d\bar{z}^\mu \otimes dz^\nu - dz^\mu \otimes d\bar{z}^\nu) \\ &= ig_{\mu\bar{\nu}}(d\bar{z}^\mu \otimes dz^\nu - dz^\nu \otimes d\bar{z}^\mu)\end{aligned}$$

So

$$\Omega = ig_{\mu\bar{\nu}} dz^\mu \wedge d\bar{z}^\nu \tag{30}$$

which we call the Kähler form. As an appetizer we would like to just mention that on a Hermitian manifold (M, g) if the Kähler form is closed i.e. $d\Omega = 0$, then M is called a Kähler manifold. Then the metric g is called a Kähler manifold. To delve deeper into that we need some tools from Dolbeault cohomology theory, which we briefly describe in the next section.

5 Complex differential forms

Upon equipping complex structure on the complexified tangent space $(T_pM_0)^{\mathbb{C}}$, we saw that there is this splitting

$$(T_pM_0)^{\mathbb{C}} = T_p^{1,0}M \oplus T_p^{0,1}M$$

in (5)
Now we take into account the dual tangent bundle $T_p^*M_0$ with the complex structure $J_p : T_p^*M_0 \to T_p^*M_0$
The dual tangent space $T_p^*M_0$ is spanned by the vectors

$$\{dx^1, ..., dx^m, dy^1, ..., dy^m\}$$

where we have omitted the subscripts p's in writing $dx^i_{|p}$
Defining

$$dz^\mu := dx^\mu + idy^\mu \tag{31}$$

$$d\bar{z}^\mu := dx^\mu - idy^\mu \tag{32}$$

we have that the complexified dual space $(T_p^*M_0)^{\mathbb{C}}$ is spanned by these vectors

$$\{dz^\mu, d\bar{z}^\mu\}$$

where $\mu = 1, ..., m$
Because of the map $J_p : T_p^*M_0 \to T_p^*M_0$ we have, as before in the case of tangent spaces, the splitting

$$(T_p^*M_0)^{\mathbb{C}} = T_p^{*1,0}M \oplus T_p^{*0,1}M \tag{33}$$

Let us explicitly construct the complex structure map in this case, as follows:

$$J(dx^\mu) = -dy^\mu \tag{34}$$

$$J(dy^\mu) = dx^\mu \tag{35}$$

(omitting the subscripts p's in J_p) from its easy to see that

$$J(dz^\mu) = -dy^\mu + idx^\mu = idz^\mu \tag{36}$$

$$J(d\bar{z}^\mu) = -dy^\mu - idx^\mu = -id\bar{z}^\mu \tag{37}$$

Similar to (3), (4) we have that

$$T_p^{*1,0}M = \{\omega \in (T_p^*M_0)^{\mathbb{C}} | J_p(\omega) = i\omega\} \tag{38}$$

$$T_p^{*0,1}M = \{\omega \in (T_p^*M_0)^{\mathbb{C}} | J_p(\omega) = -i\omega\} \tag{39}$$

Now (29),(30) implies that the space $T_p^{*1,0}M$ is spanned by the vectors dz^μ's and the space $T_p^{*0,1}M$ is spanned by the vectors $d\bar{z}^\mu$'s.
Indeed, for a $\omega^{1,0} \in T_p^{*1,0}M$ we can write

$$\omega^{1,0} = f(z^\mu)dz^\mu = \frac{\partial G(z)}{\partial z^\mu}dz^\mu + \frac{\partial G(z)}{\partial \bar{z}^\mu}d\bar{z}^\mu \tag{40}$$

iff

$$\frac{\partial G(z)}{\partial \bar{z}^\mu} = 0$$

which indicates holomorphicity.
Similarly, for a $\omega^{0,1} \in T_p^{*0,1}M$ we can write

$$\omega^{0,1} = f(\bar{z}^\mu)d\bar{z}^\mu = \frac{\partial G(\bar{z})}{\partial \bar{z}^\mu}d\bar{z}^\mu + \frac{\partial G(\bar{z})}{\partial z^\mu}dz^\mu \tag{41}$$

iff

$$\frac{\partial G(\bar{z})}{\partial z^\mu} = 0$$

which indicates anti-holomorphicity.
From now on set the notations $\Omega_p^{1,0}(M) = T_p^{*1,0}M$ and $\Omega_p^{0,1}(M) = T_p^{*0,1}M$, the holomorphic and anti-holomorphic 1-forms at point p, respectively.
Regardless of the point p, we can consider these constructions at the level of the holomorphic and anti holomorphic bundles, where the space of 1-forms are the set of smooth sections of these bundles. Any k-form is an element of the space $\Omega^k(M) = \Gamma(M, \Lambda^k T^*M)$. So we can write

$$\Omega^{1,0}(M) = \Gamma(M, T^{*1,0}M) \tag{42}$$

$$\Omega^{0,1}(M) = \Gamma(M, T^{*0,1}M) \tag{43}$$

The complex structure viewed now as a bundle endomorphism map $J : (T^*M_0)^{\mathbb{C}} \to (T^*M_0)^{\mathbb{C}}$ gives us the splitting

$$\Omega^1(M)^{\mathbb{C}} = \Omega^{1,0}(M) \oplus \Omega^{0,1}(M) \tag{44}$$

So a complex 1-form can be written as

$$\omega = \omega^{1,0} + \omega^{0,1}$$

$$= f(z^\mu)dz^\mu + f(\bar{z}^\mu)d\bar{z}^\mu$$

Before making further generalization to higher degree complex differential forms, let us study in a more concrete setting. Consider a complex 2-form, $\gamma \in \Omega^2(M)^{\mathbb{C}}$ which can be built via taking the wedge product of two complex 1-form:

$$\omega_1 = f_1(z^{\mu_1})dz^{\mu_1} + \bar{f}_2(\bar{z}^{\mu_1})d\bar{z}^{\mu_1} = f_1 dz^{\mu_1} + \bar{f}_2 d\bar{z}^{\mu_1}$$

$$\omega_2 = g_1(z^{\mu_2})dz^{\mu_2} + \bar{g}_2(\bar{z}^{\mu_2})d\bar{z}^{\mu_2} = g_1 dz^{\mu_2} + \bar{g}_2 d\bar{z}^{\mu_2}$$

$$\Rightarrow \omega_1 \wedge \omega_2 = f_1 g_1 dz^{\mu_1} \wedge dz^{\mu_2} + f_1 \bar{g}_2 dz^{\mu_1} \wedge d\bar{z}^{\mu_2} + \bar{f}_2 g_1 d\bar{z}^{\mu_1} \wedge dz^{\mu_2} + \bar{f}_2 \bar{g}_2 d\bar{z}^{\mu_1} \wedge d\bar{z}^{\mu_2}$$

Looking at this expansion above, we see that there are three types of objects here. As a product two holomorphic functions, $f_1 g_1$ is holomorphic as well, so the component

$$f_1 g_1 dz^{\mu_1} \wedge dz^{\mu_2}$$

can be called a holomorphic 2-form $\omega^{2,0}$ forming the space $\Omega^{2,0}(M)$ spanned by the basis $\{dz^{\mu_1} \wedge dz^{\mu_2}\}$
Similarly, the component

$$\bar{f}_2 \bar{g}_2 d\bar{z}^{\mu_1} \wedge d\bar{z}^{\mu_2}$$

can be called an anti-holomorphic 2-form forming the space $\Omega^{0,2}(M)$ spanned by the basis $\{d\bar{z}^{\mu_1} \wedge d\bar{z}^{\mu_2}\}$
Finally,

$$\begin{aligned} & f_1 \bar{g}_2 dz^{\mu_1} \wedge d\bar{z}^{\mu_2} + \bar{f}_2 g_1 d\bar{z}^{\mu_1} \wedge dz^{\mu_2} \\ &= f_1 \bar{g}_2 dz^{\mu_1} \wedge d\bar{z}^{\mu_2} - \bar{f}_2 g_1 dz^{\mu_2} \wedge d\bar{z}^{\mu_1} \\ &= (f_1 \bar{g}_2 - \bar{f}_2 g_1) dz^{\mu_1} \wedge d\bar{z}^{\mu_2} \end{aligned}$$

(where in the last line we made an exchange of indices) is mixed type object, having both a holomorphic and an anti-holomorphic component, we call it a mixed bidegree form, in this case its $\omega^{1,1}$ forming the space $\Omega^{1,1}(M)$ that is spanned the vectors $\{dz^{\mu_1} \wedge d\bar{z}^{\mu_2}\}$. In summary, $\gamma = \omega^{2,0} + \omega^{1,1} + \omega^{0,2}$ and we have the direct sum decomposition

$$\Omega^2(M)^{\mathbb{C}} = \Omega^{1,0}(M) \oplus \Omega^{1,1}(M) \oplus \Omega^{0,1}(M)$$

In general it holds that

$$\Omega^k(M)^{\mathbb{C}} = \bigoplus_{p+q=k} \Omega^{p,q}(M) \tag{45}$$

We now introduce the Dolbeault operators below:

$$\partial : \Omega^{p,q}(M) \to \Omega^{p+1,q}(M) \tag{46}$$

$$\bar{\partial} : \Omega^{p,q}(M) \to \Omega^{p,q+1}(M) \tag{47}$$

Take for example a $(1,1)$-form $\omega^{1,1} = f_{\mu_1 \bar{\mu}_2} dz^{\mu_1} \wedge \bar{z}^{\mu_2} \in \Omega^{1,1}(M)$
then the Dolbeault operator acts as follows:

$$\partial \omega^{1,1} = \frac{\partial f_{\mu_1 \bar{\mu}_2}}{\partial z^\nu} dz^\nu \wedge dz^{\mu_1} \wedge \bar{z}^{\mu_2} \in \Omega^{2,1}(M)$$

$$\bar{\partial} \omega^{1,1} = \frac{\partial f_{\mu_1 \bar{\mu}_2}}{\partial \bar{z}^\nu} d\bar{z}^\nu \wedge dz^{\mu_1} \wedge \bar{z}^{\mu_2} \in \Omega^{1,2}(M)$$

The exterior derivative is then acts as

$$d\omega^{1,1} = [\frac{\partial f_{\mu_1 \bar{\mu}_2}}{\partial z^\nu} dz^\nu \wedge dz^{\mu_1} \wedge \bar{z}^{\mu_2} + \frac{\partial f_{\mu_1 \bar{\mu}_2}}{\partial \bar{z}^\nu} d\bar{z}^\nu \wedge dz^{\mu_1} \wedge \bar{z}^{\mu_2}] \in \Omega^3(M)^{\mathbb{C}}$$

Thus the exterior derivative after extending as a linear operator to the complexified setting

$$d : \Omega^k(M)^{\mathbb{C}} \to \Omega^{k+1}(M)^{\mathbb{C}}$$

is essentially

$$d = \partial + \bar{\partial}$$

for which the condition $d^2 = 0$ implies that

$$\partial^2 = 0$$

$$\bar{\partial}^2 = 0$$

$$\partial\bar{\partial} + \bar{\partial}\partial = 0$$

In modern day mathematics, there is this sophisticated philosophy of Felix Klein saying that if you want to study a space (be it a geometric, algebraic, analytical or just a topological one), you should study the functions that can be defined on that space. So, not surprisingly, we study holomorphic functions on complex manifolds. Yet holomorphic functions are nothing but the 0-forms (forming the vector space $\Omega^0(M)^{\mathbb{C}} = \mathcal{F}(M)^{\mathbb{C}}$) on the manifold. For geometrical interests, we take differentials of these functions to get to the complex differential 1-forms ($\Omega^1(M)^{\mathbb{C}}$ that we just studied above) and then we deploy higher wedge products to construct higher degree forms. Let us define a holomorphic differential form below.
If an element $\gamma \in \Omega^{p,0}(M)$ on the complex manifold M (of complex dimension $dim_{\mathbb{C}} M = m$) satisfies the condition

$$\bar{\partial}\gamma = 0 \tag{48}$$

we call γ a holomorphic p-form.
For any holomorphic function h on the manifold M, it must satisfy the condition

$$\frac{\partial h}{\partial \bar{z^\mu}} = 0 \tag{49}$$

where $0 \leq \mu \leq m$, we see that this is just a generalization of the condition in equation (11) to higher (complex) dimensions. This makes the condition (38) of holomorphicity of differential forms obvious. The form γ can be written in local coordinates as

$$\gamma = \gamma_{\mu_1 \ldots \mu_p} dz^{\mu_1} \wedge \ldots \wedge dz^{\mu_p}$$

$$\Rightarrow \bar{\partial}\gamma = \frac{\partial}{\partial \bar{z}^\mu} \gamma_{\mu_1 \ldots \mu_p} dz^{\mu} \wedge dz^{\mu_1} \wedge \ldots \wedge dz^{\mu_p}$$

The condition (38) is satisfied iff

$$\frac{\partial}{\partial \bar{z}^\mu} \gamma_{\mu_1 \ldots \mu_p} = 0$$

which is nothing but the condition (39). Note that all of these constructions have already been done for complex 1-forms in our above discussion (see equation (31)).
As the last topic of this section, we describe a certain type of cohomology groups on complex manifolds, which is not quite a topological invariant of

the manifold.
We stress the fact that any element $\gamma \in \Omega^{p,0}(M)$ is not in general holomorphic (if it doesn't satisfy the condition (38), which can happen if the component function $\gamma_{\mu_1 \ldots \mu_p}$ is of mixed type). Keeping that in mind, the Dolbeault complex is constructed as follows:

$$\Omega^{p,0}(M) \xrightarrow{\bar{\partial}} \Omega^{p,1}(M) \xrightarrow{\bar{\partial}} \Omega^{p,2}(M) \xrightarrow{\bar{\partial}} ..... \xrightarrow{\bar{\partial}} \Omega^{p,m}(M)$$

This is indeed a (co)chain complex since $\bar{\partial}^2 = 0$ as a Dolbeault operator.
At the (p,q)-th position, we can thus compute cohomology as the quotient

$$\frac{ker(\bar{\partial} : \Omega^{p,q}(M) \to \Omega^{p,q+1}(M))}{Im(\bar{\partial} : \Omega^{p,q-1}(M) \to \Omega^{p,q}(M))} =: H^{p,q}_{\bar{\partial}}(M) \tag{50}$$

The kernel of the $\bar{\partial}$ at the (p,q)-th position refers to the degree (p,q) cycles, whose set we denote as

$$Z^{p,q}(M) := \{\omega \in \Omega^{p,q}(M) : \bar{\partial}\omega = 0\} \tag{51}$$

and the image to degree (p,q) boundaries

$$B^{p,q}(M) := \{\gamma \in \Omega^{p,q}(M) : \exists \eta \in \Omega^{p,q-1}(M), \bar{\partial}\eta = \gamma\} = \bar{\partial}\Omega^{p,q-1} \tag{52}$$

So we can write the Dolbeault cohomology group of the complex manifold M as

$$H^{p,q}_{\bar{\partial}}(M) := \frac{Z^{p,q}(M)}{B^{p,q}(M)} \tag{53}$$

which are vector spaces over $\mathbb{C}$ and the complex dimensions of these $\mathbb{C}$-vector spaces

$$dim_{\mathbb{C}} H^{p,q}_{\bar{\partial}}(M) =: h^{p,q} \tag{54}$$

are called Hodge numbers.
Note that the direct sum decomposition in (36) of complex differential k-forms implies the following decompositions of complex k-cycles

$$Z^k(M)^{\mathbb{C}} = \bigoplus_{p+q=k} Z^{p,q}(M) \tag{55}$$

and that of complex k-boundaries

$$B^k(M)^{\mathbb{C}} = \bigoplus_{p+q=k} B^{p,q}(M) \tag{56}$$

Taking their complex dimensions

$$dim_{\mathbb{C}} Z^k(M)^{\mathbb{C}} = \sum_{p+q=k} dim_{\mathbb{C}} Z^{p,q}(M) \tag{57}$$

$$dim_{\mathbb{C}} B^k(M)^{\mathbb{C}} = \sum_{p+q=k} dim_{\mathbb{C}} B^{p,q}(M) \tag{58}$$

$$\Rightarrow dim_{\mathbb{C}} Z^k(M)^{\mathbb{C}} - dim_{\mathbb{C}} B^k(M)^{\mathbb{C}} = \sum_{p+q=k} (dim_{\mathbb{C}} Z^{p,q}(M) - dim_{\mathbb{C}} B^{p,q}(M))$$

$$\Rightarrow dim_{\mathbb{C}} \frac{Z^k(M)^{\mathbb{C}}}{B^k(M)^{\mathbb{C}}} = \sum_{p+q=k} dim_{\mathbb{C}} \frac{Z^{p,q}(M)}{B^{p,q}(M)}$$

$$\Rightarrow H^k_{dR}(M, \mathbb{C}) \simeq \bigoplus_{p+q=k} H^{p,q}_{\bar{\partial}}(M) \tag{59}$$

As simple as it seems, the surprising fact is that the decomposition in (59) is NOT true in general complex manifolds, the subtle condition of being compact Kähler manifold ensures this decomposition, this is one of the many important significance of Kähler metric. We stress the fact that this decomposition does not depend on the Kähler structure chosen on the complex manifold. More on this in the section of harmonic forms.
We close this section by emphasizing on the fact that the complex structure on any (even dimensional) smooth manifold is not unique, each of them can induce a different splitting of the tangent bundle into holomorphic and anti-holomorphic tangent subbundles. That splitting later ends up inducing direct sum decomposition of complex differential forms (that we saw in (36)), so the Dolbeault cohomology groups (44) closely depends on the complex structure $J : TM^{\mathbb{C}} \to TM^{\mathbb{C}}$, making it an analytical object besides being a topological one as well. We will discuss on this in depth on our section on complex structure deformations and moduli space of such structures.

6 Covariant differentiation

We first consider the underlying smooth manifold (of even dimension $2m$) M_0. Let $\mathcal{V}(M_0)$ denote the set of smooth vector fields on this manifold.
The affine connection is defined as the map

$$\nabla : \mathcal{V}(M_0) \times \mathcal{V}(M_0) \to \mathcal{V}(M_0) \tag{60}$$

acting as

$$(V_1, V_2) \mapsto \nabla_{V_1} V_2$$

and satisfying the following conditions

$$\nabla_{V_1}(V_2 + V_3) = \nabla_{V_1} V_2 + \nabla_{V_1} V_3 \tag{61}$$

$$\nabla_{(V_1+V_2)} V_3 = \nabla_{V_1} V_3 + \nabla_{V_2} V_3 \tag{62}$$

$$\nabla_{(fV_1)} V_2 = f \nabla_{V_1} V_2 \tag{63}$$

$$\nabla_{V_1}(fV_2) = V_1[f] V_2 + f \nabla_{V_1} V_2 \tag{64}$$

$$\nabla_{V_1} f = V_1[f] \tag{65}$$

for vector fields $V_1, V_2, V_3 \in \mathcal{V}(M_0)$ and smooth functions $f \in \mathcal{F}(M_0)$. Here the notation $V_1[f]$ $(= \nabla_{V_1} f)$ simply means

$$\sum_{\mu=1}^{2m} V_1^{\mu}|_p \frac{\partial}{\partial x^{\mu}}|_p f$$

where the vectors $\{\frac{\partial}{\partial x^{\mu}}|_p\}_{\mu=1}^{2m}$ (which we will also denote as $\{e_\mu\}_{\mu=1}^{2m}$) spans the tangent space T_pM_0. Having set the notations thus far, we define the *connection coefficient* $\Gamma_{\nu\mu}^{\lambda}$ as follows

$$\nabla_\nu e_\mu = \nabla_{e_\nu} e_\mu = e_\lambda \Gamma_{\nu\mu}^{\lambda}$$

Let us pick two vectors from this tangent space, $U = U^\mu e_\mu$, $V = V^\nu e_\nu$. Now applying the affine connection (along with the rules (54), (55)) yields

$$\begin{aligned}
\nabla_U V &= \nabla_{U^\mu e_\mu}(V^\nu e_\nu) \\
&= U^\mu \nabla_{e_\mu}(V^\nu e_\nu) \\
&= U^\mu (V^\nu \nabla_{e_\mu} e_\nu + e_\nu \nabla_{e_\mu}(V^\nu)) \\
&= U^\mu (V^\nu e_\lambda \Gamma_{\nu\mu}^{\lambda} + e_\mu[V^\nu] e_\nu) \\
&= U^\mu (V^\nu e_\lambda \Gamma_{\nu\mu}^{\lambda} + e_\mu[V^\lambda] e_\lambda) \\
&= U^\mu (V^\nu \Gamma_{\nu\mu}^{\lambda} + \frac{\partial}{\partial x^\mu}[V^\lambda]) e_\lambda \\
&= U^\mu (\nabla_\mu V^\lambda) e_\lambda
\end{aligned}$$

where

$$\nabla_\mu V^\lambda = \frac{\partial}{\partial x^\mu}[V^\lambda] + V^\nu \Gamma_{\nu\mu}^{\lambda} \tag{66}$$

Take a 1-form $\omega \in \Omega^1(M_0)$ and a vector field $V \in \mathcal{V}(M_0)$. Clearly, their inner product would yield

$$\begin{aligned}
\langle \omega, V \rangle &= \langle \omega_\mu dx^\mu, V^\nu \frac{\partial}{\partial x^\nu} \rangle = \omega_\mu V^\nu \langle dx^\mu, \frac{\partial}{\partial x^\nu} \rangle \\
&= \omega_\mu V^\nu \delta_\nu^\mu = \omega_\mu V^\mu
\end{aligned}$$

which, as a product of smooth functions, is a smooth function on the manifold M_0 i.e. $\langle \omega, V \rangle \in \mathcal{F}(M_0)$. Now consider another vector field $W = W^\gamma \frac{\partial}{\partial x^\gamma} \in \mathcal{V}(M_0)$ and feed it the function $\langle \omega, V \rangle$ to get

$$W[\langle \omega, V \rangle] = \nabla_W[\langle \omega, V \rangle] = \langle \nabla_W \omega, V \rangle + \langle \omega, \nabla_W V \rangle$$

(here we have used (56))

$$\Rightarrow \langle \nabla_W \omega, V \rangle = W[\langle \omega, V \rangle] - \langle \omega, \nabla_W V \rangle \tag{67}$$

$$\Rightarrow \langle \nabla_{W^\gamma \frac{\partial}{\partial x^\gamma}} \omega_\mu dx^\mu, V^\nu \frac{\partial}{\partial x^\nu} \rangle = W^\gamma \frac{\partial}{\partial x^\gamma} [\langle \omega_\mu dx^\mu, V^\nu \frac{\partial}{\partial x^\nu} \rangle] - \langle \omega_\mu dx^\mu, \nabla_{W^\gamma \frac{\partial}{\partial x^\gamma}} V^\nu \frac{\partial}{\partial x^\nu} \rangle$$

$$\Rightarrow \langle \nabla_{\frac{\partial}{\partial x^\gamma}} \omega_\mu dx^\mu, V^\nu \frac{\partial}{\partial x^\nu} \rangle = \frac{\partial}{\partial x^\gamma} [\langle \omega_\mu dx^\mu, V^\nu \frac{\partial}{\partial x^\nu} \rangle] - \langle \omega_\mu dx^\mu, \nabla_{\frac{\partial}{\partial x^\gamma}} V^\nu \frac{\partial}{\partial x^\nu} \rangle$$

$$= \frac{\partial}{\partial x^\gamma} [\omega_\mu V^\mu] - \langle \omega_\mu dx^\mu, \frac{\partial V^\nu}{\partial x^\gamma} \frac{\partial}{\partial x^\nu} + V^\nu \Gamma^\lambda_{\gamma\nu} \frac{\partial}{\partial x^\lambda} \rangle$$

$$= V^\mu \frac{\partial \omega_\mu}{\partial x^\gamma} + \omega_\mu \frac{\partial V^\mu}{\partial x^\gamma} - \omega_\mu \frac{\partial V^\nu}{\partial x^\gamma} \langle dx^\mu, \frac{\partial}{\partial x^\nu} \rangle - \omega_\mu V^\nu \Gamma^\lambda_{\gamma\nu} \langle dx^\mu, \frac{\partial}{\partial x^\lambda} \rangle$$

$$= V^\mu \frac{\partial \omega_\mu}{\partial x^\gamma} + \omega_\mu \frac{\partial V^\mu}{\partial x^\gamma} - \omega_\mu \frac{\partial V^\mu}{\partial x^\gamma} - V^\nu \Gamma^\lambda_{\gamma\nu} \omega_\lambda$$

$$= V^\mu \frac{\partial \omega_\mu}{\partial x^\gamma} - V^\nu \Gamma^\lambda_{\gamma\nu} \omega_\lambda$$

$$\Rightarrow \langle \nabla_\gamma \omega_\mu dx^\mu, V^\nu \frac{\partial}{\partial x^\nu} \rangle = \langle \frac{\partial \omega_\mu}{\partial x^\gamma} dx^\mu, V^\nu \frac{\partial}{\partial x^\nu} \rangle - \langle \Gamma^\lambda_{\gamma\nu} \omega_\lambda dx^\mu, V^\nu \frac{\partial}{\partial x^\nu} \rangle$$

$$\nabla_\gamma \omega = \frac{\partial \omega_\mu}{\partial x^\gamma} dx^\mu - \Gamma^\lambda_{\gamma\nu} \omega_\lambda dx^\mu \tag{68}$$

(here $\nabla_\gamma = \nabla_{\frac{\partial}{\partial x^\gamma}}$) If the function ω_μ is constant (say = 1), then we get that

$$\nabla_\gamma dx^\mu = -\Gamma^\mu_{\gamma\lambda} dx^\lambda \tag{69}$$

For higher order mixed (p, q)-rank tensors , covariant derivatives look like

$$\nabla_\gamma T^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q} = \frac{\partial}{\partial x^\gamma} T^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_q} + \Gamma^{\mu_1}_{\gamma\theta} T^{\theta \mu_2 \dots \mu_p}_{\nu_1 \dots \nu_q} + \dots + \Gamma^{\mu_p}_{\gamma\theta} T^{\mu_1 \dots \mu_{p-1} \theta}_{\nu_1 \dots \nu_q} - \Gamma^\delta_{\gamma\nu_1} T^{\mu_1 \dots \mu_p}_{\delta \nu_2 \dots \nu_q} - \dots - \Gamma^\delta_{\gamma\nu_q} T^{\mu_1 \dots \mu_p}_{\nu_1 \dots \nu_{q-1} \delta} \tag{70}$$

As a $(0, 2)$ ranked tensor, the covariant derivative of the Riemannian metric will be

$$\nabla_\gamma g_{\mu\nu} = \partial_\gamma g_{\mu\nu} - \Gamma^\lambda_{\gamma\nu} g_{\mu\lambda} - \Gamma^\lambda_{\gamma\mu} g_{\lambda\nu} \tag{71}$$

If the covariant derivative of the metric tensor vanishes, we say that the metric is covariantly constant, for which we write

$$0 = \partial_\gamma g_{\mu\nu} - \Gamma^\lambda_{\gamma\nu} g_{\mu\lambda} - \Gamma^\lambda_{\gamma\mu} g_{\lambda\nu} \tag{72}$$

cyclic permutation of the indices (γ, μ, ν) yields two more equations

$$0 = \partial_\mu g_{\nu\gamma} - \Gamma^\lambda_{\mu\gamma} g_{\nu\lambda} - \Gamma^\lambda_{\mu\nu} g_{\lambda\gamma} \tag{73}$$

$$0 = \partial_\nu g_{\gamma\mu} - \Gamma^\lambda_{\nu\mu} g_{\gamma\lambda} - \Gamma^\lambda_{\nu\gamma} g_{\lambda\mu} \tag{74}$$

The combination -(63) + (64) + (65) gives

$$-\partial_\gamma g_{\mu\nu} + \Gamma^\lambda_{\gamma\nu} g_{\mu\lambda} + \Gamma^\lambda_{\gamma\mu} g_{\lambda\nu} + \partial_\mu g_{\nu\gamma} - \Gamma^\lambda_{\mu\gamma} g_{\nu\lambda} - \Gamma^\lambda_{\mu\nu} g_{\lambda\gamma} + \partial_\nu g_{\gamma\mu} - \Gamma^\lambda_{\nu\mu} g_{\gamma\lambda} - \Gamma^\lambda_{\nu\gamma} g_{\lambda\mu} = 0 \tag{75}$$

$$\Rightarrow -\partial_\gamma g_{\mu\nu} + \partial_\mu g_{\nu\gamma} + \partial_\nu g_{\gamma\mu} + (\Gamma^\lambda_{\gamma\nu} g_{\mu\lambda} - \Gamma^\lambda_{\nu\gamma} g_{\lambda\mu}) + (\Gamma^\lambda_{\gamma\mu} g_{\lambda\nu} - \Gamma^\lambda_{\mu\gamma} g_{\nu\lambda}) - (\Gamma^\lambda_{\mu\nu} g_{\lambda\gamma} + \Gamma^\lambda_{\nu\mu} g_{\gamma\lambda}) = 0 \tag{76}$$

$$\Rightarrow -\partial_\gamma g_{\mu\nu} + \partial_\mu g_{\nu\gamma} + \partial_\nu g_{\gamma\mu} + (\Gamma^\lambda_{\gamma\nu} - \Gamma^\lambda_{\nu\gamma}) g_{\mu\lambda} + (\Gamma^\lambda_{\gamma\mu} - \Gamma^\lambda_{\mu\gamma}) g_{\lambda\nu} - (\Gamma^\lambda_{\mu\nu} + \Gamma^\lambda_{\nu\mu}) g_{\lambda\gamma} = 0 \tag{77}$$

where we have used the symmetry of the Riemannian metric. We now define the torsion tensor as

$$T^\lambda_{\gamma\nu} = \Gamma^\lambda_{\gamma\nu} - \Gamma^\lambda_{\nu\gamma} \tag{78}$$

which is obviously anti-symmetric with respect to the lower indices.
Let us introduce some additional notations:

$$\Gamma^\lambda_{[\gamma\nu]} = \frac{1}{2}(\Gamma^\lambda_{\gamma\nu} - \Gamma^\lambda_{\nu\gamma}) \tag{79}$$

$$\Gamma^\lambda_{(\mu\nu)} = \frac{1}{2}(\Gamma^\lambda_{\mu\nu} + \Gamma^\lambda_{\nu\mu}) \tag{80}$$

$$\left\{ \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\} = \frac{1}{2} g^{\lambda\gamma}(\partial_\mu g_{\nu\gamma} + \partial_\nu g_{\gamma\mu} - \partial_\gamma g_{\mu\nu}) \tag{81}$$

Then (68) becomes

$$-\partial_\gamma g_{\mu\nu} + \partial_\mu g_{\nu\gamma} + \partial_\nu g_{\gamma\mu} + T^\lambda_{\gamma\nu} g_{\mu\lambda} + T^\lambda_{\gamma\mu} g_{\lambda\nu} - 2\Gamma^\lambda_{(\mu\nu)} g_{\lambda\gamma} = 0 \tag{82}$$

$$\Rightarrow \Gamma^\lambda_{(\mu\nu)} = \frac{1}{2} g^{\lambda\gamma}(\partial_\mu g_{\nu\gamma} + \partial_\nu g_{\gamma\mu} - \partial_\gamma g_{\mu\nu}) + \frac{1}{2}(T^\lambda_{\gamma\nu} g_{\mu\lambda} g^{\lambda\gamma} + T^\lambda_{\gamma\mu} g_{\lambda\nu} g^{\lambda\gamma}) \tag{83}$$

$$\Rightarrow \Gamma^\lambda_{(\mu\nu)} = \frac{1}{2} g^{\lambda\gamma}(\partial_\mu g_{\nu\gamma} + \partial_\nu g_{\gamma\mu} - \partial_\gamma g_{\mu\nu}) + \frac{1}{2}(T^\lambda_{\gamma\nu} \delta^\gamma_\mu + T^\lambda_{\gamma\mu} \delta^\gamma_\nu) \tag{84}$$

$$\Rightarrow \Gamma^\lambda_{(\mu\nu)} = \frac{1}{2} g^{\lambda\gamma}(\partial_\mu g_{\nu\gamma} + \partial_\nu g_{\gamma\mu} - \partial_\gamma g_{\mu\nu}) + \frac{1}{2}(T^\lambda_{\mu\nu} + T^\lambda_{\nu\mu}) \tag{85}$$

$$\Rightarrow \Gamma^\lambda_{(\mu\nu)} = \left\{ \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\} + \frac{1}{2}(T^\lambda_{\mu\nu} + T^\lambda_{\nu\mu}) \tag{86}$$

From (70),(71) we can write

$$\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{[\mu\nu]} + \Gamma^\lambda_{(\mu\nu)} \tag{87}$$

using (69)

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2} T^\lambda_{\mu\nu} + \Gamma^\lambda_{(\mu\nu)} \tag{88}$$

If all the torsion tensors vanish then we call the connection ∇ a Levi-Civita connection, in which case we have

$$\Gamma^\lambda_{\mu\nu} = \left\{ \begin{array}{c} \lambda \\ \mu\nu \end{array} \right\} = \frac{1}{2} g^{\lambda\gamma}(\partial_\mu g_{\nu\gamma} + \partial_\nu g_{\gamma\mu} - \partial_\gamma g_{\mu\nu}) \tag{89}$$

we now define the Riemann curvature tensor as

$$R : \mathcal{V}(M_0) \otimes \mathcal{V}(M_0) \otimes \mathcal{V}(M_0) \to \mathcal{V}(M_0) \tag{90}$$

such that for vector fields $X, Y, Z \in \mathcal{V}(M_0)$

$$R(X, Y, Z) = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z \tag{91}$$

where $[X, Y] = XY - YX$ is the Lie bracket. Its common to write this tensor as $R(X, Y)Z$. Let's see the tensorial property i.e. multilinearity of the Riemann curvature tensor

$$\begin{aligned} R(fX, gY)hZ &= \nabla_{fX} \nabla_{gY}(hZ) - \nabla_{gY} \nabla_{fX}(hZ) - \nabla_{[fX,gY]} hZ \\ &= f\nabla_X(g\nabla_Y(hZ)) - g\nabla_Y(f\nabla_X(hZ)) - \nabla_{[fX,gY]}(hZ) \end{aligned}$$

We should be careful about the last term since it involves a Lie bracket. Consider another smooth function j on the manifold M_0 i.e. $j \in \mathcal{F}(M_0)$. Now we compute the Lie bracket

$$\begin{aligned} [fX, gY][j] &= fX(gY[j]) - gY(fX[j]) \\ &= fX[g]Y[j] + fgXY[j] - gY[f]X[j] - gfYX[j] \\ &= fg(XY - YX)[j] - gY[f]X[j] + fX[g]Y[j] \\ \Rightarrow [fX, gY][j] &= (fg(XY - YX) - gY[f]X + fX[g]Y)[j] \\ \Rightarrow [fX, gY] &= (fg[X, Y] - gY[f]X + fX[g]Y) \end{aligned} \tag{92}$$

Therefore,

$$\begin{aligned} \nabla_{[fX,gY]}(hZ) &= \nabla_{fg[X,Y] - gY[f]X + fX[g]Y}(hZ) \\ &= \nabla_{fg[X,Y]}(hZ) - \nabla_{gY[f]X}(hZ) + \nabla_{fX[g]Y}(hZ) \\ &= fg\nabla_{[X,Y]}(hZ) - gY[f]\nabla_X(hZ) + fX[g]\nabla_Y(hZ) \end{aligned}$$

where in the second and third line we have used (53),(54) respectively.
Thus

$$\begin{aligned} R(fX, gY)hZ &= (f\nabla_X(g\nabla_Y(hZ)) - fX[g]\nabla_Y(hZ)) - (g\nabla_Y(f\nabla_X(hZ)) \\ &\quad - gY[f]\nabla_X(hZ)) - fg\nabla_{[X,Y]}(hZ) \\ &= fg\nabla_X\nabla_Y(hZ) - fg\nabla_Y\nabla_X(hZ) - fg\nabla_{[X,Y]}(hZ) \\ &= fg\nabla_X(Y[h]Z + h\nabla_Y Z) - fg\nabla_Y(X[h]Z + h\nabla_X Z) - fg[X, Y][h]Z - fgh\nabla_{[X,Y]}Z \\ &= fg(XY[h] + Y[h]\nabla_X Z + X[h]\nabla_Y Z + h\nabla_X\nabla_Y Z) \\ &\quad - fg(YX[h]Z + X[h]\nabla_Y Z + Y[h]\nabla_X Z + h\nabla_Y\nabla_X Z) \\ &\quad - fgh\nabla_{[X,Y]}Z \\ &= fgh(\nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z) \end{aligned}$$

$$= fghR(X,Y)Z$$

So it indeed possesses the tensorial property

$$R(fX,gY)hZ = fghR(X,Y)Z \tag{93}$$

Writing the vector fields as

$$X = X^\lambda e_\lambda$$
$$Y = Y^\mu e_\mu$$
$$Z = Z^\nu e_\nu$$

(where $e_i = \frac{\partial}{\partial x^i}$)(84) implies that

$$R(X,Y)Z = R(X^\lambda e_\lambda, Y^\mu e_\mu)Z^\nu e_\nu = X^\lambda Y^\mu Z^\nu R(e_\lambda, e_\mu)e_\nu$$

As R maps three vector fields to a vector field, it is a type $(1,3)$ tensor field, so its components in local coordinates should look like $R^\kappa_{\lambda\mu\nu}$ which we compute

$$R^\kappa_{\lambda\mu\nu} = \langle dx^\kappa, R(e_\mu, e_\nu)e_\lambda\rangle$$
$$= \langle dx^\kappa, \nabla_\mu\nabla_\nu e_\lambda - \nabla_\nu\nabla_\mu e_\lambda - \nabla_{[e_\mu,e_\nu]}e_\lambda\rangle$$
$$= \langle dx^\kappa, \nabla_\mu\nabla_\nu e_\lambda - \nabla_\nu\nabla_\mu e_\lambda\rangle$$
$$= \langle dx^\kappa, \nabla_\mu(\Gamma^\eta_{\nu\lambda}e_\eta) - \nabla_\nu(\Gamma^\eta_{\mu\lambda}e_\eta)\rangle$$
$$= \langle dx^\kappa, (\partial_\mu\Gamma^\eta_{\nu\lambda})e_\eta + \Gamma^\eta_{\nu\lambda}\Gamma^\xi_{\mu\eta}e_\xi - (\partial_\nu\Gamma^\eta_{\mu\lambda})e_\eta - \Gamma^\eta_{\mu\lambda}\Gamma^\xi_{\nu\eta}e_\xi\rangle$$
$$= \partial_\mu\Gamma^\eta_{\nu\lambda}\langle dx^\kappa, e_\eta\rangle - \partial_\nu\Gamma^\eta_{\mu\lambda}\langle dx^\kappa, e_\eta\rangle + \Gamma^\eta_{\nu\lambda}\Gamma^\xi_{\mu\eta}\langle dx^\kappa, e_\xi\rangle - \Gamma^\eta_{\mu\lambda}\Gamma^\xi_{\nu\eta}\langle dx^\kappa, e_\xi\rangle$$
$$= \partial_\mu\Gamma^\eta_{\nu\lambda}\delta^\kappa_\eta - \partial_\nu\Gamma^\eta_{\mu\lambda}\delta^\kappa_\eta + \Gamma^\eta_{\nu\lambda}\Gamma^\xi_{\mu\eta}\delta^\kappa_\xi - \Gamma^\eta_{\mu\lambda}\Gamma^\xi_{\nu\eta}\delta^\kappa_\xi$$
$$= \partial_\mu\Gamma^\kappa_{\nu\lambda} - \partial_\nu\Gamma^\kappa_{\mu\lambda} + \Gamma^\eta_{\nu\lambda}\Gamma^\kappa_{\mu\eta} - \Gamma^\eta_{\mu\lambda}\Gamma^\kappa_{\nu\eta}$$

Therefore

$$R^\kappa_{\lambda\mu\nu} = \partial_\mu\Gamma^\kappa_{\nu\lambda} - \partial_\nu\Gamma^\kappa_{\mu\lambda} + \Gamma^\eta_{\nu\lambda}\Gamma^\kappa_{\mu\eta} - \Gamma^\eta_{\mu\lambda}\Gamma^\kappa_{\nu\eta} \tag{94}$$

Now we turn our attention on the complex manifold M and the complex analogues of the geometric quantities for the smooth manifold M_0 thus far. We start a bit naively with equation (80). How does it look like on complex manifolds?

We have seen that a complex manifold is automatically a Hermitian manifold in section 2, and that only the mixed components of the Hermitian metric are non vanishing in equation (17). So consider the Christoffel symbol on the complex manifold M with pure indices written as follows (I am tired of typing μ, ν, λ's, so I will type α, β, γ for some moment)

$$\Gamma^\gamma_{\alpha\beta} = \frac{1}{2}g^{\gamma\bar{\delta}}(\frac{\partial}{\partial z^\alpha}g_{\beta\bar{\delta}} + \frac{\partial}{\partial z^\beta}g_{\alpha\bar{\delta}} - \frac{\partial}{\partial \bar{z}^\delta}g_{\alpha\beta})$$

$$= \frac{1}{2} g^{\gamma\bar{\delta}} \left(\frac{\partial}{\partial z^{\alpha}} g_{\beta\bar{\delta}} + \frac{\partial}{\partial z^{\beta}} g_{\alpha\bar{\delta}} - 0 \right)$$

$$= g^{\gamma\bar{\delta}} \frac{\partial}{\partial z^{\alpha}} g_{\beta\bar{\delta}} = g^{\gamma\bar{\delta}} \partial_{\alpha} g_{\beta\bar{\delta}} \tag{95}$$

Similarly,

$$\Gamma^{\bar{\gamma}}_{\bar{\alpha}\beta} = \frac{1}{2} g^{\bar{\gamma}\delta} \left(\frac{\partial}{\partial \bar{z}^{\bar{\alpha}}} g_{\bar{\beta}\delta} + \frac{\partial}{\partial \bar{z}^{\bar{\beta}}} g_{\bar{\alpha}\delta} - \frac{\partial}{\partial z^{\delta}} g_{\bar{\alpha}\beta} \right)$$

$$= g^{\bar{\gamma}\delta} \frac{\partial}{\partial \bar{z}^{\bar{\alpha}}} g_{\bar{\beta}\delta} = g^{\bar{\gamma}\delta} \partial_{\bar{\alpha}} g_{\bar{\beta}\delta} \tag{96}$$

Only these two Christoffel symbols are non vanishing, and the rest of them vanish because of the index structure of the Hermitian metric. For example,

$$\Gamma^{\gamma}_{\bar{\alpha}\bar{\beta}} = \frac{1}{2} g^{\gamma\bar{\delta}} \left(\frac{\partial}{\partial \bar{z}^{\bar{\alpha}}} g_{\bar{\beta}\bar{\delta}} + \frac{\partial}{\partial \bar{z}^{\bar{\beta}}} g_{\bar{\alpha}\bar{\delta}} - \frac{\partial}{\partial \bar{z}^{\bar{\delta}}} g_{\bar{\alpha}\bar{\beta}} \right) = 0$$

Because the complex structure J splits the complexified tangent bundle into holomorphic and anti-holomorphic subbundles, its quite obvious that vector fields on the complex manifold should split in that way too.
Taking the complexification of the space $\mathcal{V}(M_0)$ which is essentially the space of smooth sections of the tangent bundle TM_0,

$$\mathcal{V}(M_0) \otimes_{\mathbb{R}} \mathbb{C} = \mathcal{V}(M_0)^{\mathbb{C}} = \mathcal{V}(M)$$

$$= \mathcal{V}(M)^{1,0} \oplus \mathcal{V}(M)^{0,1}$$

So the complexified extension of (51) is

$$\nabla^{\mathbb{C}} : \mathcal{V}(M) \times \mathcal{V}(M) \to \mathcal{V}(M) \tag{97}$$

following rules similar to (52)-(56), here the functions should belong to the complexified space $\mathcal{F}(M_0)^{\mathbb{C}}$. And the connection splits too

$$\nabla^{\mathbb{C}} = \nabla_{\mu} + \nabla_{\bar{\mu}}$$

for which it should hold that

$$\nabla_{\mu} \frac{\partial}{\partial z^{\nu}} = \Gamma^{\lambda}_{\mu\nu} \frac{\partial}{\partial z^{\lambda}} \tag{98}$$

$$\nabla_{\bar{\mu}} \frac{\partial}{\partial \bar{z}^{\nu}} = \Gamma^{\bar{\lambda}}_{\bar{\mu}\nu} \frac{\partial}{\partial \bar{z}^{\lambda}} \tag{99}$$

and similar to what we derived in (60), in the complex settings we have

$$\nabla_{\mu} dz^{\nu} = -\Gamma^{\nu}_{\mu\lambda} dz^{\lambda} \tag{100}$$

$$\nabla_{\bar{\mu}} d\bar{z}^{\nu} = -\Gamma^{\bar{\nu}}_{\bar{\mu}\bar{\lambda}} dz^{\bar{\lambda}} \tag{101}$$

So far so good. Now we want to build the Riemann curvature tensor based these complexified covariant derivatives. But we won't be needing all the components among the ones with mixed and pure index structures.
For future needs, we have to define a quantity which is a two form called *Ricci* form, whose vanishing implies Ricci flatness of a Kähler metric. For that we will only care about the following Ricci tensor with mixed index structure

$$R_{\mu\bar{\nu}} \tag{102}$$

which is essentially the trace of the Riemann curvature tensor (in the complexified setting of course) $R^{\kappa}_{\kappa\mu\bar{\nu}}$ that we compute now.
We observe that using (86) in (85) gives

$$R^{\kappa}_{\lambda\mu\nu} = \partial_\mu \Gamma^{\kappa}_{\nu\lambda} - \partial_\nu \Gamma^{\kappa}_{\mu\lambda} + \Gamma^{\eta}_{\nu\lambda}\Gamma^{\kappa}_{\mu\eta} - \Gamma^{\eta}_{\mu\lambda}\Gamma^{\kappa}_{\nu\eta} \tag{103}$$

$$= \partial_\mu(g^{\bar{\theta}\kappa}\partial_\nu g_{\lambda\bar{\theta}}) - \partial_\nu(g^{\bar{\theta}\kappa}\partial_\mu g_{\lambda\bar{\theta}}) + g^{\bar{\theta}\eta}\partial_\nu g_{\lambda\bar{\theta}} g^{\bar{\phi}\kappa}\partial_\mu g_{\eta\bar{\phi}} - g^{\bar{\theta}\eta}\partial_\mu g_{\lambda\bar{\theta}} g^{\bar{\phi}\kappa}\partial_\nu g_{\eta\bar{\phi}} \tag{104}$$

$$= \partial_\mu g^{\bar{\theta}\kappa}\partial_\nu g_{\lambda\bar{\theta}} + g^{\bar{\theta}\kappa}\partial_\mu\partial_\nu g_{\lambda\bar{\theta}} - \partial_\nu g^{\bar{\theta}\kappa}\partial_\mu g_{\lambda\bar{\theta}} - g^{\bar{\theta}\kappa}\partial_\nu\partial_\mu g_{\lambda\bar{\theta}}$$
$$+ g^{\bar{\theta}\eta}\partial_\nu g_{\lambda\bar{\theta}} g^{\bar{\phi}\kappa}\partial_\mu g_{\eta\bar{\phi}} - g^{\bar{\theta}\eta}\partial_\mu g_{\lambda\bar{\theta}} g^{\bar{\phi}\kappa}\partial_\nu g_{\eta\bar{\phi}}$$

For a cancellation purpose, we use the following trick:

$$g^{\bar{\phi}\kappa} g_{\eta\bar{\phi}} = \delta^{\kappa}_{\eta}$$

Taking its derivative yields

$$\partial_\mu g^{\bar{\phi}\kappa} g_{\eta\bar{\phi}} = \partial_\mu \delta^{\kappa}_{\eta}$$

$$\Rightarrow g^{\bar{\phi}\kappa}\partial_\mu g_{\eta\bar{\phi}} = -g_{\eta\bar{\phi}}\partial_\mu g^{\bar{\phi}\kappa}$$

Let us look the first term in the second line of (95) a bit more closely

$$\partial_\mu g^{\bar{\theta}\kappa}\partial_\nu g_{\lambda\bar{\theta}}$$

For the first partial derivative using the above relation we have that

$$\begin{aligned} g_{\lambda\bar{\theta}}\partial_\mu g^{\bar{\theta}\kappa} &= -g^{\bar{\theta}\kappa}\partial_\mu g_{\lambda\bar{\theta}} \\ \Rightarrow g^{\bar{\theta}\beta}g_{\lambda\bar{\theta}}\partial_\mu g^{\bar{\theta}\kappa} &= -g^{\bar{\theta}\beta}g^{\bar{\theta}\kappa}\partial_\mu g_{\lambda\bar{\theta}} \\ \Rightarrow \delta^\beta_\lambda\partial_\mu g^{\bar{\theta}\kappa} &= -g^{\bar{\theta}\beta}g^{\bar{\theta}\kappa}\partial_\mu g_{\lambda\bar{\theta}} \\ \Rightarrow \partial_\mu g^{\bar{\theta}\kappa} &= -g^{\bar{\theta}\lambda}g^{\bar{\theta}\kappa}\partial_\mu g_{\lambda\bar{\theta}} \end{aligned}$$

Similarly

$$\begin{aligned} g^{\bar{\theta}\kappa}\partial_\nu g_{\lambda\bar{\theta}} &= -g_{\lambda\bar{\theta}}\partial_\nu g^{\bar{\theta}\kappa} \\ \Rightarrow g_{\beta\bar{\theta}}g^{\bar{\theta}\kappa}\partial_\nu g_{\lambda\bar{\theta}} &= -g_{\beta\bar{\theta}}g_{\lambda\bar{\theta}}\partial_\nu g^{\bar{\theta}\kappa} \\ \Rightarrow \delta^\kappa_\beta\partial_\nu g_{\lambda\bar{\theta}} &= -g_{\beta\bar{\theta}}g_{\lambda\bar{\theta}}\partial_\nu g^{\bar{\theta}\kappa} \\ \Rightarrow \partial_\nu g_{\lambda\bar{\theta}} &= -g_{\kappa\bar{\theta}}g_{\lambda\bar{\theta}}\partial_\nu g^{\bar{\theta}\kappa} \end{aligned}$$

Multiplying them yields

$$\partial_\mu g^{\bar{\theta}\kappa}\partial_\nu g_{\lambda\bar{\theta}} = \partial_\nu g^{\bar{\theta}\kappa}\partial_\mu g_{\lambda\bar{\theta}} \tag{105}$$

So we conclude that $R^\kappa_{\lambda\mu\nu} = 0$. Since (86), (87) are the only non vanishing Christoffel symbols on complex manifolds, our computations narrow down to much simpler settings. For example, using (85) we have that

$$R^\kappa_{\lambda\bar{\mu}\nu} = \partial_{\bar{\mu}}\Gamma^\kappa_{\nu\lambda} - \partial_\nu\Gamma^\kappa_{\bar{\mu}\lambda} + \Gamma^\eta_{\nu\lambda}\Gamma^\kappa_{\bar{\mu}\eta} - \Gamma^\eta_{\bar{\mu}\lambda}\Gamma^\kappa_{\nu\eta} = \partial_{\bar{\mu}}\Gamma^\kappa_{\nu\lambda} \tag{106}$$

Of particular interest to us the following:

$$R^\kappa_{\lambda\mu\bar{\nu}} = \partial_\mu\Gamma^\kappa_{\bar{\nu}\lambda} - \partial_{\bar{\nu}}\Gamma^\kappa_{\mu\lambda} + \Gamma^\eta_{\bar{\nu}\lambda}\Gamma^\kappa_{\mu\eta} - \Gamma^\eta_{\mu\lambda}\Gamma^\kappa_{\bar{\nu}\eta} \tag{107}$$

$$= -\partial_{\bar{\nu}}\Gamma^\kappa_{\mu\lambda} \tag{108}$$

Taking the trace of this component of the Riemann curvature tensor, we get the Ricci tensor

$$R_{\mu\bar{\nu}} = R^\kappa_{\kappa\mu\bar{\nu}} = -\partial_{\bar{\nu}}\Gamma^\kappa_{\mu\kappa} \tag{109}$$

Using (86),

$$R_{\mu\bar{\nu}} = -\partial_{\bar{\nu}}(g^{\kappa\bar{\theta}}\partial_\mu g_{\kappa\bar{\theta}}) \tag{110}$$

The expression under the parenthesis reminds us of the variational derivative of the determinant of the Riemannian metric. For which we first recall some relations about diagonal matrices from linear algebra, keeping in mind that since

the Riemannian metric tensor is symmetric, it can always be diagonalized.
For any square diagonal matrix A, we know the relation

$$\ln(Det(A)) = Tr(\ln(A)) \tag{111}$$

Consider the $n \times n$ square matrix with diagonal elements $a_1, a_2, ..., a_n$

$$A' = \begin{bmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \\ & & \end{bmatrix}$$

We see that the exponential of the trace of this matrix is

$$e^{Tr(A')} = \prod_{i=0}^{n} e^{a_i}$$

But now if we take the exponential of the matrix A'

$$e^{A'} = \begin{bmatrix} e^{a_1} & & \\ & e^{a_2} & \\ & & e^{a_3} \\ & & \end{bmatrix}$$

whose determinant is then

$$Det(e^{A'}) = \prod_{i=0}^{n} e^{a_i} = e^{Tr(A')}$$

So defining $A = e^{A'} \Rightarrow A' = \ln A$.
This gives

$$Det(A) = e^{Tr(\ln A)}$$

$$\ln Det(A) = Tr(\ln A) \tag{112}$$

Differentiating them,

$$\frac{\delta Det(A)}{Det(A)} = Tr(\frac{\delta A}{A}) \tag{113}$$

Let $A = g_{\mu\bar{\nu}}$, then $A^{-1} = g^{\mu\bar{\nu}}$ and denote the determinant as $Det(g_{\mu\bar{\nu}}) = g$.
Then (104) gives

$$\frac{\delta g}{g} = g^{\mu\bar{\nu}} \delta g_{\mu\bar{\nu}} \tag{114}$$

Using which we compute the following:

$$\delta\sqrt{g} = \frac{1}{2\sqrt{g}} \delta g \tag{115}$$

$$= \frac{1}{2\sqrt{g}} g g^{\mu\bar{\nu}} \delta g_{\mu\bar{\nu}}$$

$$= \frac{1}{2} \sqrt{g} g^{\mu\bar{\nu}} \delta g_{\mu\bar{\nu}}$$

$$\Rightarrow \delta \log \sqrt{g} = \frac{1}{2} g^{\mu\bar{\nu}} \delta g_{\mu\bar{\nu}}$$

$$\Rightarrow \delta \log g = g^{\mu\bar{\nu}} \delta g_{\mu\bar{\nu}}$$

Thus

$$\delta \log Det(g_{\mu\bar{\nu}}) = g^{\mu\bar{\nu}} \delta g_{\mu\bar{\nu}} \tag{116}$$

So from (101),

$$g^{\kappa\bar{\theta}} \partial_\mu g_{\kappa\bar{\theta}} = \partial_\mu \log Det(g_{\mu\bar{\nu}})$$

Hence our Ricci tensor is

$$R_{\mu\bar{\nu}} = -\partial_{\bar{\nu}} \partial_\mu \log[Det(g_{\mu\bar{\nu}})] \tag{117}$$

Finally we define the $(1,1)$ form

$$Ric_g := -\sqrt{-1} \partial_\mu \partial_{\bar{\nu}} \log[Det(g_{\mu\bar{\nu}})] dz^\mu \wedge d\bar{z}^\nu \tag{118}$$

called the Ricci form. It will play an important role in defining the Calabi Yau manifolds.

7 Kähler stuff continued

In (21) we have seen the Kähler form which a $(1,1)$ form. The Hermitian metric that it contains is called a Kähler metric the Kähler form Ω is closed and the manifold would then be called a Kähler manifold. Let us look at this closedness condition a bit more closely:

$$0 = d\Omega = (\partial + \bar{\partial}) \sqrt{-1} g_{\mu\bar{\nu}} dz^\mu \wedge d\bar{z}^\nu$$

$$= \sqrt{-1} \frac{\partial g_{\mu\bar{\nu}}}{\partial z^\lambda} dz^\lambda \wedge dz^\mu \wedge d\bar{z}^\nu + \sqrt{-1} \frac{\partial g_{\mu\bar{\nu}}}{\partial \bar{z}^\lambda} d\bar{z}^\lambda \wedge dz^\mu \wedge d\bar{z}^\nu$$

The first term:

$$\frac{\partial g_{\mu\bar{\nu}}}{\partial z^\lambda} dz^\lambda \wedge dz^\mu \wedge d\bar{z}^\nu$$

$$= -\frac{\partial g_{\mu\bar{\nu}}}{\partial z^\lambda} dz^\mu \wedge dz^\lambda \wedge d\bar{z}^\nu$$

$$= -\frac{\partial g_{\lambda\bar{\nu}}}{\partial z^\mu} dz^\lambda \wedge dz^\mu \wedge d\bar{z}^\nu$$

and similarly the second term:

$$\frac{\partial g_{\mu\bar{\nu}}}{\partial \bar{z}^\lambda} d\bar{z}^\lambda \wedge dz^\mu \wedge d\bar{z}^\nu$$

after some permuting and index exchanging ends up being :

$$= -\frac{\partial g_{\mu\bar{\lambda}}}{\partial \bar{z}^{\nu}} d\bar{z}^{\lambda} \wedge dz^{\mu} \wedge d\bar{z}^{\nu}$$

Now, back to manipulation:

$$\sqrt{-1}\frac{\partial g_{\mu\bar{\nu}}}{\partial z^{\lambda}} dz^{\lambda} \wedge dz^{\mu} \wedge d\bar{z}^{\nu} + \sqrt{-1}\frac{\partial g_{\mu\bar{\nu}}}{\partial \bar{z}^{\lambda}} d\bar{z}^{\lambda} \wedge dz^{\mu} \wedge d\bar{z}^{\nu}$$

$$= \frac{\sqrt{-1}}{2}\left(\frac{\partial g_{\mu\bar{\nu}}}{\partial z^{\lambda}} - \frac{\partial g_{\lambda\bar{\nu}}}{\partial z^{\mu}}\right) dz^{\lambda} \wedge dz^{\mu} \wedge d\bar{z}^{\nu} + \frac{\sqrt{-1}}{2}\left(\frac{\partial g_{\mu\bar{\nu}}}{\partial \bar{z}^{\lambda}} - \frac{\partial g_{\mu\bar{\lambda}}}{\partial \bar{z}^{\nu}}\right) d\bar{z}^{\lambda} \wedge dz^{\mu} \wedge d\bar{z}^{\nu}$$

since the triple wedges are linearly independent, closedness of the Kähler form then implies that

$$\frac{\partial g_{\mu\bar{\nu}}}{\partial z^{\lambda}} = \frac{\partial g_{\lambda\bar{\nu}}}{\partial z^{\mu}} \tag{119}$$

$$\frac{\partial g_{\mu\bar{\nu}}}{\partial \bar{z}^{\lambda}} = \frac{\partial g_{\mu\bar{\lambda}}}{\partial \bar{z}^{\nu}} \tag{120}$$

8 Harmonic forms

Harmonic differential forms belong to the kernel subspace of the laplacian. The purpose of this section is illustrate a phenomenon of differential forms known as the Hodge decomposition. We shall first demonstrate this for real smooth differential forms on smooth manifolds. Therefore we begin by examining the smooth compact manifold X. The space of differential forms of degree k denoted as $\Omega^k(X)$ (which is the space of the sections of the exterior product bundle $\Gamma(X, \Lambda^k T^*X)$) fit into a cochain complex whose differential operator is the exterior derivative

$$d : \Omega^k(X) \to \Omega^{k+1}(X)$$

called the exterior derivative. We define the Hodge star operator as

$$* : \Omega^k(M) \to \Omega^{n-k}(X)$$

where n is the dimension of X. Using this operator we can construct a formal adjoint of the exterior derivative.
We start that with the observation that the space $\Omega^k(X)$ is an L^2 space with the L^2 norm coming from the bilinear form

$$\langle \alpha_1, \alpha_2 \rangle_{L^2(X)} := \int_X \alpha_1 \wedge *\alpha_2 \tag{121}$$

for $\alpha_1, \alpha_2 \in \Omega^k(X)$ such that

$$||\alpha|| := \sqrt{\langle \alpha, \alpha \rangle_{L^2(X)}} \tag{122}$$

We shall denote the completion of $\Omega^k(X)$ with respect to this norm as $\Omega^k_{L^2}(X)$.

We can use Stokes' theorem for smooth forms

$$\begin{aligned}
0 &= \int_X d(\alpha_1 \wedge *\alpha_2) \\
&= \int_X d\alpha_1 \wedge *\alpha_2 + \int_X (-1)^{k-1} \alpha_1 \wedge * * ^{-1} d * \alpha_2 \\
&= \int_X d\alpha_1 \wedge *\alpha_2 - \int_X \alpha_1 \wedge *(-1)^k *^{-1} d * \alpha_2 \\
&= \langle d\alpha_1, \alpha_2 \rangle_{L^2(X)} - \langle \alpha, \delta\alpha_2 \rangle_{L^2(X)}
\end{aligned}$$

where the formal adjoint of the exterior derivative d is

$$\delta := (-1)^k *^{-1} d* : \Omega^k(X) \to \Omega^{k-1}(X)$$

The laplacian is then defined as the operator

$$\Delta := \delta d + d\delta : \Omega^k(X) \to \Omega^k(X)$$

Denoting the formal adjoint of the laplacian by Δ^*, it turns out that $\Delta = \Delta^*$ i.e. the laplacian is a self adjoint operator. To see this, pick two k-forms α_1, α_2 and compute

$$\begin{aligned}
\langle \Delta\alpha_1, \alpha_2 \rangle &= \langle (\delta d + d\delta)\alpha_1, \alpha_2 \rangle \\
&= \langle \delta d\alpha_1, \alpha_2 \rangle + \langle d\delta\alpha_1, \alpha_2 \rangle \\
&= \langle \alpha_1, d\delta\alpha_2 \rangle + \langle \alpha_1, \delta d\alpha_2 \rangle \\
&= \langle \alpha_1, (\delta d + d\delta)\alpha_2 \rangle \\
&= \langle \alpha_1, \Delta^*\alpha_2 \rangle \\
&= \langle \alpha_1, \Delta\alpha_2 \rangle
\end{aligned}$$

The space of the kernel of the laplacian

$$\mathcal{H}^k_X := \{\alpha \in \Omega^k(X) : \Delta\alpha = 0\}$$

is the space of harmonic forms and this is also a normed space with a same norm of $\Omega^k(X)$.

Now to demonstrate the L^2-decomposition of the space $\Omega^k(X)$ (we follow [8]), we need some tools from the theory of partial differential equations namely the notion of strong and weak solutions and some of their properties. The partial differential equation that we shall address will (obviously) be

$$\Delta\omega = \alpha \tag{123}$$

for the differential k-forms $\omega, \alpha \in \Omega^k(X)$. A *strong* solution to this equation (tautotologically) is $\omega \in \Omega^k(X)$. While by a *weak* solution to this equation we shall mean the linear functional

$$\ell : \Omega^k(X) \to \mathbb{R} \tag{124}$$

for which it holds that for any $\eta \in \Omega^k(X)$, we have that

$$\ell(\Delta^* \eta) = \langle \alpha, \eta \rangle$$

Having the notions of strong and weak solutions we can transit from one to the other.
First observe that if we define the linear functional (124) as

$$\ell(\eta) = \langle \omega, \eta \rangle \tag{125}$$

so then the Cauchy-Schwarz inequality gives

$$|\ell(\eta)| = |\langle \omega, \eta \rangle| \leq ||\omega|| ||\eta||$$

implying that ℓ is continuous (since it is a bounded linear operator), and since

$$\begin{aligned} \ell(\Delta^* \eta) &= \langle \omega, \Delta^* \eta \rangle \\ &= \langle \Delta \omega, \eta \rangle \\ &= \langle \alpha, \eta \rangle \end{aligned}$$

so the linear functional is a weak solution of (123). In other words, given as trong solution $\omega \in \Omega^k(X)$ of (123), the functional defined as (125) is a weak solution.

We can go the other way around too. That is, given any weak solution ℓ to the equation (123), Through the usage of Sobolev spaces and ellpitic partial differential operators it can be shown that there exists an $\omega \in \Omega^k(X)$ such that ℓ can be written as (125). This in turn implies that for any $\eta \in \Omega^k(X)$ we have

$$\begin{aligned} \langle \Delta \omega, \eta \rangle &= \langle \omega, \Delta^* \eta \rangle \\ &= \ell(\Delta^* \eta) \\ &= \langle \alpha, \eta \rangle \end{aligned}$$

So α is a strong solution to the equation (123).

As an L^2-normed space, $\Omega^k(X)$ is compact. That is, for a sequence $\{\alpha_n\}_{n \in \mathbb{N}}$ in $\Omega^k(X)$, such that there is constant $C > 0$ satsifying

$$||\alpha_n|| \leq C$$

$$||\Delta \alpha_n|| \leq C$$

then there exists a subsequence of $\{\alpha_n\}_{n \in \mathbb{N}}$ which is convergent. Proof can be found in [11].

For any k-form in the orthogonal complement of the space of harmonic k-forms $(\mathcal{H}^k_X)^\perp$, we want to make comparison between the L^2-norms of that with

and without applying the laplacian, that is, for a $\gamma \in \Omega^k(X)$ we want to compare $||\gamma||$ and $||\Delta\gamma||$. Let us make a (not so educated) guess that for any $j \in \mathbb{N}$, we have a $\gamma_j \in \Omega^k(X)$ (which are elements of sequence $\{\gamma_j\}_{j\in\mathbb{N}}$) such that

$$||\gamma_j|| > j||\Delta\gamma_j|| \tag{126}$$

Define the unit vector

$$\tilde{\gamma}_j = \frac{\gamma_j}{||\gamma_j||}$$

Thus in the sequence $\{\tilde{\gamma}_j\}_{j\in\mathbb{N}}$, every term has unit norm and thus

$$\lim_{j\to\infty} ||\Delta\tilde{\gamma}_j|| \leq \lim_{j\to\infty} \frac{1}{j} = 0$$

find out y??

But then from the compactness of $\Omega^k(X)$ we know that the sequence $\{\tilde{\gamma}_j\}_{j\in\mathbb{N}}$ has a convergent (and Cauchy) subsequence. For keeping things simple, we may take $\{\tilde{\gamma}_j\}_{j\in\mathbb{N}}$ itself as the subsequence.

For any $\psi \in \Omega^k(X)$, the sequence $\{\langle\tilde{\gamma}_j, \psi\rangle\}_{j\in\mathbb{N}}$ is Cauchy as can be seen from the following Cauchy Schwarz inequality

$$|\langle\tilde{\gamma}_j, \psi\rangle - \langle\tilde{\gamma}_k, \psi\rangle| = |\langle\tilde{\gamma}_j - \tilde{\gamma}_j, \psi\rangle| \leq ||\tilde{\gamma}_j - \tilde{\gamma}_j||.||\psi||$$

As a consequence, the linear functional

$$\ell : \Omega^k(X) \to \mathbb{R}$$

that is defined as $\psi \mapsto \lim_{j\to\infty}\langle\tilde{\gamma}_j, \psi\rangle$ is bounded and thus continuous.
Again from the inequality above we see that for any $\phi \in \Omega^k(X)$

$$\lim_{j\to\infty} |\langle\Delta\tilde{\gamma}_j, \phi\rangle| \leq \lim_{j\to\infty} ||\Delta\tilde{\gamma}_j||.||\phi|| = 0$$

we find that the linear functional ℓ satisfies

$$\begin{aligned}\ell(\Delta^*\phi) &= \lim_{j\to\infty}\langle\tilde{\gamma}_j, \Delta^*\phi\rangle \\ &= \lim_{j\to\infty}\langle\Delta\tilde{\gamma}_j, \phi\rangle \\ &= \langle 0, \phi\rangle\end{aligned}$$

Thus ℓ is a weak solution to the partial differential equation $\Delta\tilde{\gamma} = 0$. This means the k-form $\tilde{\gamma}$ is a harmonic form i.e. $\tilde{\gamma} \in \mathcal{H}^k_X$. But as we saw, given such a weak solution, we can obtain a strong solution as well. That is, there exists a $\tilde{\gamma} \in \Omega^k(X)$ such that for any $\psi \in \Omega^k(X)$ we have

$$\ell(\psi) = \langle\tilde{\gamma}, \psi\rangle$$

and $\Delta\tilde{\gamma} = 0$. Hence, $\tilde{\gamma}$ is a strong solution.

As the sequence $\{\langle\tilde{\gamma}_j, \psi\rangle\}_{j\in\mathbb{N}}$ is Cauchy, it convereges to some k-form in the L^2-completed space $\Omega^k_{L^2}(X)$, say δ. But then continuity implies that for any $\psi \in \Omega^k(X)$ we have

$$\langle\delta - \tilde{\gamma}, \psi\rangle = \lim_{j\to\infty}\langle\tilde{\gamma}_j, \psi\rangle - \lim_{j\to\infty}\langle\tilde{\gamma}_j, \psi\rangle = 0$$

which means δ is $\tilde{\gamma}$. This implies

$$||\tilde{\gamma}|| = 1$$

and

$$\tilde{\gamma} \in (\mathcal{H}^k_X)^\perp$$

But we had that

$$\Delta\tilde{\gamma} = 0$$

which says that $\tilde{\gamma}$ must be in the intersection $(\mathcal{H}^k_X)^\perp \cap \mathcal{H}^k_X = \{0\}$ which is a contradiction. So our claim (126) was false. The contrary must be true i.e. for any $\tilde{\gamma} \in (\mathcal{H}^k_X)^\perp$, there exists a constant $C > 0$ such that the following inequality

$$||\tilde{\gamma}|| \leq C||\Delta\tilde{\gamma}|| \tag{127}$$

holds.

9 Holomorphic line bundles

Vector bundles are manifolds consisting of a family vector spaces hovering above some other manifold, with a projection map that takes each of the vector spaces to a point below, thus parametrizing the family of vector spaces of a given dimension. Though the construction of a vector bundle might seem weird at first, it turns out that such spaces are quite natural and are of immense importance for doing geometry (we have already seen such things, deriving the crvature of a complex manifold using covaraint derivatives which would have been impossible to construct without the notion of tangent BUNDLES) and analyzing the topology of a manifold. Our readers are assumed to be familiar with the language of vector bundles, since in this section we are not introducing vector bundles (actually we WILL, we will look at the holomorphic ones to be precise). I am writing this section with the goal of telling you what a characteristic class is, in mind. We will be needing a specific type of characteristic classes called Chern classes that are fantastic tools for deducing the non triviality of vector bundles over a given manifold. In section 1, we have already seen the complexified tangent bundle that splits into holomorphic and anti holomophic subbundles. In this section, let me be a bit more formal, following the

beautiful book [5]. The title of this section contains the term "holomorphic", which we will see a bit lately; first let us look at complex vector bundles.
Take a C^∞ manifold M_0. We want this manifold to parametrize the family of complex vector spaces $\{E_x\}_{x\in M_0}$ of complex dimension $dim_{\mathbb{C}} E_x = k$. Forming the union

$$E = \bigcup_{x\in M_0} E_x$$

via transport of (C^∞ manifold) structure using the projection map

$$\pi : E \to M_0$$

taking E_x to x in M_0, E itself becomes a C^∞ manifold. The vector spaces E_x are called the fibers of the vector bundle, which are linearly isomorphic the vector space $\mathbb{C}^k$. This is called a C^∞ complex vector bundle (note that neither E nor M_0 possesses complex structure).
For every point $x \in M_0$ and its neighborhood $U \subset M_0$, there exists a diffeomorhism (called a trivialization map)

$$\phi_U : \pi^{-1}(U) \to U \times \mathbb{C}^k \tag{128}$$

taking the fibers E_x of E isomorphically to $\{x\} \times \mathbb{C}^k$ (which is really just $\mathbb{C}^k$).
Just as we construct a manifold via gluing together open sets from the Euclidean space with transition functions, there are brothers of those transition functions in the theory of vector bundles, that can glue vector spaces (a fixed dimension) to form a manifold which is the bundle.
For a non empty intersection $U \cap V$ in the manifold M_0, we can form a composition of the trivialization maps

$$g_{UV}(x) = (\phi_U \circ \phi_V^{-1})(x) : \{x\} \times \mathbb{C}^k \to \{x\} \times \mathbb{C}^k$$

for any point $x \in U \cap V$. We can see that this composed map should be an element of the group $GL_k(\mathbb{C})$, of $k \times k$ matrices with complex entries. Thus

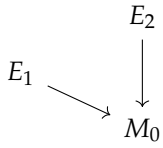
$$g_{UV} : U \cap V \to GL_k(\mathbb{C}) \tag{129}$$

And necessarily the transition functions satisfy the conditions (that are too to check)
(i) $g_{UV} \circ g_{UV}(x) = Id$ for all $x \in U \cap V$
(ii) $g_{UV} \circ g_{VW} \circ g_{WU}(x) = Id$ for all $x \in U \cap V \cap W$
called the cocycle conditions of the transition functions.
And quite interestingly, the converse also holds. That is, if you have a cover $\{U_\alpha\}$ of the C^∞ manifold M_0 along with maps

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to GL_k(\mathbb{C})$$

satisfying the cocycle conditions on the non empty intersections of the covers, there exists a unique C^∞ complex vector bundle $E \to M_0$ whose transition

functions are precisely the maps $\{g_{\alpha\beta}\}$!
Over the same C^∞ manifold M_0, suppose we have two vector bundles E_1, E_2 whose fibres are $\mathbb{C}^k, \mathbb{C}^l$ respectively



with transition functions

$$g^{(1)}_{\alpha\beta} : U_\alpha \cap U_\beta \to GL(\mathbb{C}^k)$$

$$g^{(2)}_{\alpha\beta} : U_\alpha \cap U_\beta \to GL(\mathbb{C}^l)$$

we can blend them together to form a new map

$$j_{\alpha\beta} : U_\alpha \cap U_\beta \to GL(\mathbb{C}^k \oplus \mathbb{C}^l)$$

SO there must exist a bundle over M_0 whose transition functions should the maps $\{j_{\alpha\beta}\}$, that we denote as $E_1 \oplus E_2$.
And similarly we can form the tensor product bundle $E_1 \otimes E_2$ whose transitions maps are

$$j_{\alpha\beta} : U_\alpha \cap U_\beta \to GL(\mathbb{C}^k \otimes \mathbb{C}^l)$$

The C^∞ structure on E is induced from the inclusion map $i : U_\alpha \times \mathbb{C}^k \hookrightarrow E$.
Well, the covers can have non empty intersections. Lets consider such a case: The fibers over points in $U_\alpha \cap U_\beta$ are "transformed" versions of each other, where the transformations come from the group $GL_k(\mathbb{C}^k)$. So we can find an equivalence ($\sim$) relation with respect to which the points $(x, \gamma) \in U_\beta \times \mathbb{C}^k$ and $(x, g_{\alpha\beta}(x)\gamma) \in U_\alpha \times \mathbb{C}^k$ are the same.
Then the manifold E as a set is just the quotient

$$\frac{\bigcup_\alpha U_\alpha \times \mathbb{C}^k}{\sim}$$

So over the same base manifold, using different transition functions, we can construct different vector bundles. If we build a continuous (smooth) family of transition functions, we would have a continous (smooth) family of vector bundles over the particular base manifold.
Of particular interest to us is the notion of a dual bundle

$$\pi : E^* \to M_0$$

whose transition functions are just the inverse matrices coming from those of the bundle $E \to M_0$.

Now as holomorphic bundles over the complex manifold $\mathbb{CP}^m$ we take the two important two examples: tautological line bundle that I will denote by $O_{\mathbb{CP}^m}(-1)$ (inspired by [3]) and its dual called the hyperplane line bundle denoting which by $O_{\mathbb{CP}^m}(1)$.
The line bundle

$$\pi_{O_{\mathbb{CP}^m(-1)}} : O_{\mathbb{CP}^m}(-1) \to \mathbb{CP}^m \tag{130}$$

really is tautological in the sense that over a point $[l] \in \mathbb{CP}^m$ (where l is a line through the origin in $\mathbb{C}^{m+1}$) the fiber $\pi^{-1}_{O_{\mathbb{CP}^m}}([l])$ is just l.

On the other hand, the fiber over a point $[l] \in \mathbb{CP}^m$ is just the space of linear functionals on the line $l \in \mathbb{C}^{m+1}$ through the origin, l^*. Its name comes from the important observation that each hyperplane in $\mathbb{CP}^m$ parametrized by some complex coordinate $a = (a_1, ..., a_{n+1}) \in \mathbb{C}^{n+1}$

$$H_a = \{[z_1, ..., z_{n+1}] \in \mathbb{CP}^m : \sum_{j=1}^{n+1} a_j z_j = 0\} \tag{131}$$

determines a line bundle that are all holomorphically isomorphic to each other.

We know that we can blend these line bundles together using the tensor product recipe to get new line bundles $O_{\mathbb{CP}^m}(-1)^{\otimes k}$ that will be denoted as $O_{\mathbb{CP}^m}(-k)$; similarly for its dual (the hyperplane line bundle) we can form $O_{\mathbb{CP}^m}(1)^{\otimes k}$ that will be denoted as $O_{\mathbb{CP}^m}(k)$.
The crucial result that connects the vector space of global sections $\Gamma(O_{\mathbb{CP}^m}(k))$ of the tensor powered line bundle $O_{\mathbb{CP}^m}(k)$ to algebraic hypersurfaces in $\mathbb{CP}^m$ is that elements of $\Gamma(O_{\mathbb{CP}^m}(k))$ have 1-1 correspondence with homogenous polynomials in variables $z_1, ..., z_{n+1}$ with complex coefficients i.e. elements of the polynomial ring $\mathbb{C}[z_1, ..., z_{n+1}]$ of degree k.

10 Connections, Curvatures and characteristic classes

Recall from (51), the covariant derivative was really a map from a pair of sections of the tangent bundle TM_0 to another bundle. In this section we attempt to build the notion of connections and curvature for arbitrary vector bundles over the C^∞ manifold M_0, then we will look at complex or holomorphic cases. First things first, for the tangent bundle

$$\pi : TM_0 \to M_0$$

connection is nothing but the covariant derivative (also called the affine connection) that we saw in (51), generalizing that notion to arbtrary C^∞ vector bundles of rank k

$$\pi : E \to M_0$$

the connection is the map

$$\nabla : \mathcal{V}(M_0) \times \Gamma(E) \to \Gamma(E) \tag{132}$$

where $\mathcal{V}(M_0)$ denotes the space of smooth vector fields on the manifold M_0, $\Gamma(E)$ denotes the space of smooth sections of the vector bundle E. The connection satisfies the properties: (i) $\nabla_X s$ is $C^\infty(M)$-linear and $\mathbb{R}$-linear in the sections $s \in \Gamma(E)$
(ii) for all $f \in C^\infty(M)$ there is a Leibniz rule

$$\nabla_X(fs) = X(f)s + f\nabla_X s$$

This Leibniz property indicates to the fact that we can't just linearly combine any two or more connections to get another connection on a particular smooth vector bundle.
Say we have two connections $\nabla^{(1)}$, $\nabla^{(2)}$ on the smooth vector bundle $\pi : E \to M_0$ and by definition they satisfy

$$\nabla^{(1)}_X(fs) = X(f)s + f\nabla^{(1)}_X s$$

$$\nabla^{(2)}_X(fs) = X(f)s + f\nabla^{(2)}_X s$$

their combination then gives

$$(\nabla^{(1)}_X + \nabla^{(2)}_X)(fs) = 2X(f)s + f(\nabla^{(1)}_X + \nabla^{(2)}_X)s$$

which is not quite the Leibniz propert because of the extra factor of 2. But a slight modification in the linear combination yields the Leibniz rule, which is the convex combination of connections:

$$(1-t)\nabla^{(1)}_X(fs) = (1-t)X(f)s + (1-t)f\nabla^{(1)}_X s$$

$$t\nabla^{(2)}_X(fs) = tX(f)s + tf\nabla^{(2)}_X s$$

$$\Rightarrow [(1-t)\nabla^{(1)}_X + t\nabla^{(2)}_X](fs) = X(f)s + f((1-t)\nabla^{(1)}_X + t\nabla^{(2)}_X)s$$

So we have the important general fact that any convex linear combination of connections on a smooth vector bundle is again a connection on that bundle.
Now lets make a similar generalization for the curvature that we defined in (81), (82). For the smooth vector bundle E over the smooth manifold M_0 we should have $\mathbb{R}$ linear map

$$\mathcal{V}(M_0) \times \mathcal{V}(M_0) \times \Gamma(E) \to \Gamma(E) \tag{133}$$

explicitly, for vector fields $X, Y \in \mathcal{V}(M_0)$ and a section $s \in \Gamma(E)$ the generalized curvature acts as

$$R(X,Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X,Y]} s \tag{134}$$

Consider the trivializing cover $U \subset M_0$ i.e. for U we must have the diffeomorphism

$$E_U := \pi^{-1}(U) \simeq U \times \mathbb{R}^k$$

Now suppose we have a frame $e_1, .., e_k$ for the bundle E over U i.e. E_U (recall that a frame is a finite set of sections that can generate the fibres of a vector bundle over a trivializing open set). Over U, there are vector fields $X \in \mathcal{V}(U)$ and also local sections $s \in \Gamma(U, E)$ that can be (with respect to this frame) written as

$$s = \sum_{j=1}^{k} a^j e_j \tag{135}$$

So because of (116) we must have another section $\nabla_X s \in \Gamma(U, E)$ that should look like (119) if we impose:

$$\nabla_X e_j = \sum_{i=1}^{k} \omega_j^i(X) e_i \tag{136}$$

where the coefficients $\omega_j^i(X)$ are functions and the matrix $[\omega_j^i]$ contains 1-forms and is called the connection matrix relative to the frame $e_1, ..., e_k$.
For the curvature in (118), we should have a section $R(X, Y)e_j$ that can be expressed as a linear combination of the sections of the frame i.e. relative to the frame $e_1, ..., e_k$ we have

$$R(X, Y)e_j = \sum_{i=1}^{k} \Omega_j^i(X, Y) e_i \tag{137}$$

the matrix $\Omega = [\Omega_j^i]$ of the 2-forms Ω_j^i is called the curvature matrix of the connection ∇ relative to the frame $e_1, ..., e_k$ on the smooth vector bundle.
Recall the explicit construction of the exterior derivative acting on an n-form $\omega \in \Omega^n(M_0)$ is

$$d\omega(X_1, ..., X_{n+1}) := \sum_{i=1}^{n} (-1)^{i+1} X_i(\omega(X_1, .., \hat{X}_i, .., X_{n+1}))$$

$$+ \sum_{i<j} (-1)^{i+j} \omega([X_i, X_j], X_1, .., \hat{X}_i, .., \hat{X}_j, .., X_{n+1})$$

S for a 1-form α, the exterior derivative acting on it takes the form

$$(d\alpha)(X, Y) = X\alpha(Y) - Y\alpha(X) - \alpha([X, Y]) \tag{138}$$

Take another 1-form β, their wedge gives

$$(\alpha \wedge \beta)(X, Y) := \alpha(X)\beta(Y) - \alpha(Y)\beta(X) \tag{139}$$

Armed with the tools above, let us compute the curvature (121) using (118) for local sections $s \in \Gamma(U, E)$ and local vector fields $X, Y \in \mathcal{V}(U)$ where (we recall that) $U \subset M_0$ is a trivializing open set for which we have been considering the frame $e_1, ..., e_k$.
So the first term (of (118)), following (120) is :

$$\nabla_X \nabla_Y e_j = \nabla_X \sum_{l=1}^{k} (\omega_j^l(Y) e_l)$$

Applying the Leibniz rule

$$= \sum_{l=1}^{k} X\omega_j^l(Y) e_l + \sum_{l=1}^{k} \omega_j^l(Y) \nabla_X e_l$$

$$= \sum_{l=1}^{k} X\omega_j^i(Y) e_i + \sum_{i,l=1}^{k} \omega_j^l(Y) \omega_l^i(X) e_i$$

For the second term (of (118)) we simply interchange $X \leftrightarrow Y$:

$$\nabla_Y \nabla_X e_j = \sum_{i=1}^{k} Y\omega_j^i(X) e_i + \sum_{i,l=1}^{k} \omega_j^l(X) \omega_l^i(Y) e_i$$

Since the space of vector space is closed under the Lie bracket, (120) further yields

$$\nabla_{[X,Y]} e_j = \sum_{i=1}^{k} \omega_j^i([X, Y]) e_i \tag{140}$$

Putting them together in (118) and suppressing the sums via the Einstein summation convention:

$$R(X, Y) e_j = \nabla_X \nabla_Y e_j - \nabla_Y \nabla_X e_j - \nabla_{[X,Y]} e_j$$

$$= X\omega_j^i(Y) e_i + \omega_j^l(Y) \omega_l^i(X) e_i - Y\omega_j^i(X) e_i - \omega_j^l(X) \omega_l^i(Y) e_i - \omega_j^i([X, Y]) e_i$$

$$= (X\omega_j^i(Y) - Y\omega_j^i(X) - \omega_j^i([X, Y])) e_i + (\omega_j^l(Y) \omega_l^i(X) - \omega_j^l(X) \omega_l^i(Y)) e_i$$

Applying (122),(123)

$$= d\omega_j^i(X, Y) e_i + \omega_l^i \wedge \omega_j^l(X, Y) e_i$$

$$= (d\omega_j^i + \omega_l^i \wedge \omega_j^l)(X, Y) e_i$$

$$= \Omega_j^i(X, Y) e_i$$

In summary, we have the second structural formula as it is known:

$$\Omega_j^i = d\omega_j^i + \sum_l \omega_l^i \wedge \omega_j^l \tag{141}$$

So far whatever we have written is for a particular frame, but there could other frames and they all can be transformed into each other via the conventional linear algebra. We want to know how the connection matrix and the curvature matrix transform under a change of the frame of the vector bundle over a trivializing open set. For that, using matrix notation is convenient.
Writing (120) in the matrix notation, it looks like

$$\nabla_X e = e\omega(X) \tag{142}$$

where we have set the frame as a row vector i.e. $e = [e_1, ..., e_k]$. The vector field X can be omitted and we will write

$$\nabla e = e\omega \tag{143}$$

for the moment.
Consider another frame $\bar{e} = [\bar{e}_1, ..., \bar{e}_k]$. At a point $p \in M_0$, there is a change of basis matrix via which the basis vector $\bar{e}_q(p)$ can be written as a linear combination of the sections $e_1(p), ..., e_k(p)$ from the previous frame:

$$\bar{e}_q(p) = \sum_{l=1}^{k} a_q^l(p) e_l(p) \tag{144}$$

Considering them as sections of the open set U, we can suppress the points from the expression

$$\bar{e}_q = \sum_{l=1}^{k} a_q^l e_l \tag{145}$$

The change of basis matrix $a(p) = [a_q^l(p)]$ defines at each point $p \in U$ a map

$$a : U \to GL(k, \mathbb{R}) \tag{146}$$

Writing this transformation of frames in the matrix notation gives

$$[\bar{e}_1, ..., \bar{e}_1] = [e_1, ..., e_k][a_q^l] \tag{147}$$

or

$$\bar{e} = ea \tag{148}$$

$$\Rightarrow \bar{e}a^{-1} = e \tag{149}$$

Let the connection matrix relative to frame $[\bar{e}_1, ..., \bar{e}_1]$ be $\overline{\omega}$ for which (127) reads

$$\nabla \bar{e} = \bar{e}\bar{\omega} \tag{150}$$

Now following (132),(133) and (127)

$$\nabla \bar{e} = \nabla(ea)$$

$$= (\nabla e)a + e(da)$$

$$= (e\omega)a + e(da)$$
$$= \bar{e}a^{-1}\omega a + \bar{e}a^{-1}da$$
$$= \bar{e}(a^{-1}\omega a + a^{-1}da)$$

So (134) implies

$$\overline{\omega} = a^{-1}\omega a + a^{-1}da \tag{151}$$

which is how the connection matrix transforms under a change of frame.
Now its curvature matrix's turn. Writing (121) in the matrix notation

$$R(X,Y)e = R(X,Y)[e_1,...,e_k] = [e_1,...,e_k][\Omega^i_j(X,Y)] = e\Omega(X,Y) \tag{152}$$

for local vector fields $X, Y \in \mathcal{M}_l$. Suppressing the vector fields

$$R(e) = e\Omega \tag{153}$$

for the other frame this should look like

$$R(\bar{e}) = \bar{e}\overline{\Omega} \tag{154}$$

Now

$$R(\bar{e}) = R(ea)$$
$$= R(e)a$$
$$= e\Omega a$$
$$= \bar{e}a^{-1}\Omega a$$

where in the second line we have used the fact that the curvature is $C^\infty(M_0)$-linear and the change of basis matrix a is indeed a matrix of smooth functions on M_0. Lastly, (138) implies that

$$\overline{\Omega} = a^{-1}\Omega a \tag{155}$$

which is how the curvature matrix transforms under the change of frame. This is conjugacy relation of the matrices Ω and $\overline{\Omega}$. This conjugacy relation has a central implication, almost the starting point of Chern-Weil theory of characteristic classes.
The crucial observation is the following:
If we have an invariant matrix polynomial i.e. a polynomial for the matrix $X \in \mathfrak{gl}(k,\mathbb{R}) = \mathbb{R}^{k\times k}$ such that

$$P(X) = P(A^{-1}XA) \tag{156}$$

where $A \in GL(k,\mathbb{R})$, then such a polynomial for the curvature matrix Ω is a global closed form on the base manifold M_0. Moreover, the cohomology class of this global closed form does not even depend on the choice of the connection. This brilliant result was obtained by Shiing Shen Chern and Andre Weil.
But here we will make a shift to the complexified settings. In which we shall

consider a complex vector bundle $E^{\mathbb{C}}$ over the smooth manifold M_0. We will rewrite things a bit differently and rederive many things. Then, in the complexified settings, we will construct the Chern-Weil homomorphism and study its properties.
The complex vector bundle

$$\pi : E^{\mathbb{C}} \to M_0$$

has fibers $E_p^{\mathbb{C}}$ over some point $p \in M_0$ (which is just $\mathbb{C}^k$) and for some open subset $U \subset M_0$, we have the trivialization map

$$\varphi_U : \pi^{-1}(U) \to U \times \mathbb{C}^k \tag{157}$$

On the neighborhood of p, i.e. on U, the fibers are generated by the family of smooth sections $\{e_j\}_{j=1}^k$ (as before). We now write the map (125) as

$$\nabla : \Gamma(E^{\mathbb{C}}) \to \Gamma(T^*M_0^{\mathbb{C}} \otimes E^{\mathbb{C}}) \tag{158}$$

(we are writing $\Gamma(E^{\mathbb{C}})$ for $\Gamma(U, E^{\mathbb{C}})$ and likewise for $\Gamma(T^*M_0^{\mathbb{C}})$) where $T^*M_0^{\mathbb{C}}$ denotes the complexified dual tangent bundle of M_0. The difference here will be that every smooth fucntion and differential form will be complex valued. We see that (129) can be written as

$$\nabla e_i = \omega_i^j \otimes e_j \tag{159}$$

(omitting the summation) where now ω_i^j are complex valued 1-forms. For any complex valued smooth function f and and smooth section s, the Leibniz rule is rewritten as

$$\nabla(fs) = df \otimes s + f\nabla s$$

The goal is to let you know that there is really tensor product of bundles here that comes to play secretly. The space $\Gamma(T^*M_0^{\mathbb{C}} \otimes E^{\mathbb{C}})$ means sections of the complexified dual tangent bundle i.e. 1- forms taking values in $E^{\mathbb{C}}$. Similarly, we can look at differential forms of higher rank that take values in $E^{\mathbb{C}}$; they form the space $\Gamma(\bigwedge^k T^*M_0^{\mathbb{C}} \otimes E^{\mathbb{C}})$. Then the map (151) can be further streched to

$$\nabla : \Gamma(\bigwedge^k T^*M_0^{\mathbb{C}} \otimes E^{\mathbb{C}}) \to \Gamma(\bigwedge^{k+1} T^*M_0^{\mathbb{C}} \otimes E^{\mathbb{C}}) \tag{160}$$

with the Leibniz relation being

$$\nabla(\psi\xi) := (d\psi) \wedge \xi + (-1)^k \psi \wedge \nabla\xi \tag{161}$$

where ψ is a complex valued k-form and ξ is a smooth section of $E^{\mathbb{C}}$. We will omit the tensor product notation in formulas like (152) to keep things look familiar as before.

Instead of (126), we now define the curvature as the map

$$R := \nabla^2 : \Gamma(E^{\mathbb{C}}) \to \Gamma(\bigwedge^2 T^*M_0^{\mathbb{C}} \otimes E^{\mathbb{C}}) \tag{162}$$

For a smooth section $\xi \in \Gamma(E^{\mathbb{C}})$ and a smooth complex valued function f, we see that

$$R(f\xi) = \nabla(\nabla(f\xi)) = \nabla(df\xi + f\nabla\xi)$$

$$= d(df)\xi - df \wedge \nabla\xi + df \wedge \nabla\xi + f\nabla^2\xi = fR(\xi)$$

where in the second line we used (154). Recall (128) and (130), let a smooth section $s \in \Gamma(E^{\mathbb{C}})$, $s = f^i e_i$ (suppressing the summation). For two set of frames $\{e_i\}$, $\{e_j\}$

$$R(e_i) = \Omega_i^j e_j$$

where Ω_i^j are locally defined complex valued smooth 2-forms. Then we write

$$R(s) = R(f^i e_i)$$

$$= f^i R(e_i) = \Omega_i^j f^i e_j = \Omega_j^i f^j e_i$$

we can rederive (134) using the above as follows:

$$\Omega_i^j e_j = \nabla(\nabla e_i)$$

$$= \nabla(\omega_i^l e_l)$$

$$= d(\omega_i^l e_l) - \omega_i^l \wedge \nabla(e_l)$$

$$= d(\omega_i^l e_l) - \omega_i^l \wedge \omega_l^j e_j$$

$$= d(\omega_i^j e_j) - \omega_i^l \wedge \omega_l^j e_j$$

Thus we get

$$\Omega_j^i = d\omega_j^i + \omega_l^i \wedge \omega_j^l$$

So (126) and (155) are essentially the same.
We can equip with the complex vector bundle $E^{\mathbb{C}}$ a hermitian metric, which is just an assignment of hermitian inner products

$$h_p(.,.) = \langle .,. \rangle_p$$

at each fiber $E_p^{\mathbb{C}}$ where p is a point from a local trivialization U. Recall that it has a frame $\{e_i\}_{i=1}^k$ which are sections generating the fibers over the open set U. For any two sections s_1, s_2 over U, the hermitian inner product gives a smooth complex valued function on U, i.e.

$$h_p(s_1, s_2) \in C^\infty(U, \mathbb{C})$$

so by defining smooth hermitian matrix valued functions

$$h_{i\bar{j}} = h(e_i, e_j) \tag{163}$$

As the sections over U can be expressed via the frames,

$$s_1 = f^i e_i$$

$$s_2 = f^j e_j$$

thus $h(s_1, s_2) = h(f^i e_i, f^j e_j) = h_{i\bar{j}} f^i \overline{f^j}$ because of the sesquilinearity of hermitian inner products.

Having a hermitian metric on a smooth complex vector bundle, we can ask whether we can connection on it with the desired properties. It turns that there is a natural connection that we can find on such vector bundles if the bundle is holomorphic. Such a connection is called a Chern connection. A Chern connection is compatible with the hermitian metric, i.e. for two holomorphic sections (since the bundle is now holomorphic) s_1, s_2,

$$dh(s_1, s_2) = h(\nabla s_1, s_2) + h(s_1, \nabla s_2) \tag{164}$$

and also compatible with the complex structure of the (now holomorphic) bundle $E^{\mathbb{C}}$, i.e. for any holomorphic section s of $E^{\mathbb{C}}$, ∇s is $E^{\mathbb{C}}$ valued $(1,0)$-form. So we see that the Chern connection is essentially the complexification of the Levi Civita connection on the real smooth case. But the crucial condition is that the vector bundle should not only be complex but also holomorphic, so the base smooth manifold M_0 must have integrable almost complex structure, and the "transport of structure" then makes the total space manifold $E^{\mathbb{C}}$ a complex manifold as well.
Now the exterior derivative is really the sum of Dolbeault operators i.e.

$$d = (\partial + \overline{\partial})$$

For a local holomorphic frame $\{e_i\}_{i=1}^k$, (157) reads

$$dh_{i\bar{j}} = h(\nabla e_i, e_j) + h(e_i, \nabla e_j)$$

$$\Rightarrow (\partial + \overline{\partial}) h_{i\bar{j}} = h(\omega_i^l e_l, e_j) + h(e_i, \omega_j^l e_l)$$

$$\Rightarrow \partial h_{i\bar{j}} + \overline{\partial} h_{i\bar{j}} = \omega_i^l h_{l\bar{j}} + \overline{\omega_j^l} h_{i\bar{l}}$$

Comparison then yields

$$\partial h_{i\bar{j}} = \omega_i^l h_{l\bar{j}}$$

$$= (\omega_i^l)^t h_{i\bar{j}}$$

Removing the indices, we may write

$$\omega^t = \partial(\log h) \tag{165}$$

11 Algebraic Varieties

Varieties are at the same time generalizations of manifolds and special cases of manifolds. Generalizations in the sense that they allow singularities, many of their geometric properties can be captured using pure algebra without even bothering about the metrics on them. Special cases in the sense that they are manifolds (with sharp corners i.e. singularities) defined by zeros algebraic equations i.e. polynomials on some ambient affine or projective space. We have already seen the definition at the beginning of section 1, what the defintion of varieties is. Similar to those, in the realm of complex analysis, there is something called analytic varieties that are, as the keen reader might have guessed, zero sets of holomorphic functions in some affine space $\mathbb{C}^n$. Note that we are compelled to consider their embeddings into some "bigger" ambient space. This is not quite like doing manifold theory. So we are heading towards an abstract definition of varieties, that doesn't require the ambient space.
We warn the reader that the adjective "affine" means something different when used before the space $\mathbb{C}^n$, which is of course the cartesian product of $\mathbb{C}$, n times; but when we are studying this space as an "affine" space we don't view it as simply the Euclidean space or a vector space, we equip it with a strange kind of topology known as the Zariski topology, which we will define in this section.

First we recall some classical definitions and facts (theorems without proofs) about (affine) algebraic varieties. As an algebraically closed field, we take $\mathbb{K} = \mathbb{C}$.
An algebraic set is the zero set of bunch polynomials. If we take the subset $S = \{f_1, ..., f_m\} \subset \mathbb{C}[z_1, ..., z_n]$, an algebraic set X is

$$X = Z(f_1, ..., f_m) := \{(z_1, ..., z_n) \in \mathbb{C}^n : f_1(z_1, ..., z_n) = ... = f_m(z_1, ..., z_n) = 0\}$$

We observe that since the zero polynomial vanishes everywhere in $\mathbb{C}^n$, we can write

$$\mathbb{C}^n = Z(\{0\})$$

and the polynomial ring $\mathbb{C}[z_1, ..., z_n]$ contains 1 which vanishes nowhere we have that

$$\varnothing = Z(\mathbb{C}[z_1, ..., z_n])$$

Consider a bunch of collections of polynomials $S_1, ..., S_m \subset \mathbb{C}[z_1, ..., z_n])$ and define the product set as

$$S_1...S_m = \{f_1...f_m : f_i \in S_i\}$$

If $(\lambda_1, ..., \lambda_n) \in Z(S_i)$ then clearly this point is in $Z(S_1..., S_m)$. This implies that

$$Z(S_i) \subset Z(S_1...S_m)$$

$$\Rightarrow Z(S_1) \cup ... \cup Z(S_m) \subset Z(S_1...S_m)$$

Conversely if $(\lambda_1, ..., \lambda_n) \in Z(S_1...S_m)$, if $(\lambda_1, ..., \lambda_n) \notin Z(S_i)$, then for every i, there exists a $g_i \in S_i$ such that $g_i(\lambda_1, ..., \lambda_n) \neq 0$. Then for every such non zero polynomials, if we take the product

$$g_1...g_m(\lambda_1, ..., \lambda_n) \neq 0$$

But that's a contradiction since the point $(\lambda_1, ..., \lambda_n) \in Z(S_1...S_m)$. So it has to be that $(\lambda_1, ..., \lambda_n) \in Z(S_i)$ for all i. Thus $Z(S_1...S_m) \subset Z(S_1) \cup ... \cup Z(S_m)$ and so $Z(S_1...S_m) = Z(S_1) \cup ... \cup Z(S_m)$.
Next consider finitely many polynomials $f_1, ..., f_l$ and a point μ in their zero set i.e.

$$\mu = (\mu_1, ..., \mu_n) \in Z(f_1, f_2, ..., f_l)$$

so it must also lie in the intersection

$$\mu \in \bigcap_{i=1}^{l} Z(f_i)$$

The converse is obviously true as well. So we can write

$$\bigcap_i Z(S_i) = Z(\bigcup_i S_i)$$

Note that the algebraic sets $Z(S)$ satisfy of the properties of closed sets of a given topology. So we get a topology on the affine space $\mathbb{C}^n$ whose closed sets are precisely the algebraic sets.
Once we have a topological space, we can think about continuous functions on this space. Here in our case these continuous functions are polynomials.
So far, we have on the geometric side the algebraic sets which turned out to be the closed sets of the Zariski topology; while on the commutative algebraic side we just had a subset of a polynomial ring. But with just sets we cannot to algebra, we need sets equipped algebraic structures such as ring or ideal structure. Naively, if we assume that the subsets $S \subset \mathbb{C}[z_1, ..., z_n]$ are subrings, then S will contain 1 as the ring unit element. But then the algebraic set will be empty since 1 doesn't vanish anywhere.
We know that an ideal is nontrivial iff it doesn't contain a unit 1, if it does contain 1 (which is an invertible element) then the ideal has to be the whole ring. So now, how do we go from the subset S to ideals? Whenever we have an object of a certain type (or living in a certain category) and a subset of that object, to get the subobject we just consider the object (of that category) generated by that subset.
So, considering our subset S of the ring $\mathbb{C}[z_1, ..., z_n]$, to get a non trivial ideal from it, we consider the ideal generated by the set S, which we denote as $\langle S \rangle$ which as a set is a superset of S i.e. $S \subset \langle S \rangle$. This is the smallest ideal containing S, where the adjective "smallest" means that this is the ideal formed from the intersection of all the ideals containing S.
The ideal $\langle S \rangle$ can alternatively be described as the ring of $\mathbb{C}[z_1, ..., z_n]$-linear

combinations of finitely many elements of S i.e.

$$\langle S\rangle := \{\sum_{i=1}^{l} f_i g_i : f_i \in S, g_i \in \mathbb{C}[z_1, ..., z_n]\}$$

Containing the vanishing polynomials f_i in these combinations, it so happens (which is obvious) that

$$Z(\langle S\rangle) = Z(S)$$

The funny thing is that even though the set S contains finitely many polynomials, the ideal $\langle S\rangle$ contains inifitely many polynomials; so it might seem like that the zero set $Z(\langle S\rangle)$ is empty, but that's not what happens.

Similar to the above, we can show that for ideals $\{I_\alpha\}_{\alpha\in A}$ of $\mathbb{C}[z_1, ..., z_n]$: $Z(I_1) \cup Z(I_2) = Z(I_1 \cap I_2), \bigcap_\alpha Z(I_\alpha) = Z(\sum_\alpha I_\alpha)$

So far what we have seen is that, given a subset $S \subset \mathbb{C}[z_1, ..., z_n]$, we have an algebraic set $Z(\langle S\rangle) \subset \mathbb{C}^n$. So Z is a mapping from $\mathbb{C}[z_1, ..., z_n]$ to $\mathbb{C}^n$. We want this mapping to be a bijection. For that we will be needing an inverse. Let's just do that. Define a mapping from $\mathbb{C}^n$ to $\mathbb{C}[z_1, ..., z_n]$ as follows: Given a subset $X \subset \mathbb{C}^n$, define the ideal of X as

$$\mathcal{I}(X) := \{f \in \mathbb{C}[z_1, ..., z_n] : f(x) = 0, \forall x \in X\} \tag{166}$$

(it can be trivially shown that this set is indeed an ideal of the ring $\mathbb{C}[z_1, ..., z_n]$). We defined the Zariski topology by declaring as it's closed sets the algebraic sets $Z(I)$ for any given ideal I of $\mathbb{C}[z_1, ..., z_n]$. Obviously not any subset of $\mathbb{C}^n$ is closed with respect to the Zariski topology since every such subset is not necessarily algebraic. But the mappings Z and $\mathcal{I}$ can give us a closed set (closed with respect to the Zariski topology) for any subset of $\mathbb{C}^n$ as we will see below.

Let X be any subset of $\mathbb{C}^n$ and H be an algebraic set that contains the set X, $X \subset H$. It then follows that $Z(\mathcal{I}(X)) \subset H$.
Since the set $Z(\mathcal{I}(X))$ is already closed (by definition), this set is the smallest closed set that contains the set X. Whenever $Z(\mathcal{I}(X))$ is contained in any other closed sets like H, H will also contain $Z(\mathcal{I}(X))$, by the minimality of the closure of a set. This implies that $Z(\mathcal{I}(X))$ is the closure of X, i.e.

$$Z(\mathcal{I}(X)) = \overline{X}$$

And if the set X is already closed, then

$$Z(\mathcal{I}(X)) = X$$

Given any ideal J of a ring, its radical is defined as the ideal

$$\sqrt{J} := \{j : j^n \in J\}$$

for some $n \in \mathbb{N}$. An ideal whose radical is itself i.e. $\sqrt{J} = J$ is called a radical ideal. Let I be an ideal of the polynomial ring $\mathbb{C}[z_1, ..., z_n]$, then $\mathcal{I}(Z(I)) \subset \sqrt{I}$. This result known as (the strong form of) the Hilbert's Nullstellensatz in commutative algebra.
To prove the converse inclusion, let $f \in \sqrt{I}$. This implies that, for some $m \in \mathbb{N}$, $f^m \in I$. Which means that f^m vanishes on the algebraic set $Z(I)$ and this means that f vanishes on $Z(I)$ as well. So, $f \in \mathcal{I}(Z(I))$ and thus $\sqrt{I} \subset \mathcal{I}(Z(I))$. Therefore,

$$\sqrt{I} = \mathcal{I}(Z(I))$$

Finally, if we further impose the condition that the ideals I are radical ideals i.e. $I = \sqrt{I}$, then we see that the mapping Z and $\mathcal{I}$ become inverse of each other

$$I = \mathcal{I}(Z(I))$$

Now we have a bijective correspondence between the algebraic sets of $\mathbb{C}^n$ and the radical ideals of the polynomial ring $\mathbb{C}[z_1, ..., z_n]$
Recall that a subset U of a topological space X is called irreducible if it cannot be written as a union of two proper closed subsets of U. To define a variety, just put the further restriction that the algebraic sets are irreducible with respect to the Zariski topology.
That is, *irreducible algebraic sets are called (affine) algebraic varieties*.
Thus far, we have seen that to get a bijective correspondence between geometric and commutative algebraic sides, we need to consider the radical ideals on the algebraic side. Since now we have further restricted the geometric side to varieties via putting the irreducibility constraint in the Zariski topology, let us investigate what happens on the algebraic side.
Take a closed $X \in \mathbb{C}^n$ and avoid the irreducibility condition. Then it can be written as a union of two proper closed sets say $Z(J_1), Z(J_2)$, i.e.

$$X = Z(\mathcal{I}(X)) = Z(J_1) \cup Z(J_2) = Z(J_1 J_2) = Z(J_1 \cap J_2)$$

for some radical ideals J_1, J_2 such that $Z(J_1), Z(J_2) \subset X$. From the strong Nullstellensatz we get

$$\sqrt{J_1 \cap J_2} = \mathcal{I}(Z(J_1 \cap J_2)) = \mathcal{I}(X)$$

thus $J_1 \cap J_2 \subseteq \mathcal{I}(X)$. Consider an element $x \in X - V(J_1)$. Then $\exists f_1 \in J_1$ that does not vanish at x, i.e. $f_1(x) \neq 0$, then we conclude that it does not belong to the ideal of X, $f_1 \notin \mathcal{I}(X)$ (since the elements of the ideal $\mathcal{I}(X)$ vanishes everywhere on the algebraic set X).
Similarly, we can again consider $y \in X - V(J_2)$ for which $\exists f_2 \in J_2$ such that $f_2(y) \neq 0$, which implies that $f_2 \notin \mathcal{I}(X)$.
Now consider the product $f_1 f_2$ which is an element of the ideal $J_1 \cap J_2$. Observe that $(f_1 f_2)(x) = (f_1 f_2)(y) = 0$, so $f_1 f_2 \in \mathcal{I}(X)$ (since $J_1 \cap J_2 \subseteq \mathcal{I}(X)$).
We recall from commutative algebra that an ideal J is called a prime ideal if $\forall a_1 a_2 \in J$ implies that either $a_1 \in J$ or $a_2 \in J$. From this we conclude that if a prime ideal contains a finite product of ideals, then it has to contain at least

one of them.
From the above we see that although $f_1 f_2 \in \mathcal{I}(X)$, neither of them ($f_1, f_2$ individually) are contained in the ideal $\mathcal{I}(X)$. Hence the for an algebraic set X, without irreducibility, its ideal $\mathcal{I}(X)$ is not prime.
Conversely, let us assume that the ideal $\mathcal{I}(X) \subset \mathcal{C}[z_1, ..., z_n]$ is not prime. So we can have polynomials f_1, f_2 outside the ideal i.e. $f_1, f_2 \in \mathbb{C}[z_1, ..., z_n] - \mathcal{I}(X)$ such that $f_1 f_2 \in \mathcal{I}(X)$.
Then, since X is closed,

$$X = Z(\mathcal{I}(X)) \subseteq Z(f_1 f_2) = Z(f_1) \cup Z(f_2)$$

Thus we can write

$$X = X \cap Z(f_1 f_2) = (X \cap Z(f_1) \cup (X \cap Z(f_2))$$

since $Z(f_1 f_2) = Z(f_1) \cup Z(f_2)$. But the sets $Z(f_1), Z(f_2), X$ are all closed sets and the intersection of closed sets are closed as well. So we see above that X can be written as the union of two proper closed subsets of X, hence X is not a variety.
So from the above we conclude that
(affine) algebraic varieties correspond to prime ideals .
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The coordinate ring of a variety V is the quotient ring

$$\mathbb{C}[V] := \frac{\mathbb{C}[z_1, ..., z_n]}{\langle f_1, ..., f_m \rangle} \tag{167}$$

that contains all the polynomial functions on the variety V, which is the defined to be the zero set of the ideal generated by the polynomials $f_1, ..., f_m$. Since V is a variety, from the above we can say that the ideal $\langle f_1, ..., f_m \rangle$ must be a prime ideal. So the coordinate ring must be an integral domain. Its a trivial observation that the zero set of a bunch of polynomials is the same as the zero set of the ideal generated by those polynomials. The notion of coordinate ring of an affine algebraic variety stems from the relation between the affine space $\mathbb{C}^n$ and the polynomial ring $\mathbb{C}[z_1, ..., z_n]$, which is also the ring of polynomial functions on the space $\mathbb{C}^n$.
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So far we have defined affine varieties as "something embedded in the affine space $\mathbb{C}^n$". But get rid of this "embedding dependent" definition, we move forward a more abstract one, for which we need to define an important tool in algebraic geometry called *sheaf* theory. Sheaves allows us to *glue* (rings of) functions, that are defined on some topological space, compatibly with the topological gluing.
Let Y be a topological space. A *presheaf* $\mathcal{F}$ on Y is an assignment of algebraic objects on open sets of Y. For any open set

12 Chern class and a crucial computation

blah blah blah

On the manifold $\mathbb{CP}^m$, the holomorphic bundle $T^{1,0}\mathbb{CP}^m$ is spanned by the vectors $s_i(z)\frac{\partial}{\partial z_i}$ where s_i are the sections of the hyperplane line bundle $O_{\mathbb{CP}^m}(1)$ i.e. $s_i \in \Gamma(O_{\mathbb{CP}^m}(1))$, here the index $i = 0, 1, .., m$. There exists a surjection

$$O_{\mathbb{CP}^m}(1)^{\oplus(m+1)} \to T^{1,0}\mathbb{CP}^m \tag{168}$$

that has the explicit form

$$(z_0, ..., z_m) \mapsto \sum_{i=0}^{m} z_i \frac{\partial}{\partial z_i}$$

since the homogenous coordinates of the complex projective space are also sections of the hyperplane line bundle over that space. The kernel of this map are constants, so we have got the short exact sequence

$$0 \to \mathbb{C} \to O_{\mathbb{CP}^m}(1)^{\oplus(m+1)} \to T^{1,0}\mathbb{CP}^m \to 0 \tag{169}$$

which indicates the direct sum of the holomorphic vector spaces

$$T^{1,0}\mathbb{CP}^m \oplus \mathbb{C} \simeq O_{\mathbb{CP}^m}(1)^{\oplus(m+1)}$$

For which the total chern class are

$$c(T^{1,0}\mathbb{CP}^m \oplus \mathbb{C}) = c(O_{\mathbb{CP}^m}(1)^{\oplus(m+1)})$$

$$\Rightarrow c(T^{1,0}\mathbb{CP}^m)c(\mathbb{C}) = [c(O_{\mathbb{CP}^m}(1))]^{m+1}$$

Here, multiplication really means taking wedge products. As $\mathbb{C}$ is a trivial line bundle, $c(\mathbb{C}) = 1$. For the hyperplane line bundle we wont get chern classes higher than that of the first order, so

$$c(O_{\mathbb{CP}^m}(1)) = 1 + c_1(O_{\mathbb{CP}^m}(1))$$

Denoting the closed two form $c_1(O_{\mathbb{CP}^m}(1)) \in H^2(\mathbb{CP}^m)$ by x, we write

$$c(T^{1,0}\mathbb{CP}^m) = [c(O_{\mathbb{CP}^m}(1)]^{m+1} = (1+x)^{m+1} \tag{170}$$

If we have a complex projective hypersurface $X \in \mathbb{CP}^m$ that is there is homogeneous polynomial of degree d

$$P \in \mathbb{C}[z_0, ..., z_m]$$

whose zero set $Z(P)$ is X, let us try to compute its total chern class.
As a degree d homogeneous polynomial, we recall the correspondence (that we saw in section 7) between homogeneous polynomials of degree k and the global sections of the k-times tensor powered hyperplane line bundle $O_{\mathbb{CP}^m}(k)$, we will view the polynomial P as an element of $\Gamma(O_{\mathbb{CP}^m}(d))$.
By Chow's theorem, the projective algebraic variety X is also a complex manifold, as a submanifold of $\mathbb{CP}^m$. So it has a metric, a Kähler to be precise; induced from the Fubini-Study metric of $\mathbb{CP}^m$. But its quite hard to determine that metric on the variety X. Why do we bother knowing about the metric on such hypersurfaces? It turns out that if we want to check the crucial Ricci flatness criterion for the hypersurface, we need the metric since the Ricci form (109) contains the metric. Well its common knowledge, how can do geometry without the metric?
Quite remarkably, algebraic geometry gives us a way out: without actually determining the metric we can still check Ricci flatness of X just by looking at the degree of the homogeneous polynomial P and the dimension of the complex projective space $\mathbb{CP}^m$! That's a powerful result. Let us see how that is done.
Since X is also a complex manifold, it has a holomorphic tangent bundle over it

$$T^{1,0}X \to X \tag{171}$$

and there is also the subbundle of the holomorphic tangent bundle over $\mathbb{CP}^m$, $T^{1,0}\mathbb{CP}^m$ which is just its restriction to the hypersurface X: $T^{1,0}\mathbb{CP}^m|_X$
We define the normal bundle to X as

$$N_X = \frac{T^{1,0}\mathbb{CP}^m|_X}{T^{1,0}X} \tag{172}$$

which is essentially the restriction of the d-tensor powered hyperplane line bundle $O_{\mathbb{CP}^m}(d)$ to the variety X i.e. $O_{\mathbb{CP}^m}(d)|_X$ its because on the variety X the section P maps points of X to 0 in the fibers of $O_{\mathbb{CP}^m}(d)$. So we get the following short exact sequence

$$0 \to T^{1,0}X \to T^{1,0}\mathbb{CP}^m|_X \to O_{\mathbb{CP}^m}(d)|_X \to 0 \tag{173}$$

As before

$$T^{1,0}\mathbb{CP}^m|_X \simeq T^{1,0}X \oplus O_{\mathbb{CP}^m}(d)|_X$$
$$\Rightarrow c(T^{1,0}\mathbb{CP}^m|_X) = c(T^{1,0}X)c(O_{\mathbb{CP}^m}(d)|_X) \tag{174}$$

We have seen above that $c(O_{\mathbb{CP}^m}(1)) = (1+x)$; to compute the chern class of the d-tensor powered hyperplane line bundle, we look at the chern character of the bundle $O_{\mathbb{CP}^m}(1)$. Since $c_1(O_{\mathbb{CP}^m}(1)) = x$, the chern character will be

$$ch(O_{\mathbb{CP}^m}(1)) = e^x$$
$$\Rightarrow ch(O_{\mathbb{CP}^m}(d)) = ch(O_{\mathbb{CP}^m}(1) \otimes ... \otimes O_{\mathbb{CP}^m}(1)) = e^{dx}$$
$$\Rightarrow 1 + c_1(O_{\mathbb{CP}^m}(1)) + \frac{1}{2}(c_1(O_{\mathbb{CP}^m}(1))^2 - c_2(O_{\mathbb{CP}^m}(1))) + ... = 1 + dx + ...$$

Since the rank of the bundle $O_{\mathbb{CP}^m}(1)$ is 1, from the properties of the chern classes, we conclude that only the first chern class remains and the rest of them vanish, where $c_1(O_{\mathbb{CP}^m}(1)) = dx$. So the total chern class is

$$c(O_{\mathbb{CP}^m}(d)) = 1 + dx \tag{175}$$

Now (147) gives (using (143))

$$(1+x)^{m+1} = c(T^{1,0}X)(1+dx)$$

$$\Rightarrow 1 + (m+1)x + ... = (1 + c_1(T^{1,0}X) + ..)(1+dx)$$

$$c_1(T^{1,0}X) = [(m+1) - d]x \tag{176}$$

Which tells us that X is Ricci flat iff

$$d = m + 1 \tag{177}$$

13 Kähler Ricci flow

A family of Kähler metrics $g_{i\bar{j}}(t)$ parametrized by some continuous parameter t (that can be thought of as *time*), that evolve at the opposite direction of the complex Ricci curvature of the Kähler manifold under the PDE

$$\frac{\partial}{\partial t} g_{i\bar{j}}(x,t) = -R_{i\bar{j}}(x,t) \tag{178}$$

is called a Kähler Ricci flow.

Given an initial Kähler metric $g_{i\bar{j}}(t=0) = g_{i\bar{j}}(0)$, under the evolution equation (178) the Kähler manifold evolves to a more uniform geometric structure where the Ricci curvature is *less varied on the manifold*. We recal that the Ricci curvature tensor in this case is

$$R_{i\bar{j}} = \partial\overline{\partial}\log(\det(g_{i\bar{j}}))$$

On a Kähler manifold we know (shall prove it in section on "Kähler stuff continued") that the Laplacian satisfies

$$\frac{1}{2}\Delta = \partial\overline{\partial} = \overline{\partial}\partial \tag{179}$$

So the Kähler Ricci flow

$$\frac{\partial}{\partial t} g_{i\bar{j}}(x,t) = -\frac{1}{2}\Delta\log(\det(g_{i\bar{j}}))$$

is really a heat type equation for metrics.

It's obvious that

$$\begin{aligned}
&\frac{\partial}{\partial t} g_{i\bar{j}}(t) = -R_{i\bar{j}} \\
&\Rightarrow \frac{\partial}{\partial t} \sqrt{-1} g_{i\bar{j}}(t) dz^i \wedge d\bar{z}^j = -\sqrt{-1} R_{i\bar{j}}(t) dz^i \wedge d\bar{z}^j \\
&\Rightarrow \frac{\partial}{\partial t} \omega(t) = -\operatorname{Ric}(\omega(t)) \\
&\Rightarrow \frac{\partial}{\partial t} [\omega(t)] = -c_1(M)
\end{aligned}$$

where in the last line, we didn't bother about the factor of $\frac{\sqrt{-1}}{2\pi}$. Thus in terms of the first Chern class and the Kähler class (both of which are elements of $\mathrm{H}^{1,1}(M, \mathbb{Z})$) we have the initial value problem

$$\begin{cases} \frac{\partial}{\partial t}[\omega(t)] = -c_1(M) \\ [\omega(0)] = [\omega_0] \end{cases}$$

Since the Chern class is a topological invariant it doesn't change under this continuous variation of the metric, to the right hand side of the above differential equation doesn't depend on "time" really. Thus it is an ODE whose solution is

$$[\omega(t)] = [\omega(0)] - t c_1(M) \tag{180}$$

14 Un tour en cohomologie et la théorie des schémas

In this section we discuss a very general way of viewing cohomologies that is attributed to Grothendieck mostly. The goal of Grothendieck and his school was to prove a number of beautiful conjectures of Weil, on which we will discuss on a later section. But the machinery that was developed made everyone look at cohomologies in a *functorial style*. We all know the good old definition of cohomologies: starting with a cochain complex where the differential operator squares to zero, take the quotient, at the i-th, of the kernel by the image to get the i-th cohomology. But Grothendieck's perspective generalizes this to a great extent from which we get powerful results about cohomologies for objects on which it's hard to capture what *the cochain complex* would even mean.
After having defined and discussed sheaves we want to spend a brief amount of time with ringed spaces.
The story starts with sheaf theory and essentially Grothendieck was viewing cohomologies as something called the sheaf cohomology theory. So we start with the definitions of sheaves. A sheaf of rings is a presheaf of rings on a topological space X which is given by the following:

Definition: a contravariant functor from the category $\mathbf{OP}_X$ of open sets of X to the category of rings **Ring**.

We stress the fact that we can replace the category **Ring** with other categories of algebraic objects like **Group**,**Set**,**Module** etc. Let us stick with rings for the moment as we want to assign rings on topological spaces.

So to any open set $U \subset X$, a presheaf $\mathcal{F}$ of rings is an assignment of a ring $\mathcal{F}(U)$. The contravariant functoriality is seen as following: in the category $\mathbf{OP}_X$ the morphisms are inclusions of open sets, $V \subset U$, for which there exists a ring homomorphism $\rho_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$, which are called *restriction* maps. The elements $s \in \mathcal{F}(U)$ are called sections of the presheaf $\mathcal{F}$ over the open set U. The image under the restriction map of this section is really a section of the target ring i.e. $\rho_{UV}(s) =: s|_V \in \mathcal{F}(V)$.
A **Sheaf** of rings is a presheaf of rings satisfying the following two conditions:

(i) For any open covering $\{U_i\}_{i\in I}$ of an open set $U \subset X$ and for all sections $s \in \mathcal{F}(U)$ such that $s|_{U_i} = 0$ for all i, we have that the section s of $\mathcal{F}$ over U must be zero i.e. $s = 0$. This is called the local identity of sheaves.
(ii) if the sections over the covers $s_i \in \mathcal{F}(U_i)$ satisfy $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ for all pairs $i, j \in I$, then there exists a section $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i. This is called the *gluing* condition of sheaves.
What about equipping rings to points of X? Can we do that? It turns out that indeed we can. The rings over points are known as *stalks* of the (pre)sheaf of rings on X, they contain *germs* of sections of the (pre)sheaf.
Thus, let us consider a point $x \in X$ and two of its neighborhoods U, V over which we pick two sections $s \in \mathcal{F}(U), t \in \mathcal{F}(V)$. These two sections will be called equivalent if and only if there exists a smaller neighborhood $W \subseteq U \cap V$ such that $x \in W$ and $s|_W = \rho_{UW}(s) = \rho_{VW}(t) = t|_W$. It's a straightforwad thing to check that this is indeed an equivalence relation. The equivalence classes of which are defined as the *germs* of sections of the (pre)sheaf $\mathcal{F}$ at x

$$[s]_x := \{t : s \sim_x t\}$$

Such germs at x form the ring $\mathcal{F}_x$ is called the *stalk* of the (pre)sheaf of $\mathcal{F}$ at x.
Given two such (pre)sheaves $\mathcal{F}, \mathcal{G}$ of rings on the topological space X, we can have a morphism between them

$$\varphi : \mathcal{F} \to \mathcal{G}$$

such that for all open sets $U \subset X$, we have a ring homomorphism

$$\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$$

such that for a smaller open set $V \subset U$ the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\rho_{UV}} & \mathcal{F}(V) \\ \downarrow{\scriptstyle \varphi_U} & & \downarrow{\scriptstyle \varphi_V} \\ \mathcal{G}(U) & \xrightarrow{\rho'_{UV}} & \mathcal{G}(V) \end{array}$$

commutes. One of the subtle differences of using presheaves and sheaves is for two sheaves to be isomorphic, their stalks at all of the points of topological space X should be isomorphic i.e. $\forall x \in X$, it must hold

$$\varphi_x : \mathcal{F}_x \xrightarrow{\simeq} \mathcal{G}_x$$

to the sheaves $\mathcal{F}, \mathcal{G}$ to be isomorphic. But if $\mathcal{F}, \mathcal{G}$ are two presheaves, then it must hold that $\forall U \subseteq X$

$$\varphi_U : \mathcal{F}(U) \xrightarrow{\simeq} \mathcal{G}(U)$$

for the two presheaves $\mathcal{F}, \mathcal{G}$ to be isomorphic.
Since not all presheaves are sheaves, all presheaves can at least admit the so called *associated sheaves* with the presheaves. Such sheaves are also called sheafifications of presheaves.
Given a presheaf $\mathcal{F}$ on X, there exists a sheaf $\tilde{\mathcal{F}}$ on X and a morphism of presheaves $\theta : \mathcal{F} \to \tilde{\mathcal{F}}$ satisfying a universal property: if $\mathcal{G}$ is a sheaf on X with a morphism of presheaves $\phi : \mathcal{F} \to \mathcal{G}$ there exists a unique morphism of sheaves $\psi : \tilde{\mathcal{F}} \to \mathcal{G}$ such that $\phi = \psi \circ \theta$. The sheaf $\tilde{\mathcal{F}}$ is the sheafification of the presheaf $\mathcal{F}$.

(Pre)sheaves can be of various other algebraic objects like (pre)sheaves of abelian groups/modules over some ring/sets etc. Above we discussed (pre)sheaves of rings. The reason is we want to develop the notion of *ringed spaces*. The notions of algebraic varieties and manifolds can be thought as ringed spaces, after doing which we can import tools from commutative algebra in studying their geometry.
Having defined what (pre)sheaves and their morphisms, we see that they fit in a category, a category of (pre)sheaves of rings on X that we shall denote by $\mathbf{Sh}_R(X)$. If we have a homeomorphism of two topoloical spaces

$$f : X \to Y$$

we can talk of pushforwards and pullbacks of sheaves. The pushforward is a functor between the associated category of sheaves (we will only be considering the categories of sheaves)

$$f_* : \mathbf{Sh}_R(X) \to \mathbf{Sh}_R(Y)$$

such that for any $\mathcal{F} \in \mathbf{Sh}_R(X)$, there is a sheaf $f_*\mathcal{F} \in \mathbf{Sh}_R(Y)$ that is defined as the following. For any open set $V \subseteq Y$ we have $f_*\mathcal{F}(V) := \mathcal{F}(f^{-1}(V))$ for sheaves $\mathcal{F} \in \mathbf{Sh}_R(X)$. This functor is called the direct image functor of categories of sheaves. For sheaves, pullbacks are not directly defined as the functor $\mathbf{Sh}_R(Y) \to \mathbf{Sh}_R(X)$. Instead, it is obtained from the functor

$$f^{-1} : \mathbf{Sh}_R(Y) \to \mathbf{Sh}_R(X)$$

called the inverse image functor.

This also applies to sheaves of other algebraic objects as well.

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The cohomology groups are abelian groups. So we shift our focus on sheaves of abelian groups on a topological space. There is a peculiar feature of the category of sheaves of abelian groups on a topological space X that we will denote as $\mathbf{Sh}_{Ab}(X)$ that is it behaves like the category of abelian groups. In fact, such categories are known by the name *Abelian categories* that we define below (following the appendix of [1]). This nomenclature stems from generalizing the properties of the category of abelian groups **Ab**.
Here we list the essential ones among the properties:

1. $A, B \in \mathbf{Ab}$, the set of morphisms (that are group homorphisms) $\mathrm{Hom}_{\mathbf{Ab}}(A, B)$ is itself an abelian group under the operation of composition of group composition.
2. There exists $\mathbb{Z}$-bilinearity of composition of morphisms i.e.

   $$\mathrm{Hom}_{\mathbf{Ab}}(A, B) \times \mathrm{Hom}_{\mathbf{Ab}}(B, C)] \to \mathrm{Hom}_{\mathbf{Ab}}(A, C)$$

   given by $(f, g) \mapsto g \circ f$ satisfying

   $$g \circ (f_1 + f_2) = g \circ f_1 + g \circ f_2$$

   $$(g_1 + g_2) \circ f = g_1 \circ f + g_2 \circ f$$

   .
3. finite direct sums exists.
4. every morphism has a kernel and a cokernel.
5. every monomorphism (generalization of injective maps) is the kernel of it's cokernel.
6. every epimorphism (generalization of surjective maps) is the cokernel of it's kernel.
7. every morphism can be factored into an epimorphism and a monomorphism after that.

Turns out that $\mathbf{Sh}_{Ab}(X)$ is also an abelian category.

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————**Add further examples**

————————

The global section functor is defined as

$$\Gamma(X, -) : \mathbf{Sh}(X) \to \mathbf{Ab} \tag{181}$$

such that for any such sheaf $\mathcal{F} \in \mathbf{Sh}_{Ab}(X)$ the functor acts as $\Gamma(X, \mathcal{F}) := \mathcal{F}(X)$ whose elements are called global sections. A nontrivial fact about this functor is that it's a left exact functor i.e. on a short exact sequence of sheaves

$$0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$$

if we apply the global section functor we get a sequence of rings that is exact only on the left

$$0 \to \Gamma(X, \mathcal{F}) \to \Gamma(X, \mathcal{G}) \to \Gamma(X, \mathcal{H})$$

The right derived functors of the global section functor are the sheaf cohomology groups $\mathrm{H}^i(X, \mathcal{F})$.

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———**Add more on this**

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As (pre)sheaves, we defined them as certain contravariant functors from the category of open sets $\mathbf{OP}_X$ of the topological space X to the category of some algebraic objects. To construct some very specific type of cohomologies (to be precise Weil cohomologies), this construction is generalized via replacing $\mathbf{OP}_X$ with any category admitting finite limits, with a certain type of topology known as the Grothendieck topology. Such categories are called *Sites*.
So let $\mathcal{C}$ be a category with finite limits. A Grothendieck topology on $\mathcal{C}$ is a family of maps $\{f_i : U_i \to X\}$ that are called coverings (motivated from topological coverings of course) that satsifies the following axioms:

1. for a covering $\{f_i : U_i \to X\}_{i \in I}$ and any other morphism $g : X \to Y$ in $\mathcal{C}$, then Y has a covering coming from the fibred product

$$\left\{ \begin{array}{ccc} U_i \times_X Y & \xrightarrow{p_Y} & Y \\ \downarrow{\scriptstyle p_{U_i}} & & \downarrow \\ U_i & \longrightarrow & X \end{array} \right\}_{i \in I}$$

2. Each of the U_i has a subcovering $\{g_{ij} : V_{ij} \to U_i\}_{j \in I}$ that is also the covering of X i.e. composing them we get the covering $\{f_i \circ g_{ij} : V_{ij} \to X\}_{i \in I}$.

3. If some of the morphisms from the family $\{f_i : U_i \to X\}$ have sections $\sigma_i : X \to U_i$, then $\{f_i : U_i \to X\}$ is a covering.

Such categories are called Sites.

We can see that $U \in \mathbf{OP}_X$, we have coverings $\{U_i \to U\}$ with $\bigcup_{i \in I} U_i = U$, if we further consider the partial ordering of objects of $\mathbf{OP}_X$ with respect to the inclusions, we see that it is a Site. Depending on what kind of morphisms $\{f_i : U_i \to X\}$ we consider, we get different sites like the étale site, crystalline site etc. Then the sheaves are thought of a contravariant functors from these

sites to **Ab**. Finally, the right derived functors of the global section functor for such sites are the corresponding cohomologies e.g. for the étale site we get the étale cohomology, for the crystalline site we get crystalline cohomology. We will discuss more on this in later sections. But for now, we shall only focus on sheaf cohomologies of spaces that are of algebro-geometric nature. Our aim is to tell you about the famous GAGA theorem in algebraic geometry.

It's immensely useful to replace algebraic varieties by their generalization to discuss cohomologies in algebraic geometry. Originally introduced for arithmetic applications of algebraic geometry, schemes are the central objects in the modern version of the subject.

Schemes are a vast generalization of algebraic varieties. We note that on the commutative algebraic side of the story, the coordinate ring of a variety is of a very specific type. Putting this the other way around, to a specific type geometric objects (which are in this case the varieties) we can attach a specific kind of rings. The idea of scheme theory is to extend that class of rings to the most general case, namely the commutative rings with unity.
So in the story of classical algebraic geometry, when we replace coordinate rings with any commutative ring with unity, we have schemes on the geometric side of the story.
We know from basic theory of algebraic varieties that subvarieties of a variety correspond to prime ideals of the coordinate ring of the variety and the points of the variety correspond to the maximal ideals of the coordinate ring of the variety.
To start with schemes, we begin with considering rings rather than a geometric set. That is, start with a commutative ring R with unity, the *spectrum* of R is defined as the set

$$\mathrm{Spec}(R) := \{\mathfrak{p} \subset R\}$$

of prime ideals $\mathfrak{p}$ of R. This is the *Scheme* associated to the ring R, or at least the underlying set of that *Scheme*.
Adding a topology in this set which is also called the Zariski topology we get Schemes. Let's not worry about the topology for the moment and just try to see the sets.
The most basic example might the affine space over some field k or even some ring R i.e. $\mathbb{A}^n_k$ or $\mathbb{A}^n_R$. As maximal ideals are prime ideals too, we see that the set of points has a bijective correspondence the set of maximal ideals. So (a bit naively I must say) we might write

$$\mathbb{A}^n_k = \mathrm{Spec}(k[x_1, ..., x_n])$$

and

$$\mathbb{A}^n_R = \mathrm{Spec}(R[x_1, ..., x_n])$$

but the strange thing is that the spectrum of the polynomial rings then contains *extra points*. Bucause, the spectrum has prime ideals that correspond to subvariaties of the affine space. These are called *fat/generic points* of the scheme.

Generic points have the property that their closure is the whole topological space. That's one of the fundamental differences between $\mathbb{A}^n_k$ or $\mathbb{A}^n_R$ as affine space the scheme and affine space the variety.

.....

.....**Describe the Zariski topology of schemes via their Distinguished open sets**

.....

So the shift in our view towards the geometric objects is that rather than emphasizing on the zero sets of a bunch of polynomials $f_1, ..., f_r \in k[x_1, ..., x_n]$ we look the spectrum

$$\text{Spec}(\frac{k[x_1, ..., x_n]}{\langle f_1, ..., f_r \rangle})$$

A very useful property of schemes is that they come in families. We know that the spectrum of rings is essentially a contravariant functor

$$\text{Spec} : \textbf{Ring} \to \textbf{Set}$$

Thus applying the Spec functor on the morphism

$$\mathbb{Z} \to \frac{\mathbb{Z}[x_1, ..., x_n]}{\langle f_1, ..., f_r \rangle}$$

in the category of commutative ring, we arrive at the morphism

$$\varphi : \text{Spec}(\frac{\mathbb{Z}[x_1, ..., x_n]}{\langle f_1, ..., f_r \rangle}) \to \text{Spec}(\mathbb{Z})$$

in the category of sets. This is actually family of schemes or a fibration of schemes. The base space here is $\text{Spec}(\mathbb{Z}) = \{\langle 0 \rangle, \langle 2 \rangle, \langle 3 \rangle, ...\}$. For each prime $\langle p \rangle$, there is a *subscheme* (a subspace in the Zariski topology of schemes) $\text{Spec}(\mathbb{F}_p[x_1, ..., x_n]/\langle f_1, ..., f_r \rangle) = \varphi^{-1}(\langle p \rangle)$. This scheme has the fat/generic point corresponding to the zero ideal. So we see that schemes over the ring of integers contain all the reduction modulo p subschemes in them.

By many such observations, Grothendieck told us to study morphisms of schemes $X \to S$ rather than a particular scheme X. Such (families of) schemes are called S-schemes for any other base scheme S. This is called *the relative perspective* of Grothendieck.

Truth be told, the above schemes that we mentioned are called affine schemes really. In general, schemes can be very complicated. Affine schemes are the building blocks of schemes just as open balls of the Euclidean space are the building blocks of manifolds. But the major difference between schemes and manifolds is that for manifolds the open balls are all the same. But for schemes the affine schemes that are *glued* together to construct the scheme can be wildly different from each other. To perform these gluing of affine scheme, it is useful to define schemes as locally ringed spaces.

An affine scheme is the ringed space $(\text{Spec}(R), \mathcal{O}_{\text{Spec}(R)})$ where the structure

sheaf is the sheaf of coordinate ring over the whole underlying topological space i.e. $\mathcal{O}_{\mathrm{Spec}(R)}(\mathrm{Spec}(R)) = R$. And over any open set U of $\mathrm{Spec}(R)$, $\mathcal{O}_{\mathrm{Spec}(R)}(U)$ is just the ring of rational functions on U.
For any point $\mathfrak{p} \in \mathrm{Spec}(R)$, the stalk $\mathcal{O}_{\mathrm{Spec}(R),\mathfrak{p}}$ is the local ring at that point with it's unique maximal ideal $\mathfrak{m}_\mathfrak{p}$ and residue field $\mathcal{O}_{\mathrm{Spec}(R),\mathfrak{p}}/\mathfrak{m}_\mathfrak{p}$.

Now if the ring R is some polynomial ring over some ground field k, then the residue field at some point $\mathfrak{p}$ is an extension of k. The degree of this extension is called the degree of that point. These degrees (of points) are used to define zeta functions attached to schemes that we will describe in section(??).

To glue things in geometry, we need *gluing data*. Let us step back to ringed spaces for a while. For a collection of locally ringed spaces $\{X_i\}_{i\in I}$ (we keep their structure sheaves $\mathcal{O}_{X_i}$ implicit), suppose for each pair of indices $i, j \in I$ there exists open sets $U_{ij} \subset X_i$ and $U_{ji} \subset X_j$ such that for each $i, j, k \in I$,
(a)the morphism $\psi_{ij} : U_{ij} \to U_{ji}$ of locally ringed spaces is an isomorphism. We are calling these open sets locally ringed spaces in the sense of the restriction $(U_{ij}, \mathcal{O}_{X_i}|_{U_{ij}})$.
(b)$\psi_{ij}^{-1}(U_{ji} \cap U_{jk}) = U_{ij} \cap U_{ik}$
(c) $\psi_{ik} = \psi_{jk} \circ \psi_{ij}$ on the open set $U_{ij} \cap U_{jk} \cap U_{ik}$ called the cocycle condition.
For $i = j$, we assume that $U_{ii} = X_i$ and $\psi_{ii} = \mathrm{Id}_{X_i}$. Packing all of these up, we get the collection $\{(I, X_i, U_{ij}, \psi_{ij})\}_{i\in I}$, called the gluing data for the family of locally ringed spaces $\{X_i\}_{i\in I}$.
Given such a gluing data for locally ringed spaces $\{X_i\}_{i\in I}$, we can prove (which we will later) that there exists a locally ringed space X with a family of open subspaces $\{U_i\}_{i\in I}$ such that each of them are isomorphic to the X_i's. In other words, the X's are *glued* to form X. Putting it a bit more precisely, we have a family of isomorphisms of locally ringes spaces

$$\{\psi_i : U_i \xrightarrow{\sim} X_i\}$$

such that
(a) for $U_{ij} \subset X_i$, $\psi_i(U_{ij}) = U_i \cap U_j$.
(b) $\psi_j^{-1}|_{U_j \cap U_i} \circ \psi_i|_{U_{ij}} =: \psi_{ij} : U_{ij} \xrightarrow{\sim} U_{ji}$.
Now if we have a family of commutative rings $\{R_i\}_{i\in I}$, their associated schemes form a family of locally ringed space $\{X_i\}_{i\in I}$ where $X_i = \mathrm{Spec}(R_i)$. Thus, there exists a scheme $X = \mathrm{Spec}(R)$ such that the schemes $X_i = \mathrm{Spec}(R_i)$'s will be it's subschemes.
In fact we can glue schemes via modding out by some equivalence relation. Let us now do just that.
Given a family $\{X_i\}_{i\in I}$ of such ringed space (or let's be more specific by assuming that they are schemes), on their union $\coprod X_i$ an equivalence relation is defined as: for $x_i \in U_{ij} \subset X_i$, $x_j \in U_{ji} \subset X_j$, if the isomorphism map $\psi_{ij} : U_{ij} \xrightarrow{\sim} U_{ji}$ maps x_i to x_j i.e. $\psi_{ij}(x_i) = x_j$, then $x_i \sim x_j$.
As an equivalence relation the reflexivity and symmetry of this relation are ob-

vious. Transitivity can be seen as follows:
Assume that $x_i \sim x_j$, $x_j \sim x_k$. Now the first equivalence $x_i \sim x_j$ implies that $\psi_{ij}(x_i) = x_j$ for $x_i \in U_{ij} \subset X_i$, $x_j \subset U_{ji} \in X_j$. But then the second equivalence $x_j \sim x_k$ implies $\psi_{jk}(x_j) = x_k$ for $x_j \in U_{jk} \subset X_j$, $x_k \in U_{kj} \subset X_k$. So the common point is x_j i.e. $x_j \in U_{ji} \cap U_{jk}$.
Observe that since $\psi_{ij}(x_i) = x_j$ is an isomorphism, we can invert that

$$x_i = \psi_{ij}^{-1}(x_j) \in \psi_{ij}^{-1}(U_{ji} \cap U_{jk}) = U_{ij} \cap U_{ik}$$

we see that $x_i \in U_{ik}$ and similarly

$$x_k = \psi_{jk}(x_j) \in \psi_{jk}(U_{ji} \cap U_{jk}) = U_{ki} \cap U_{kj}$$

implies $x_k \in U_{ki}$, we can apply the cocycle condition:

$$\psi_{ik}(x_i) = (\psi_{jk} \circ \psi_{ij})(x_i) = \psi_{jk}(x_j) = x_k$$

so $x_i \sim x_k$. Hence we see the transitivity.
So the glued space X is the quotient $\coprod_i X_i / \sim$.

15 Too early for Calabi Yaus?

A celebrity is known by many names they say. A Calabi Yau manifold has a number of equivalent definitions, we will choose a particular one to state at first:
Definition *:*A Kähler manifold whose first chern class vanishes is called a Calabi Yau manifold.*
Let us try to unfold this definition bit by bit. Let M be a Kähler manifold. The definition $*$ uses the notion of the first chern class of the Kähler manifold M that will be called Calabi Yau; what we actually mean is the chern class of the holomorphic tangent bundle $T^{1,0}M$ of M. As a characteristic class, the chern class has Ricci forms (as traces of the curvature two forms) in it. We have seen the Ricci form Ric_g in (118) and know that the first chern class of a Kähler manifold is

$$c_1(M) = c_1(T^{1,0}M) := \frac{1}{2\pi}[Ric_g] \tag{182}$$

for the Kähler metric g, whose Kähler form will be denoted as Ω (given by (30)). So if

$$c_1(M) = 0 \tag{183}$$

this means the Kähler manifold should be Ricci flat i.e. it's Ricci form vanishes. So we can say

Definition **:*A Ricci flat Kähler manifold is called a Calabi Yau manifold.*

We warn the reader that the equivalence between the two definitions above

is a subtle matter. We have oversimplified that. This deserves some discussion, that will lead us to its historical root known as the Calabi conjecture.
From (160), we see that if the Kähler manifold admits a Ricci flat metric, (161) will hold. But the converse (that (161) implies Ricci flatness) was a form of the famous Calabi conjecture. To understand the statement of this conjecture, we will require the $\partial\bar{\partial}$-lemma that says:
On a compact Kähler manifold M, two Kähler forms Ω and Ω' are cohomologous iff $\Omega - \Omega' = \sqrt{-1}\partial\bar{\partial}f$, where f is a real valued function on M.

We recall that on a Kähler manifold, the Kähler form is required to be closed as $(1,1)$ form, so we can get cohomology classes $[\Omega] \in H^2(M, \mathbb{R})$ (called the Kähler class) as the form Ω is also a real form. We say that the two Kähler metrics

$$g_{\mu\bar{\nu}} dz^\mu \otimes d\bar{z}^\nu$$

$$g'_{\mu\bar{\nu}} dz^\mu \otimes d\bar{z}^\nu$$

are cohomologous iff the associated forms

$$\Omega = i g_{\mu\bar{\nu}} dz^\mu \wedge d\bar{z}^\nu \tag{184}$$

$$\Omega' = i g'_{\mu\bar{\nu}} dz^\mu \wedge d\bar{z}^\nu \tag{185}$$

are cohomologous according to the $\partial\bar{\partial}$-lemma. We are ready to state the Calabi conjecture. In 1954 Eugenio Calabi conjectured:
For a compact Kähler manifold M with Kähler metric $g_{\mu\bar{\nu}} dz^\mu \otimes d\bar{z}^\nu$ and the Ricci form $\sqrt{-1}R'_{\mu\bar{\nu}} dz^\mu \wedge d\bar{z}^\nu$ representing the first chern class of M, there exists a unique Kähler metric $g'_{\mu\bar{\nu}} dz^\mu \otimes d\bar{z}^\nu$ cohomologous to $g_{\mu\bar{\nu}} dz^\mu \otimes d\bar{z}^\nu$ such that its Ricci tensor is $R'_{\mu\bar{\nu}} dz^\mu \otimes d\bar{z}^\nu$

proved by Shiing Tung Yau in 1978. In the equivalence class $[g_{\mu\bar{\nu}}]$, Calabi claimed that we can always find a unique Kähler $g'_{\mu\bar{\nu}}$ metric for which the Ricci form represents the first chern class. So now if we further impose the condition that the first chern class $c_1(M) = 0$, the conjecture says that we should be able to find a unique metric in the same Kähler class that will be Ricci flat. Later after Yau's proof, H.D. Cao gave an alternative proof, in which he "deforms" the metric $g_{\mu\bar{\nu}}$ under a partial differential equation

$$\frac{\partial g_{\mu\bar{\nu}}(t)}{\partial t} = -R_{\mu\bar{\nu}} \tag{186}$$

known as the Kähler Ricci flow (with the initial condition $g_{\mu\bar{\nu}}(t=0) = g_{\mu\bar{\nu}}$), to the metric $g'_{\mu\bar{\nu}}$.

Ricci flat Kähler metrics are thus called Calabi Yau metrics. The bad news is, even though we know the existence and uniqueness of such metrics, only a few explicit Calabi Yau metrics are known among which the only non trivial Calabi

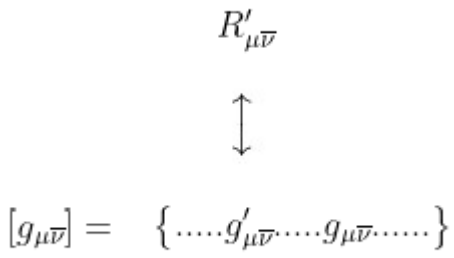


Figure 1: Rough "picture" of the Calabi Conjecture

Yau metric is that on a complex torus!
The way out of this difficulty is to use other equivalent definitions that don't require the explicit construction of the Kähler metric. We note that our **Definition *** has the significant consequence that it implies that the canonical bundle of M, denoted K_M must be trivial. Recall that the canonical bundle K_M is the highest wedge product of the holomorphic cotangent bundle $T^{1,0*}M$ i.e.

$$K_M := \bigwedge^m T^{1,0*}M \tag{187}$$

(where $dim_{\mathbb{C}}M = m$) whose sections are $(m,0)$ forms on M i.e. $\Gamma(K_M) = \Omega^{m,0}(M)$. We know that the dimesion of this complex vector space $\Omega^{m,0}(M)$ is $dim_{\mathbb{C}}\Omega^{m,0}(M) = 1$ making it a line bundle. Because of the antisymmetry of wedge products, it's bundle structure further implies that K_M is the determinant bundle of the holomorphic cotangent vector bundle $T^{1,0*}M$ since its transition matrices are the determinants of those of $T^{1,0*}M$. Using some properties of chern classes (e.g. the first chern class of a trivial bundle is zero and the first chern class of a finite rank complex vector bundle is equal to the first chern class of the determinant line bundle), a non trivial implication is that triviality of the canonical bundle K_M gives

$$c_1(T^{1,0*}M) = -c_1(T^{1,0}M) = 0$$

by our **Definition *** with the converse also being true. So we get

Definition ***: *A Kähler manifold with trivial canonical bundle is called a Calabi Yau manifold.*

We stress the fact that this is the algebro geometric definition of a Calabi Yau manifold because of the following. Chow's theorem says that any complex analytic submanifold X of the complex projective space $\mathbb{CP}^m$ is automatically a

non singular projective variety. The Study Fubini metric is induced on X, making it a Kähler manifold too. So, if the canonical bundle is trivial for such a variety, then it's a Calabi Yau variety.
Now this triviality of the canonical line bundle K_M has the further implication that there must exist a globally non vanishing holomorphic top form on M. As a trivial line bundle, the total space of K_M should be $M \times \mathbb{C}$ as it's also a complex line bundle. So (??) in the complex vector space of its sections $\Gamma(K_M)$, to the unit section from M to $M \times \{1\}$ (i.e. the constant function 1), corresponds a globally non vanishing holomorphic $(m, 0)$ form Θ on M. Such a form can be written as $f\Theta$ for some function f on M. If M is assumed to be compact, then the extended maximum modulus theorem implies that the function f is constant. So all the other elements of the space of holomorphic $(m, 0)$ forms, $\Omega^{m,0}(M)$ are just scalings of each other. Hence

$$dim_{\mathbb{C}}\Omega^{m,0}(M) = 1$$

Conversely, the existence of such a form implies that the bundle K_M is trivial. The existence of such a form is implied by the Ricci flatness condition too. We use the fact that an $(n, 0)$ form $\Theta \in \Omega^{m,0}(M)$ is proportional to the m dimensional alternating symbol

$$\eta_{i_1,\ldots,i_m} = \frac{1}{g}\epsilon_{i_1,\ldots,i_m} \tag{188}$$

(where g is the determinant of the metric $g_{\mu\bar{\nu}}$) which we will be needing because of the total antisymmetry of wedge products. Pick a globally defined holomorphic function $f(z)$ on the manifold M and define

$$\Theta_{i_1,\ldots,i_m} = f(z)\eta_{i_1,\ldots,i_m} \tag{189}$$

Using a Hermitian metric on the Kähler manifold M, we can write the norm of this form

$$||\Theta||^2 = \frac{1}{n!}\Theta_{i_1,\ldots,i_m}g^{i_1\bar{j}_1}\ldots g^{i_m\bar{j}_m}\overline{\Theta}_{\bar{i}_1,\ldots,\bar{i}_m} = \frac{|f|^2}{g}$$

Which is globally defined as a scalar. Therefore

$$g = \frac{|f|^2}{||\Theta||^2} \tag{190}$$

Recall the complex Ricci form from (118), we see that

$$Ric_g := -\sqrt{-1}\partial_\mu\partial_{\bar{\nu}}\log[g]dz^\mu \wedge d\bar{z}^\nu \tag{191}$$

$$= -\sqrt{-1}\partial_\mu\partial_{\bar{\nu}}\log[\frac{|f|^2}{||\Theta||^2}]dz^\mu \wedge d\bar{z}^\nu$$

$$= \sqrt{-1}\partial_\mu\partial_{\bar{\nu}}\log[||\Theta||^2]dz^\mu \wedge d\bar{z}^\nu$$

But Θ being a globally defined coordinate scalar, the Ricci form vanishes and thus the first chern class vanishes too. Hence we get

Definition ****: *A compact Kähler manifold M admitting a globally non vanishing holomorphic top form is a called a Calabi Yau manifold*.
The reason for looking at different but equivalent definitions of Calabi Yau manifolds is the fact that it's extremely hard to write down a Calabi Yau metric. We know it's existence and uniqueness on some manifolds, but we don't know it's explicit form. The only non trivial example of a compact Calabi Yau manifold of which we know the metric is the complex torus.

16 Moduli space: geometer's love

Geometers of all type are sort of obssessed with the notion of moduli spaces as these spaces are so important, hiding wealths of information about objects that they are constructed to study. We won't give formal definitions at first, but only show some examples. Informally, moduli spaces are geometric spaces whose points parametrize isomorphism classes of certain structures. Often and quite surprisingly, the geometry of the moduli space reveals many delicate secrets about the objects whose isomorphism classes it parametrizes. But the overall structure, the geometry and the topology of moduli spaces is most of the time very difficult, some properties are almost impossible to compute.
To motivate the theory of moduli spaces, we briefly discuss some examples. First, we study plane conics in $\mathbb{R}^2$. These are basically just zero sets of the second degree polynomial

$$f(x_1, x_2) = a_1 x_1^2 + a_2 x_1 x_2 + a_3 x_2^2 + a_4 x_1 + a_5 x_2 + a_6 \tag{192}$$

where the coefficients $a_1, a_2, ..., a_6 \in \mathbb{R}$ and $(x_1, x_2) \in \mathbb{R}^2$. We see that plane conics are essentially parametrized by these coefficients a_i's where $i = 1, ..., 6$. These are points in the 6-space i.e. $(a_1, ..., a_6) \in \mathbb{R}^6$. Obviously the point $(0, ..., 0) \in \mathbb{R}^6$ represents no conic and any scalar multiple

$$\lambda.(a_1, ..., a_6) := (\lambda a_1, ..., \lambda a_6)$$

represents the same conic as that is represented by the polynomial (170). Clearly, this "representing the same conic" is an equivalence relation. So the moduli space for plane conics should be

$$\frac{\mathbb{R}^6 - \{0\}}{\sim} =: \mathbb{RP}^5 \tag{193}$$

Remarkably, the intersection theory of this moduli space of plane conics answers enumerative questions about conics.
A very intellectually appealing fact is that, the moduli space of algebraic varieties can itself turn out to be a variety. As we can see, the moduli space $\mathbb{RP}^5$ is a projective variety. Well, if we are worry about the algebraic non closure of the field $\mathbb{R}$, we can think of the moduli space of complex plane conics $\mathbb{CP}^5$.

The space $\mathbb{CP}^5$ is a complex projective variety.
Same goes for Riemann surfaces i.e. the moduli space of Riemann surfaces is itself a Riemann surface as we will see below.
Moduli spaces arises in different branches of mathematics, in different contexts. So general theory of moduli spaces is still underdeveloped. Also it is because of this variety of contexts in which the problem of constructing a good moduli space that classifies objects of interest that we see tools from various fields are play in extracting analytic, topological and geometrical properties of the moduli space. But some attempts have been made to make general definitions of moduli spaces if they exist, when they exist and how they exist.
Grothendieck first gave a general definition of moduli problems in a functorial langauage. Actually, Grothedieck probably had seen the power in the language of categories and functors better than any other mathematician at that time. Perhaps that is the reason why we see the usage of this language in his work so often. He was very good at solving problems using categories and functors.
First we review bits of category theory that will be necessary for us in this section and thereafter. Arguably, the most significant theorem in all of category theory is the famous Yoneda lemma. This lemma roughly tells you that if you want to study an object an any category, then study the morphisms from and to that object from other objects in that category. The story starts from the so called *functor of points*. Why it's called so is(will be) explained in section discussing schemes (section ??).
Let X be an object from a locally small category (i.e. a category whose collection of objects forms a set) $\mathbf{C}$ and **Set** be the category of sets. A functor of points of X denoted h_X is a contravariant functor

$$h_X : \mathbf{C} \to \mathbf{Set}$$

defined as $h_X(Y) = \mathrm{Hom}_{\mathbf{C}}(Y, X)$ for objects Y in the category $\mathbf{C}$. To all objects of the category $\mathbf{C}$ we can associate functor of points. These functors form a category whose morphisms are natural transformations among these functors. This is called a functor category, we shall denote the functor category for our present case as $\mathbf{Set}^{\mathbf{C}}$.
There exists this embedding of the categories

$$\mathbf{C} \to \mathbf{Set}^{\mathbf{C}}$$

called the Yoneda embedding functor. The advantage of this embedding functor is just that studying X is equivalent to studying h_X. Problems like proving the isomorphism $X \simeq Y$ in $\mathbf{C}$ is equivalent to proving the natural isomorphism of the associated functor of points i.e. $h_X \simeq h_Y$ in the functor category $\mathbf{Set}^{\mathbf{C}}$.
Functors are in general very abstract. But the Yoneda lemma which states that there exists the isomorphism:

$$Hom_{\mathbf{Set}^{\mathbf{C}}}(h_X, F) \simeq F(X)$$

That is, there exists bijective correspondence between a natural isomorphism $\eta : h_X \to F$ and objects $F(X) \in \mathbf{Set}$ for functors $h_X, F \in \mathbf{Set}^{\mathbf{C}}$. Now if we look at the stronger case where we have a natural isomorphism $h_X \simeq F$, we say the that the functor F from a locally small category $\mathbf{C}$ to the category of sets $\mathbf{Set}$ is *represented by the object* X of the category $\mathbf{C}$. Then Yoneda lemma ensures that if such an object $X \in \mathbf{C}$ exists representing some other functor, then it must be unique up to a unique isomorphism.
Now we will see that *moduli problems* can be interpreted by a functor called the *moduli functor*, the existence of a moduli space for any moduli problem is equivalent to the existence of representability of the moduli functor in the category of schemes/manifolds/complex analytic spaces etc.
We have seen collections of objects that are *labelled* by elements of some other set e.g. $\{X_i\}_{i\in I}$. This is also called a *family of the objects* X_i *parameterized by the set* I. In general, families of objects show functorial behavior. Roughly, changing the parametrizing set changes the family. We will see more on this functorial behavior of families when we will discuss deformations in the next section. For now, let us elaborate what a family functor is. It's a functor $\mathbb{FAM} : \mathbf{C} \to \mathbf{Set}$ from the category $\mathbf{C}$ of the objects that we want to classify upto some equivalence defined by the particular kind of isomorphisms of these objects. The category $\mathbf{C}$ should have a number of useful properties, and we it call the parameter category.
.....
Properties of the parameter category
.....
So for any object $T \in \mathbf{C}$, under the family functor it is mapped as $T \mapsto \mathbb{FAM}(T)$ where $\mathbb{FAM}(T)$ is the set of families of the objects that we want to classify upto isomorphism parameterized by T for which there is an equivalence of these objects $\sim_T$ which is just the isomorphism of families parameterized by T
The moduli functor is the family functor where in the image of this functor we mod out by the equivalence relation, $\mathbf{F} : \mathbf{C} \to \mathbf{Set}$ such that $T \mapsto \mathbf{F}(T) := \mathbb{FAM}(T)/\sim_T$.
If there exists an object $X \in \mathbf{C}$ that represents the moduli functor i.e. if the isomorphism

$$h_X \simeq \mathbf{F} \tag{194}$$

then X will be the moduli space for our moduli problem defined by the moduli functor $\mathbf{F}$. Yoneda lemma then tells us that the moduli space X, if it exists, is unique upto unique isomorphism. More precisely, X is called a fine moduli space. In fact, fine moduli spaces rarely exists. If the objects of our interest that we want to classify upto isomorphism admit nontrivial automorphisms, then the moduli functor fails to be representable. Unfortunately, most of the objects that geometers are curious to classify do admit nontrivial automorphisms. Thus, it is useful if we *weaken* our moduli problem if such automorphisms exist. The weaking is done by relaxing the condition that the moduli functor should be naturally isomorphic the functor of points. Instead, we just require that there should exist a natural transformation $\eta : h_X \to \mathbf{F}$ between the functors.

Then X is called a *coarse* moduli space.
Fine moduli spaces, when they exist, admit a family hovering above them cause moduli spaces themselves are parameter spaces of (isomorphism classes (because we have performed the modding out by the equivalence relation of the families) of) objects. So the fibres of the points are members of a family which is called the universal family. Universal in the sense that every other possible family over different parameterizing objects that admit a morphism to the fine moduli space X, can be obtained via the pullback of that family. This is a universal property in the terminology of category theory. Hence the family parameterized by the fine moduli space is called the universal family.
Indeed, from the isomorphism (183) we can see that such family should exist. Let $Y \in \mathbf{C}$ be any other object. When functors on both sides of (183) feed Y

$$h_X(Y) := \mathrm{Hom}_{\mathbf{Set}}(Y, X) \simeq \mathbf{F}(Y) =: \mathbb{FAM}(Y)/\sim_Y$$

we see that for every morphism $f : Y \to X$, there should exist a family over Y such that the following diagram commutes.

$$\begin{array}{ccc} f^*\mathbb{U} = \mathbb{FAM}(Y) & \longrightarrow & \mathbb{U} \\ \downarrow & & \downarrow \\ Y & \xrightarrow{\quad f \quad} & X \end{array}$$

where $\mathbb{U}$ is the universal family over the fine moduli space X. But for coarse moduli spaces, there are no such bijective correspondences since the natural isomorphism in (183) is replaced by a natural transformation. So a coarse moduli space doesn't have a universal family.

17 Deformations of complex structures

Complex manifolds of complex dimension n have $2n$ real dimensional smooth manifolds underlying them. What's fascinating is that on given (even dimensional) smooth manifolds, the complex structure that can be equipped with it is not unique, instead complex structures come in families on the underlying smooth manifold. This brilliant observation was recorded in the work of Bernhard Riemann on Riemann surfaces that are the most primitive class of complex manifolds. He calculated that on a genus $g > 1$ topologically *nice* surface, there are $6g - 6$ parameters on which the complex structures that can be equipped on the surface depend. Which is basically the real dimension of the moduli space $\mathcal{M}_g$.
We can really have many different complex manifolds that are as smooth manifolds diffeomorphic. Roughly speaking, different complex manifolds whose underlying smooth manifolds are diffeomorphic are called **Deformations** of each other. These deformations are parametrized by another complex analytic space namely the moduli space of complex manifolds. In general it is really hard to study the topology and geometry of moduli space globally of higher

dimensional complex manifolds. So geometers shifted their emphasis on the *local moduli* spaces, the fibres of whose points are called *local deformations* of complex manifolds.
This study started with the pioneering work of Kunihiko Kodaira and Donald C. Spencer. Their work concerning deformations of complex manifolds are recorded in a beautiful and comprehensive text [6]. Recall how we equip analytic structures on topological manifolds, the guiding principle was to declare the transition functions on local charts to be analytic. So for complex manifolds the transition functions must be holomorphic functions of several complex variables. So *holomorphically* varying these transition functions would mean that the complex coordinate charts are being varied as transition functions are gotten through composing chart maps. This is, in effect, actually varying the *complex structures* on the manifold. This is the fundamental philosophy of Kodaira and Spencer. They claimed that the *local deformations* are gotten through varying the transition functions with respect to some parameters or *moduli*. Then the complex manifolds will be local deformations of each other upon the variation of the transition maps. This vantange point makes *visualizing* the process of locally deforming complex structures a bit easier for us. Since the transition function essentially glues open (poly)disks of $\mathbb{C}^n$ to form the complex manifold. Varying the transition function would then mean then the same (poly)disks are being glued differently. Thus, locally deforming complex structures boils down to gluing (poly)disks differently.
They built a mapping that made it possible to think of such complex deformations as elements in the first sheaf cohomology group of a particular space (that we will define here in this section) where the coefficient sheaf is a sheaf of (germs of) holomorphic vector fields.

18 An interlude on p-adic numbers

The p-adic number field $\mathbb{Q}_p$ (for a prime p) is obtained by completing the field $\mathbb{Q}$ by a non-archimedean absolute value called the p-adic absolute value. Why do we want to introduce a different number field? Well, it turns out that such number field and their ring of integers are not only of immense importance to number theory but also they help us make deep connections of number theoretic i.e. *arithmetic* objects with objects in complex geometry, that we will show in the next section.
We will construct $\mathbb{Q}_p$ from two point of view, namely the analytic point of view and the algebraic one. In the analytic approach, we mimic the construction of $\mathbb{R}$ from $\mathbb{Q}$ that is done via completing the field $\mathbb{Q}$ with the usual absolute value which is an *archimedean* absolute value. In case the notion of *archimedean* absolute value is unfamiliar to the reader, an absolute value is called *nonarchimedean* if it satisfies the so called *ultrametric* inequality instead of the good old triangle inequality i.e. for $x, y \in \mathbb{Q}$,

$$|x + y| \leq \max\{|x|, |y|\}$$

which is a stronger condition than the triangle inequality.
The p-adic valuation of a rational number a is defined as $\nu_p(a) = r$ where $a = p^r \frac{m}{n}$ with p being coprime with m, n. Then the p-adic valuation of a is given as

$$|a|_p := p^{-\nu_p(a)}$$

Armed with absolute values, we can define *metric* on $\mathbb{Q}$ with this absolute value. Recall that from $\mathbb{Q}$ we can get to the field $\mathbb{R}$ via the following ring isomorphism

$$C/I \simeq \mathbb{R}$$

where C is the ring of Cauchy sequences in $\mathbb{Q}$ and I is the maximal ideal C consisting of all the Cauchy sequences that converge to zero with respect to the usual absolute value.
Now replace that absolute value with the p-adic one, we get a field

$$C/I \simeq \mathbb{Q}_p$$

Because of the ultrametric property of the p-adic absolute value, analysis on the field $\mathbb{Q}_p$ is wildly different from the typical archimedean analysis (on the fields $\mathbb{R}, \mathbb{C}$) that we learned when we were kids.
Like any number field, the nonarchimedean field $\mathbb{Q}_p$ has got a ring integers that is defined as

$$\mathbb{Z}_p := \{x \in \mathbb{Q}_p : |x|_p \leq 1\} \tag{195}$$

whose field of fractions is exactly $\mathbb{Q}_p$. Interestingly it has got $\mathbb{F}_p$ as it's residue field, namely

$$\mathbb{Z}_p/p\mathbb{Z}_p \simeq \mathbb{F}_p$$

As we will show later in this section, we can go the other way around. That is, we can start with the finite field $\mathbb{F}_p$ and then forming the ring of Witt vectors for this field i.e. $W(\mathbb{F}_p)$, it turns out that $W(\mathbb{F}_p)$ is exactly $\mathbb{Z}_p$. Then forming the field of fraction we get to $\mathbb{Q}_p$. We will get to this later.
But now, we look at the mere algebraic construction. Essentially, the ring of p-adic integers $\mathbb{Z}_p$ can be built from the ring $\mathbb{Z}$ via taking *inverse limit of the inverse system of quotients of* $\mathbb{Z}$ *by the ideals* $\langle p^n \rangle$ *for varying* $n \in \mathbb{N}$ which we explain below.
We start with the notion of inverse limits which are defined for *inverse systems* of algebraic objects (in this we will be considering rings). Consider a collection of rings and ring homomorphisms $(R_n, f_n)_{n \in \mathbf{N}}$ such that they fit into the following sequence

$$.... \to R_{n+1} \overset{f_n}{\to} R_n \overset{f_{n-1}}{\to} R_{n-1} \overset{f_{n-2}}{\to} .... \overset{f_1}{\to} R_1 \overset{f_0}{\to} R_0 \tag{196}$$

We call this collection an inverse system of rings. For such an inverse system the inverse limit is defined as

$$\varprojlim_n R_n := \{\prod_n R_n : f_n(r_{n+1}) = r_n\}$$

which is still a ring (of infinite sequence of elements from the rings of the inverse system) but with different addition and multiplication operations.
As a concrete example, we can think of the ring of formal power series in a variable x with coefficients in the field k, $k[[x]]$ is built by taking the inverse limit of the inverse system

$$.... \to k[x]/\langle x^n \rangle \xrightarrow{f_{n-1}} k[x]/\langle x^{n-1} \rangle \xrightarrow{f_{n-2}} .... \xrightarrow{f_1} k[x]/\langle x \rangle \xrightarrow{f_0} k[x]$$

Then clearly taking their product respecting the ring homomorphisms we get the ring $k[[x]]$.
Now let us construct another (which will be central in this section) concrete inverse system that looks like the following

$$.... \to \mathbb{Z}/p^{n+1}\mathbb{Z} \xrightarrow{f_n} \mathbb{Z}/p^n\mathbb{Z} \xrightarrow{f_{n-1}} \mathbb{Z}/p^{n-1}\mathbb{Z} \xrightarrow{f_{n-2}} .... \xrightarrow{f_1} \mathbb{Z}/p\mathbb{Z} \xrightarrow{f_0} \{0\} \tag{197}$$

where the ring homomorphism acts as

$$f_n : z + p^{n+1}\mathbb{Z} \mapsto z + p^n\mathbb{Z}$$

so the homomorphism merely decreases the prime powers in the ideals i.e. $\langle p^{n+1} \rangle \mapsto \langle p^n \rangle$
Then taking the inverse limit

$$\mathbb{Z}_p := \varprojlim_n \mathbb{Z}/p^n\mathbb{Z} = \{\prod_n \mathbb{Z}/p^n\mathbb{Z} : f_n(z + \langle p^{n+1} \rangle) = (z + \langle p^n \rangle)\} \tag{198}$$

is essentially a ring of finite sequences from each of the sequence from the inverse system and each term in the sequences are related to their adjacent terms via the homomorphisms of the inverse system. These are called p-adic integers. It is curious that on this ring $\mathbb{Z}_p$ a topology can be equipped that is generated by sets of the form

$$z + p^n\mathbb{Z}$$

We say that we have p-adically completed the ring $\mathbb{Z}$ and ended up with a *topological ring* $\mathbb{Z}_p$.
But how do these strange integers look like? Well, they are like p-ary (as in bi-nary) integers that have *infinitely many digits*. Let's dive deep into that.

A p-adic digit d is a natural number less than p. A p-adic integer is then an infinite sequence of such digits $(d_i)_{i=0}^{\infty}$ which has infinite power series (aka formal power series) expansion

$$\sum_{i=0}^{\infty} d_i p^i = ...d_2 d_1 d_0$$

where on the right hand side is how p-adic integers are conventionally written, the dots on the left represents the infinitely many other digits. Looking at the

construction of $\mathbb{Z}_p$ in (185), the infinite product of the pieces $\mathbb{Z}/p^n\mathbb{Z}$ with the *power decreasing ring homomorphisms* now seems to make sense and we can say that what we really have is the set

$$\mathbb{Z}_p := \{\sum_{i=0}^{\infty} d_i p^i = ...d_2 d_1 d_0 : 0 \leq d < p\} \tag{199}$$

with termwise addition and multiplication modulo p *with carrying digits*, for which $\mathbb{Z}_p$ is a ring. Let us elaborate their strange addition and multiplication.

Take two p-adic integers $....a_2a_1a_0$and $....b_2b_1b_0$. Their sum will be another p-adic integer i.e. $....c_2c_1c_0 = ...a_2a_1a_0 + ...b_2b_1b_0$ that is done modulo p. Indeed, the rightmost digits

$$a_0 + b_0 \equiv c_0 (\text{mod } p)$$

and the rest of the digits

$$a_i + b_i + \text{carry}_{i-1} \equiv c_i (\text{mod } p)$$

Below two 5-adic integers are considered

$$\begin{array}{r} \cdots 34121 \\ +\cdots 03432 \\ \hline \cdots 43103 \end{array}$$

Multiplication is done similarly, below a 5-adic multiplication is shown

$$\begin{array}{r} \cdots 20441 \\ \times\cdots 12334 \\ \hline \cdots 33776 \\ \cdots 4988 \\ \cdots 612 \\ \cdots 82 \\ \cdots 1 \\ \hline \cdots 42444 \end{array}$$

.....
Write about $\mathbb{Q}_p$
.....
Let's ask how we can get to finite fields from the ring of p-adic integers as it's residue fields. Basically there exists a short exact sequence

$$0 \to \mathbb{Z}_p \xrightarrow{p^m} \mathbb{Z}_p \xrightarrow{\epsilon_m} \mathbb{Z}/p^m\mathbb{Z} \to 0$$

where the map p^m is multiplication by p^m and ϵ_m is the projection map.
proof of exactness:
Proof. (taken from [9]) First let us check that multiplication-by-p on $\mathbb{Z}_p$ is injective. Suppose that $a = (a_n) \in \mathbb{Z}_p$ is in the kernel. Then $pa = 0$, so $pa_n = 0$ in

$\mathbb{Z}/p^n\mathbb{Z}$ for all n. In particular, $pa_{n+1} = 0$ in $\mathbb{Z}/p^{n+1}\mathbb{Z}$. That means that $a_{n+1} = p^n y_{n+1}$ for some $y_{n+1} \in \mathbb{Z}/p^{n+1}\mathbb{Z}$. But then $a_n = f_n(a_{n+1}) = p^n f_n(y_{n+1}) = 0$ in $\mathbb{Z}/p^n\mathbb{Z}$. This holds for all n, so $a = 0$.
Exactness on the left: Since multiplication-by-p is injective, composing this with itself m times shows that multiplication-by-p^m is injective.
Exactness on the right: Given an element $\beta \in \mathbb{Z}/p^m\mathbb{Z}$, choose an integer b that represents β. Then the constant sequence b represents an element of $\mathbb{Z}_p$ mapping to β.
Exactness in the middle: If $a \in \mathbb{Z}_p$, then $\epsilon_m(p^m a) = p^m \epsilon(a) = 0$ in $\mathbb{Z}/p^m\mathbb{Z}$. Thus the image of the incoming arrow (multiplication-by-p^m) is contained in the kernel of the outgoing arrow (ϵ_m).

Conversely, suppose that $x = (x_n)$ is in the kernel of ϵ_m. So $x_m = 0$. Then for all $n \geq m$, we have $x_n \in p^{n-m}\mathbb{Z}/p^n\mathbb{Z}$. So there is a unique y_{n-m} mapping to x_n via the isomorphism

$$\frac{\mathbb{Z}}{p^{n-m}\mathbb{Z}} \xrightarrow{\cdot p^m} \frac{p^m\mathbb{Z}}{p^n\mathbb{Z}} \xrightarrow{\text{id}} \frac{\mathbb{Z}}{p^n\mathbb{Z}}$$

These y_{n-m} are compatible (because the x_n are), so as n ranges through integers $\geq m$, they form an element $y \in \mathbb{Z}_p$ such that $p^m y = x$. So x is in the image of multiplication-by-p^m. □

So for $m = 1$ we get the isomorphism

$$\mathbb{Z}_p/p\mathbb{Z}_p \simeq \mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$$

Vector spaces over the field $\mathbb{Q}_p$ are of special interest because of their usage in representations of absolute Galois groups of fields. Namely, if we have a field K then the absolute Galois group of extensions of K, $G_K = \text{Gal}(\bar{K}/K)$ consisting of automorphisms of (the separable closure) $\bar{K}$ fixing elements of K, then we have the representation

$$\rho : G_K \to \text{Aut}_{\mathbb{Q}_p}(V) \tag{200}$$

for a $\mathbb{Q}_p$ vector space V. As such these are called p-adic Galois representations. The field of rational numbers $\mathbb{Q}$ has the interesting property that it's Galois group $G_{\mathbb{Q}} = \text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ has a representation to the automorphism group of the cyclic group of p^n-th roots of unity. With varying n, we can construct an inverse system from which, taking the inverse limit, we can get to another p-adic representation of the Galois group $G_{\mathbb{Q}}$.

Fix a primitive p^n-th root ζ_{p^n} that generates the following cyclic group

$$\mu_{p^n} := \{\zeta \in \bar{\mathbb{Q}}^\times : \zeta^{p^n} = 1\} = \langle \zeta_{p^n} \rangle \tag{201}$$

For any $x \in \mathbb{Q}$, the so called Galois conjugates of x are elements of the separable closure $\bar{\mathbb{Q}}$ which are in the orbit of x under the group action of the absolute Galois group $G_{\mathbb{Q}}$ i.e. elements of the orbit $G_{\mathbb{Q},x} := \{\sigma \cdot x : \sigma \in G_{\mathbb{Q}}\}$. So

all primitive roots in the group μ_{p^n} are Galois conjugates, the absolute Galois group acts on the group of roots of unity by automorphisms

$$G_{\mathbb{Q}} \times \mu_{p^n} \to \mu_{p^n}$$

such that $\sigma \cdot \zeta := \zeta_{p^n}^{a(\sigma,n)}$ for some $a(\sigma, n) \in (\mathbb{Z}/p^n\mathbb{Z})^\times$ depending on the Galois automorphism $\sigma \in G_{\mathbb{Q}}$ and p.
We have the following chain isomorphisms

$$\mathrm{Aut}(\mu_{p^n}) \simeq (\mathbb{Z}/p^n\mathbb{Z})^\times \simeq \mathrm{GL}_1(\mathbb{Z}/p^n\mathbb{Z}) \tag{202}$$

using which we view the group homomorphism

$$\rho_{p^n} : G_{\mathbb{Q}} \to (\mathbb{Z}/p^n\mathbb{Z})^\times \tag{203}$$

as to any Galois automorphism $\sigma \in G_{\mathbb{Q}}$ we can assign an $a(\sigma, n) \in (\mathbb{Z}/p^n\mathbb{Z})^\times$, called the *mod p^n cyclotomic character*, as a Galois representation

$$G_{\mathbb{Q}} \to \mathrm{GL}_1(\mathbb{Z}/p^n\mathbb{Z})$$

In this sense, the mod p^n cyclotomic character can be seen as a character of this representation.
Now if we vary only n while keeping the Galois automorphism σ and the prime p we get the inverse system

$$\ldots \to \left\{ \ \rho_{p^{n+1}} : G_{\mathbb{Q}} \to (\mathbb{Z}/p^{n+1}\mathbb{Z})^\times \ \right\} \to \left\{ \ \rho_{p^n} : G_{\mathbb{Q}} \to (\mathbb{Z}/p^n\mathbb{Z})^\times \ \right\} \to \ldots$$

taking the inverse limits

$$\varprojlim_n \rho_{p^n} : G_{\mathbb{Q}} \to \varprojlim_n (\mathbb{Z}/p^n\mathbb{Z})^\times$$

we get the p-adic cyclotomic character

$$\rho_p : G_{\mathbb{Q}} \to (\mathbb{Z}_p)^\times \simeq \mathrm{GL}_1(\mathbb{Z}_p)$$

19 Comparisons with arithmetic analogues

In 1949 something outstanding were conjectured in community of number theorists. What's striking is that these conjectures and their proofs in case of curves which were the evidences for these conjectures suggested that methods from algebraic geometry and algebraic topology can find their applications in solving extremely hard number theory problems! This is nothing short of extraordinary and at the same not so surprising since number theory is the branch of mathematics that borrows tools from other branches of mathematics in oftentimes extremely unexpected ways. Our hero is called André Weil, a french mathematician who conjectures these beautiful and surprising statements all

of which are now proved to be theorems. This sparked perhaps the most beautiful episode of mathematical research in the history of the discipline.
The Weil conjectures connect point counting of projective varieties over finite fields to the topology of smooth projective varieties over $\mathbb{C}$. It says that if the variety over finite field is the *reduction modulo p* (for some prime number p) of a complex projective variety (whose analytification is a compact complex manifold in the spirit of Serre's GAGA theorem), then from the number of points of former variety, we can determine the topological genus of the latter and vice versa!
The starting point of these conjectures was deeply rooted in the so called zeta function of any projective nonsingular variety $V/\mathbb{F}_q$, of dimension d defined over a finite field $\mathbb{F}_q$ where $q = p^m$ for some prime number p.
Let $(\mathbb{F}_q)_r$ be the r-degree extension of the field $\mathbb{F}_q$ i.e. $[(\mathbb{F}_q)_r : \mathbb{F}_q] = r$. Let N_r be the number of points of $V_r = V/(\mathbb{F}_q)_r$ i.e. $N_r = |V_r|$. Then the following formal power series

$$Z_{V/\mathbb{F}_q}(T) := \exp(\sum_{r=0}^{\infty} \frac{N_r}{r} T^r) \tag{204}$$

is called the Weil zeta function attached to the variety $V/\mathbb{F}_q$.
The first of the Weil conjectures states that this function is a rational function, that is there are polynomials $P_0(T), P_1(T), ..., P_{2d}(T) \in \mathbb{Z}[T]$ such that

$$Z_{V/\mathbb{F}_q}(T) = \frac{P_1(T)...P_{2d-1}(T)}{P_0(T)...P_{2d}(T)} \in \mathbb{Q}[[T]]$$

Now here's the connection with topology of complex manifolds, which was also conjectured by Weil. If the variety $V/\mathbb{F}_q$ is the reduction modulo p of some complex projective variety $\mathcal{V}/\mathbb{C}$, whose analytification is the complex manifold $\mathcal{V}^{an}$, then the degree of the i-th polynomial $P_i(T)$ is exactly the i-th Betti number of $\mathcal{V}^{an}$ i.e.

$$\deg(P_i(T)) = b_i(\mathcal{V}^{an})$$

This hinted a powerful bridge between arithmetic (number theory) and complex topology.
The polynomials $P_i(T)$ are themselves very peculiar. Like it must be that $P_0(T) = 1 - T$ and $P_{2d}(T) = 1 - q^d T$ for some $q \in \mathbb{Z}$. Quite amazingly, these polynomials also have complex factorization:

$$P_i(T) = \prod_j (1 - \alpha_{ij} T) \tag{205}$$

where $\alpha_{ij} \in \mathbb{C}$ such that $|\alpha_{ij}| = q^{i/2} \in \mathbb{Z}$ for $1 \leq i \leq 2d - 1$. Note that this implies that the complex zeros of the polynomial $P_i(T)$ *lives in a restricted region*. That's why this last statement was called the Riemann hypothesis for varieties over finite fields. In fact, it really coincides with the famous Riemann hypothesis when we consider the *arithmetic* variety $\mathrm{Spec}(\mathbb{Z})$ as then the zeta function becomes the Riemann zeta function, for this we need to develop some parts

of scheme theory that we will do in the last part of this section. We will not show detailed proofs of the Weil conjectures, we will demonstrate some example cases and some of the mechanisms of the machinery and specifically the interplay with complex manifolds.
As the first case for demonstration, we consider elliptic curves over some $\mathbb{F}_q$ that are nonsingular (over finite fields, nonsingularity is peculiar notion that we will discuss later) projective varieties and their analytification is the complex torus $T(\mathbb{C})$. Since an elliptic curve $E/\mathbb{F}_q$ is a curve first of all, it's zeta function looks like

$$Z_{E/\mathbb{F}_q}(T) = \frac{P_1(T)}{P_0(T)P_2(T)} \tag{206}$$

where $P_0(T) = 1 - T, P_2(T) = 1 - qT$. The connection with complex topology of the zeta function implies that

$$\deg(P_0(T)) = b_0(T(\mathbb{C})) = 1$$

$$\deg(P_1(T)) = b_1(T(\mathbb{C})) = 2$$

$$\deg(P_2(T)) = b_2(T(\mathbb{C})) = 1$$

since any g-genus compact surface Σ_g has the Betti numbers $b_0(\Sigma_g) = 1, b_1(\Sigma_g) = 2g, b_2(\Sigma_g) = 1$.
So we conclude that the polynomial $P_1(T)$ must be quadratic in T. It's complex factorization is

$$P_1(T) = (1 - \alpha_1 T)(1 - \alpha_2 T)$$

and the Riemann hypothesis states that $|\alpha_1| = |\alpha_2| = q^{1/2}$ where $q \in \mathbb{Z}$. To maintain the condition that $P_1(T) \in \mathbb{Z}[T]$, we see that the complex numbers α_1, α_2 must be complex conjugates of each other and their real parts must be an integer. That means

$$P_1(T) = 1 - \mathrm{Re}(\alpha_1)T + |\alpha_1|^2 T^2$$

Recall from (183) that the zeta function is essentially the generating function of the point -counts of the variety over various degree field extensions, so the number of points of the variety V_r (over the extended field $(\mathbb{F}_q)_r$) of degree r is

$$N_r = \frac{1}{(r-1)!}\frac{d^r}{dT^r} Z_{V/\mathbb{F}_q}(T)|_{T=0} \tag{207}$$

So for a given extension of $\mathbb{F}_q$ of some degree r, we see that the number of points of our elliptic curve $E/\mathbb{F}_q$ is

$$N_r = \frac{1}{(r-1)!}\frac{d^r}{dT^r}\frac{1 - \mathrm{Re}(\alpha_1)T + |\alpha_1|^2 T^2}{(1-T)(1-|\alpha_1|^2 T)}|_{T=0}$$

where the integer $\mathrm{Re}(\alpha_1)$ depends on the prime p for which $p^r = q$. We will later address this dependence in case of eilliptic curves. For the moment we

will focus on analogy of Weil's conjectures namely the Riemann hypothesis for complex manifolds. This was proven by Jean P. Serre.
The setup goes as follows. Let X be a projective nonsingular variety over $\mathbb{C}$ and $X(\mathbb{C})^{an}$ be it's analytification i.e. the associated complex manifold. Recall that this means that we are applying the GAGA analytification functor that merely changes the topology from the stronger complex topology to the weaker Zariski topology. Also, as Chow's theorem implies, any compact complex submanifold of the complex projective space is a projective variety over $\mathbb{C}$. But in the present discussion we can identify the variety and the manifold, thinking of them as the same. That won't bother us in our reasoning. Anyway, we also consider the hyperplane class on the complex manifold which is the cohomology class $[H] \in H^2(X(\mathbb{C})^{an}; \mathbb{Z})$.
Suppose on the variety exists an endomorphism (later in the cases of varieties over finite fields we will see that this endomorphism corresponds to Frobenius endomorphism)

$$F : X \to X$$

such that the pullback on the hyperplane class is $F^*[H] = q[H]$ for some positive integer q. The analogue of the Riemann hypothesis in this case is that

The eigenvalues of F^* on $H^i(X, \mathbb{C})$ all have absolute value $q^{i/2}$.

To prove ([2]) this, we need several cohomological tools that we introduce below.
The Lefschetz operator is the map

$$L : H^i(X, \mathbb{C}) \to H^{i+2}(X, \mathbb{C}) \tag{208}$$

such that for any class $\alpha \in H^i(X, \mathbb{C})$, we have $L(\alpha) := \alpha \smile [H]$. Then the complex singular cohomology decomposes

$$H^i(X, \mathbb{C}) \simeq \mathrm{im}(L) \oplus H^i_{prim} \tag{209}$$

which is a fact from **Hard Lefschetz Theorem**.
.....
Write about primitive cohomology
.....
And the primitive cohomology itself satisfies the canonical decomposition

$$H^i_{prim} = \bigoplus_{p+q=j} H^{p,q}_{prime}$$

For each of the primitive cohomology, there exists a bilinear form

$$\langle \alpha_1, \alpha_2 \rangle = (\sqrt{-1})^k \int_X \alpha_1 \wedge \bar{\alpha_2} \wedge [H]^{n-k}$$

for $\alpha_1, \alpha_2 \in H^k_{prim}(X)$.

We want to show that the eigenvalues of $F^* \cong H^k(X, \mathbb{C})$ have absolute value $q^{k/2}$. Note that it suffices to do this for H^k_{prim}. Indeed, if $\alpha \in H^{k-2}(X, \mathbb{C})$, we can inductively assume its eigenvalue has absolute value $q^{(k-2)/2}$, and then

$$F^*(\alpha \smile [H]) = F^*\alpha \smile F^*[H] = \lambda\alpha \smile q[H] = q\lambda(\alpha \smile [H])$$

so $\alpha \smile [H]$ has eigenvalue with absolute value $q^{k/2}$.

This reduces us to the primitive case. Let $\alpha \in H^k_{\text{prim}}$ be an F^*-eigenvector. Since F^* preserves the Hodge decomposition, we may assume that $\alpha \in H^{p,q}_{\text{prim}}$ s.t. $p + q = k$. This pairing is sesquilinear, so

$$|\lambda|^2(\alpha, \alpha) = (F^*\alpha, F^*\alpha) = i^k \int_X F^*\alpha \wedge \overline{F^*\alpha}[H]^{n-k}$$

$$= i^k q^n \int_X \alpha \wedge \overline{\alpha}[H]^{n-k} = q^n i^k \int_X \alpha \wedge \overline{\alpha}[H]^{n-k} = q^k(\alpha, \alpha).$$

Above, we have used the fact that $F^* \cong H^{2n}(X, \mathbb{C})$ (top degree cohomology) via multiplication by q^n. You can show this using Poincaré duality (top cohomology is generated by $[H]^n$) or Lefschetz fixed point. Since $(\alpha, \alpha) \neq 0$ (the form is definite on $H^{p,q}_{\text{prim}}$), we conclude that $q^k = |\lambda|^2$ as desired.

20 Arithmetic deformations

21 Weil cohomology

Let us denote the category of smooth projective varieties as **SmProj** and the category of graded vector spaces as .... A Weil cohomology theory is contravariant functor from the former to the latter category satisfying the number axioms.

22 On ℓ-adic Cohomology & Getting Crystallized

Recall from our brief tour through the theory of cohomology groups. We first replace the topological space X on which we are computing (sheaf) cohomology groups $\mathrm{H}^i(X, \mathcal{F})$ with the category $\mathbf{OP}_X$ whose objects are open sets of X and morphisms are inclusion maps. Then sheaves of abelian groups are contravariant functors $\mathcal{F} : \mathbf{OP}_X \to \mathbf{Ab}$, **Ab** where is the category of abelian groups. An important point to keep in mind is the category of abelian groups on a topological space X, $\mathbf{Sh}_{Ab}(X)$ is an example of an abelian category (which are categories that satisfy the properties of the category of abelian groups **Ab**). Then the right derived functors of the global section functor

$$\Gamma : \mathbf{Sh}_{Ab}(X) \to \mathbf{Ab}$$

which itself is a left exact functor, are the sheaf cohomology groups $\mathrm{H}^i(X, \mathcal{F})$. Now ℓ-adic cohomologies of varieties over finite fields is the inverse limit over

varying n of the étale cohomologies with coefficients in the locally constant sheaf $\mathbb{Z}/\ell^n\mathbb{Z}$, while the étale cohomologies are the sheaf cohomologies obtained via replacing the category $\mathbf{OP}_X$ with the so called étale site $\acute{\mathbf{E}}\mathbf{t}_X$ and considering étale (pre)sheaves. It should be stressed that ℓ should be different from the characteristic p or else the constructions yield weird results.

23 On fields of Characteristic 0 & p

There have been a number of outstanding progress in the comparative studies of objects over fields of characteristic 0 and those of characteristic p. As for characteristic zero field, we pick the field $\mathbb{Q}_p$ whose elements are formal power series over the variable being p or p-ary numbers of *infinite size* if you like,

$$\sum_{i=0}^{\infty} a_i p^i$$

where $a_i \in \{0, 1, ..., p-1\}$. These are reminiscent to Laurent series expansions over finite fields which are elements of the field $\mathbb{F}_p[[t]]$

$$\sum_{j=0}^{\infty} c_j t^j$$

where the coefficients are elements of $\mathbb{F}_p$. Definitely, the two fields $\mathbb{Q}_p$ and $\mathbb{F}_p[[t]]$ are far from being isomorphic. But a striking result known as the Fontaine-Wintenberger theorem claims that the following two fields

$$\mathbb{Q}_p(p^{1/p^\infty})$$

and

$$\mathbb{F}_p[[t^{1/p^\infty}]]$$

have isomorphic absolute Galois groups. The notations here $\mathbb{Q}_p(p^{1/p^\infty})$ means $\bigcup_{n\geq 0} \mathbb{Q}_p(p^{1/p^n})$ and similarly for the latter field.

Moreover, p-adically completing the field $\mathbb{Q}_p(p^{1/p^\infty})$ and t-adically completing the field $\mathbb{F}_p[[t^{1/p^\infty}]]$ does not have any effect on their Galois group.

24 Why you shouldn't hate an Algebraic Geometer

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